Liu, Chapter 3: Morphisms and base change

Solutions to the exercises

Conventions. Rings are commutative with 1. Results of the text are cited by their numbers in Liu ("Prop. 3.9", "Cor. 3.32", ...), and results of Chapters 1 and 2 as "Prop. 2.3.12" etc. Exercise labels are those of the book, numbered within each section of the chapter: for instance Exercise 2.11 is the eleventh exercise at the end of Section 3.2. Each exercise is restated before its solution; sub-items that the book displays as nested lists are written as running prose.

Honest notes. Several printed statements need a correction or an interpretation, and each solution says what is proved: Ex. 1.9(b) (the intersection condition is read as nonempty), Ex. 1.10(e) (false for finite k; proved with the needed hypothesis), Ex. 2.21 (needs finite presentation), Ex. 3.5 (needs a nonempty scheme), Ex. 3.13(b) (false for a non-separated base, with the doubled-origin counterexample; the corrected version is proved), and Ex. 3.25 (false as printed, with a counterexample over $\mathbf{Z}$). Some solutions cite standard commutative-algebra facts (for instance the dimension formula for polynomial rings) which are named where used.

3.1 The technique of base change

Throughout, results of the text are cited by their numbers: Prop. 1.2, Prop. 1.4, Prop. 1.16, Prop. 1.23 are those of Section 3.1; references to earlier chapters carry the chapter, for instance Lemma 2.1.7 (spectra and ring homomorphisms), Prop. 2.1.9 (Noether normalization), Cor. 2.1.12 (residue fields of finitely generated algebras), Prop. 2.3.25 (morphisms to an affine scheme), Lemma 2.3.40 (Proj and graded homomorphisms), Prop. 2.4.12 (generic points). Several exercises use Exercise 1.7; its proof is given in full below.

Exercise 1.1

> **Problem.** Let S, X, Y be sets, $\varphi : X \to S$, $\psi : Y \to S$ maps, $X \times_S Y = \{(x, y) \in X \times Y \mid \varphi(x) = \psi(y)\}$, and let $p : X \times_S Y \to X$, $q : X \times_S Y \to Y$ be the restrictions of the projections. Show that for any set Z and any pair of maps $f : Z \to X$, $g : Z \to Y$ with $\varphi \circ f = \psi \circ g$ there is a unique map $(f, g) : Z \to X \times_S Y$ with $p \circ (f, g) = f$ and $q \circ (f, g) = g$.

Solution. *Existence.* Put $(f, g)(z) = (f(z), g(z))$. Since $\varphi(f(z)) = \psi(g(z))$, this pair lies in $X \times_S Y$, and clearly $p \circ (f, g) = f$, $q \circ (f, g) = g$.

Uniqueness. Let $h : Z \to X \times_S Y$ satisfy $p \circ h = f$ and $q \circ h = g$. For $z \in Z$ write $h(z) = (x, y)$. Then $x = p(h(z)) = f(z)$ and $y = q(h(z)) = g(z)$, so $h(z) = (f(z), g(z))$. Hence $h = (f, g)$. $\square$

Exercise 1.2

> **Problem.** Let $f : X \to Y$ be a morphism of schemes. For a scheme T let $f(T) : X(T) \to Y(T)$ be the map $g \mapsto f \circ g$ (here $X(T) = \mathrm{Mor}(T, X)$). Show that $f(T)$ is bijective for every T if and only if f is an isomorphism.

Solution. If f is an isomorphism with inverse h, then $g \mapsto h \circ g$ is a two-sided inverse of $f(T)$, because $h \circ f = \mathrm{Id}_X$ and $f \circ h = \mathrm{Id}_Y$ (so $h \circ (f \circ g) = g$ and $f \circ (h \circ g') = g'$).

Conversely, assume $f(T)$ is bijective for all T. Take $T = Y$. Surjectivity of $f(Y)$ applied to $\mathrm{Id}_Y \in Y(Y)$ gives $h : Y \to X$ with $f \circ h = \mathrm{Id}_Y$. Now take $T = X$. Both $h \circ f$ and Id_X belong to $X(X)$ and

$$f(X)(h \circ f) = f \circ h \circ f = f = f \circ \mathrm{Id}_X = f(X)(\mathrm{Id}_X).$$

By injectivity of $f(X)$, $h \circ f = \mathrm{Id}_X$. So h is a two-sided inverse of f. □

Exercise 1.3

> **Problem.** Let $f : X \to Y$ be a dominant morphism of irreducible schemes. Show that the generic fiber of f is dense in X.

Solution. Let ξ be the generic point of Y and η that of X (Prop. 2.4.12(a): an irreducible scheme has a unique generic point). By Prop. 1.16 the first projection $X_\xi \to X$ is a homeomorphism onto $f^{-1}(\xi)$, so the statement means that $f^{-1}(\xi)$ is dense in X.

By continuity, $f(X) = f(\overline{\{\eta\}}) \subseteq \overline{\{f(\eta)\}}$. Since f is dominant, $\overline{f(X)} = Y$, hence $\overline{\{f(\eta)\}} = Y$. So $f(\eta)$ is a generic point of Y, and by uniqueness $f(\eta) = \xi$. Thus $\eta \in f^{-1}(\xi)$. As η is dense in X, so is $f^{-1}(\xi)$. □

Exercise 1.4

> **Problem.** Let $f : X \to Y$ be a morphism of affine schemes. Show that for any affine open subscheme V of Y, $f^{-1}(V)$ is affine.

Solution. Write $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$. The inclusion $j : V \to Y$ is an open immersion and V is an affine scheme, say $V = \operatorname{Spec} C$. Consider $X \times_Y V$, where V is a Y-scheme via j. By Prop. 1.2, $X \times_Y V = \operatorname{Spec}(B \otimes_A C)$ is affine.

On the other hand, Prop. 1.4(d), applied to the Y-schemes $X \to Y$ and $\mathrm{Id}_Y : Y \to Y$ and to the open subschemes $X \subseteq X$ and $V \subseteq Y$, shows that $X \times_Y V$ is isomorphic to $p^{-1}(X) \cap q^{-1}(V) \subseteq X \times_Y Y$, where p, q are the projections. Under the canonical isomorphism $X \times_Y Y \simeq X$ of Prop. 1.4(a), p becomes Id_X and q becomes f. Hence $X \times_Y V \simeq f^{-1}(V)$ as schemes (this is the argument of Prop. 1.23 for open immersions). Therefore $f^{-1}(V) \simeq \operatorname{Spec}(B \otimes_A C)$ is affine. □

Exercise 1.5

Problem. Let X, Y be algebraic varieties over a field k. Show that $\dim(X \times_k Y) = \dim X + \dim Y$ (use Prop. 2.1.9).

Solution. We only use that X, Y are nonempty schemes of finite type over k (separatedness is not needed). The proof has three steps. For a ring R we use the standard facts: if $R_0 \to R$ is injective and finite then $\dim R = \dim R_0$ (lying over, going-up, incomparability), and $\dim k[T_1, \ldots, T_m] = m$.

Step 1: dimension of an open cover. Let X be a topological space covered by opens U_i. Then $\dim X = \sup_i \dim U_i$. Indeed $\geq$ is clear (a chain of irreducible closed subsets of U_i gives one in X by taking closures, which stay distinct since $Z = \overline{Z} \cap U_i$ for Z closed in U_i and the closure of an irreducible set is irreducible). For $\leq$, take a chain $Z_0 \subsetneq \cdots \subsetneq Z_n$ of irreducible closed subsets of X, and choose i with $U_i \cap Z_0 \neq \emptyset$. Each $Z_j \cap U_i$ is a nonempty open subset of the irreducible space Z_j, hence irreducible and dense in Z_j; so $Z_j = \overline{Z_j \cap U_i}$ and the $Z_j \cap U_i$ form a strictly increasing chain of irreducible closed subsets of U_i.

Step 2: the affine case. Let A, B be nonzero finitely generated k-algebras. We claim $\dim(A \otimes_k B) = \dim A + \dim B$.

First note: if A', B' are nonzero finitely generated k-algebras, $a = \dim A'$, $b = \dim B'$, then $\dim(A' \otimes_k B') = a + b$. Indeed, by Prop. 2.1.9 there are finite injective maps $A_0 = k[x_1, \ldots, x_{a'}] \to A'$ and $B_0 = k[y_1, \ldots, y_{b'}] \to B'$, and by invariance of dimension under finite injective maps $a' = a$, $b' = b$. The induced map $A_0 \otimes_k B_0 \to A' \otimes_k B'$ is injective (tensor products over a field of injective linear maps are injective: factor it as $A_0 \otimes B_0 \to A' \otimes B_0 \to A' \otimes B'$ and use that vector spaces are flat) and finite (if e_i generate A' over A_0 and f_j generate B' over B_0, then the $e_i \otimes f_j$ generate $A' \otimes B'$ over $A_0 \otimes B_0$). Since $A_0 \otimes_k B_0 = k[x_1, \ldots, x_a, y_1, \ldots, y_b]$, we get $\dim(A' \otimes B') = a + b$.

Lower bound. Choose a minimal prime $\mathfrak{p}$ of A with $\dim A/\mathfrak{p} = \dim A$ (the first member of a chain of maximal length), and likewise $\mathfrak{q} \subset B$. The surjection $A \otimes B \to A/\mathfrak{p} \otimes B/\mathfrak{q}$ shows $\dim(A \otimes B) \geq \dim(A/\mathfrak{p} \otimes B/\mathfrak{q}) = \dim A + \dim B$ (the quotient is nonzero, being a tensor product of two nonzero vector spaces).

Upper bound. Let $P_0 \subsetneq \cdots \subsetneq P_n$ be a chain of primes of $A \otimes B$. Let $\mathfrak{p} \subset A$ and $\mathfrak{q} \subset B$ be the preimages of P_0 under $a \mapsto a \otimes 1$, $b \mapsto 1 \otimes b$. The map $A \otimes B \to (A \otimes B)/P_0$ kills $a \otimes b = (a \otimes 1)(1 \otimes b)$ whenever $a \in \mathfrak{p}$ or $b \in \mathfrak{q}$, so it factors through a surjection $A/\mathfrak{p} \otimes B/\mathfrak{q} \to (A \otimes B)/P_0$. The chain P_i/P_0 pulls back to a chain of length n of primes of $A/\mathfrak{p} \otimes B/\mathfrak{q}$, so $n \leq \dim(A/\mathfrak{p} \otimes B/\mathfrak{q}) = \dim A/\mathfrak{p} + \dim B/\mathfrak{q} \leq \dim A + \dim B$.

Step 3: general case. Cover X and Y by affine opens $U_i = \operatorname{Spec} A_i$, $V_j = \operatorname{Spec} B_j$; the A_i, B_j are finitely generated k-algebras. By Prop. 1.4(d) and Prop. 1.2, the open subsets $p^{-1}(U_i) \cap q^{-1}(V_j)$ of $X \times_k Y$ are isomorphic to $U_i \times_k V_j = \operatorname{Spec}(A_i \otimes_k B_j)$ and cover $X \times_k Y$. By Step 1 (used for $X \times_k Y$, X and Y) and Step 2,

$$\dim(X \times_k Y) = \sup_{i,j} \dim(A_i \otimes B_j) = \sup_{i,j} (\dim A_i + \dim B_j) = \sup_i \dim A_i + \sup_j \dim B_j = \dim X + \dim Y.$$

(All these numbers are finite by Prop. 2.1.9.) □

Exercise 1.6

Problem. Let $\pi : T \to S$ be a morphism of schemes. Suppose that π is an open or closed immersion, or that $S = \operatorname{Spec} A$ and π is induced by a localization $A \to F^{-1}A$ for a multiplicative set $F \subset A$.

(a) Let $f, g : Z \to T$ be two morphisms of schemes such that $\pi \circ f = \pi \circ g$. Show that $f = g$.

(b) Let X, Y be T-schemes. Show that the canonical morphism $X \times_T Y \to X \times_S Y$ is an isomorphism.

(c) Show that (b) can be false if the hypothesis on $T \to S$ is not satisfied.

Solution. (a) We use the following fact. *Two morphisms of schemes $f, g : Z \to T$ coincide as soon as they have the same underlying continuous map and the same stalk maps $f_z^\# = g_z^\# : \mathcal{O}_{T,f(z)} \to \mathcal{O}_{Z,z}$ for all $z \in Z$.* Indeed, for an open $W \subseteq T$ and $s \in \mathcal{O}_T(W)$, the sections $f^\#(s)$ and $g^\#(s)$ of $\mathcal{O}_Z(f^{-1}W)$ have germs $f_z^\#(s_{f(z)}) = g_z^\#(s_{f(z)})$ at every point z, hence are equal.

Next, in each of the three cases the map π is injective on points, and every stalk map $\pi_t^\# : \mathcal{O}_{S,\pi(t)} \to \mathcal{O}_{T,t}$ is surjective. For an open immersion, π is a homeomorphism onto an open set and $\pi_t^\#$ is an isomorphism. For a closed immersion, π is a homeomorphism onto a closed set, $\mathcal{O}_S \to \pi_*\mathcal{O}_T$ is a surjective map of sheaves, and $(\pi_*\mathcal{O}_T)_{\pi(t)} = \mathcal{O}_{T,t}$, so $\pi_t^\#$ is surjective. For a localization, π is injective by Lemma 2.1.7(c), and for a prime $\mathfrak{p}$ of A with $\mathfrak{p} \cap F = \emptyset$ the stalk map is $A_\mathfrak{p} \to (F^{-1}A)_{F^{-1}\mathfrak{p}} = A_\mathfrak{p}$, an isomorphism.

Now let $\pi \circ f = \pi \circ g$. As maps of sets, $\pi(f(z)) = \pi(g(z))$ gives $f(z) = g(z)$ by injectivity. For the stalk maps, $(\pi \circ f)_z^\# = f_z^\# \circ \pi_{f(z)}^\#$ and likewise for g, so $f_z^\# \circ \pi_{f(z)}^\# = g_z^\# \circ \pi_{f(z)}^\#$, and surjectivity of $\pi_{f(z)}^\#$ gives $f_z^\# = g_z^\#$. By the fact above, $f = g$.

(b) Let $a : X \to T$, $b : Y \to T$ be the structure morphisms, so X, Y are S-schemes via $\pi \circ a$, $\pi \circ b$. Let $W = X \times_T Y$ with projections p_W, q_W, and $Z = X \times_S Y$ with projections p, q. The canonical morphism $\varphi : W \to Z$ is (p_W, q_W) (it exists since $\pi a p_W = \pi b q_W$). Since $\pi \circ (a \circ p) = \pi \circ (b \circ q)$ (definition of Z), (a) gives $a \circ p = b \circ q$. Hence p, q are morphisms of T-schemes if Z is given the structure $c = a \circ p : Z \to T$, and the universal property of W gives $\psi = (p, q) : Z \to W$ with $p_W \psi = p$, $q_W \psi = q$.

Then $p \circ (\varphi\psi) = p_W \psi = p$ and $q \circ (\varphi\psi) = q$; the uniqueness in the universal property of Z gives $\varphi\psi = \mathrm{Id}_Z$. Also $p_W \circ (\psi\varphi) = p\varphi = p_W$ and $q_W \circ (\psi\varphi) = q_W$. Moreover $\psi\varphi$ is a T-morphism: the structure morphism of W is $a \circ p_W$ and $a \circ p_W \circ \psi\varphi = a \circ p \circ \varphi = a \circ p_W$. By the uniqueness in the universal property of W over T, $\psi\varphi = \mathrm{Id}_W$. So φ is an isomorphism.

(c) Take $S = \operatorname{Spec} \mathbf{R}$, $T = X = Y = \operatorname{Spec} \mathbf{C}$, with the canonical morphism $T \to S$ (it is neither an open immersion, nor a closed immersion, nor a localization). Then $X \times_T Y = \operatorname{Spec} \mathbf{C}$ is a single point, whereas $X \times_S Y = \operatorname{Spec}(\mathbf{C} \otimes_\mathbf{R} \mathbf{C})$ has two points: see Exercise 1.10(c) below, where $\mathbf{C} \otimes_\mathbf{R} \mathbf{C} \simeq \mathbf{C} \times \mathbf{C}$ is proved. So $X \times_T Y \to X \times_S Y$ is not an isomorphism (not even surjective). □

Exercise 1.7

Problem. Let X, Y be S-schemes, p, q the projections from $X \times_S Y$ to X, Y. Fix $s \in S$. Show that for any $x \in X_s$ and any $y \in Y_s$ there is a natural homeomorphism

$$\operatorname{Spec}(k(x) \otimes_{k(s)} k(y)) \to \{z \in X \times_S Y \mid p(z) = x,\ q(z) = y\}.$$

Solution. Here $x \in X_s$ means that $x \in X$ lies over s (Prop. 1.16), so $k(s) \to k(x)$ and $k(s) \to k(y)$ are defined. Put $K = k(x) \otimes_{k(s)} k(y)$ and $F = \{z \mid p(z) = x,\ q(z) = y\}$.

The map. Let $i_x : \operatorname{Spec} k(x) \to X$ and $i_y : \operatorname{Spec} k(y) \to Y$ be the canonical morphisms. The structure morphism of X composed with i_x equals $\operatorname{Spec} k(x) \to \operatorname{Spec} k(s) \to S$, and similarly for y. Hence the compositions $\operatorname{Spec} K \to \operatorname{Spec} k(x) \to X$ and $\operatorname{Spec} K \to \operatorname{Spec} k(y) \to Y$ are S-morphisms (both S-structures on $\operatorname{Spec} K$ factor through $\operatorname{Spec} k(s)$), and give $u : \operatorname{Spec} K \to X \times_S Y$. The images of $p \circ u$ and $q \circ u$ are $\{x\}$ and $\{y\}$, so $u(\operatorname{Spec} K) \subseteq F$. We show that u is a homeomorphism onto F.

Reduction to the affine case. Let $V = \operatorname{Spec} A$ be an affine open neighborhood of s in S, and $X' = \operatorname{Spec} B$, $Y' = \operatorname{Spec} C$ affine open neighborhoods of x, y in the preimages of V in X, Y. By Prop. 1.4(d), $X' \times_S Y' \simeq p^{-1}(X') \cap q^{-1}(Y')$ is an open subscheme W of $X \times_S Y$, it contains F, and F is the set of points of W whose images under the projections of W are x and y. By Exercise 1.6(b) (applied to the open immersion $V \to S$; the only property used there is 1.6(a)), $X' \times_S Y' \simeq X' \times_V Y' = \operatorname{Spec}(B \otimes_A C)$ (Prop. 1.2). The morphism u factors through the open subscheme W since its image lies in $F \subseteq W$, and the resulting morphism $\operatorname{Spec} K \to W = \operatorname{Spec}(B \otimes_A C)$ is, by the universal property, the one induced by the ring homomorphism $B \otimes_A C \to k(x) \otimes_A k(y)$, $b \otimes c \mapsto \bar{b} \otimes \bar{c}$. Note $k(x) \otimes_A k(y) = K$: the ring map $A \to k(s)$ is a surjection followed by a localization, hence an epimorphism of rings, and so for $k(s)$-vector spaces M, N every A-bilinear map $M \times N \to L$ (L a $k(s)$-vector space) is $k(s)$-bilinear. (If $\lambda = \bar{a}_1/\bar{a}_2$ with $a_i \in A$, $\bar{a}_2 \neq 0$, then $a_2\beta(\lambda m, n) = \beta(a_1 m, n) = a_1\beta(m, n)$, and a_2 acts invertibly on L.) So $M \otimes_A N = M \otimes_{k(s)} N$.

Algebra. Let $P \subset B$, $Q \subset C$ be the prime ideals of x, y, and $R = B \otimes_A C$, $\bar{B} = B/P$, $\bar{C} = C/Q$, with fraction fields $k(x)$, $k(y)$. The points of F are the primes $\mathfrak{r}$ of R whose preimages in B and C (under $b \mapsto b \otimes 1$, $c \mapsto 1 \otimes c$) are P and Q. Such a prime contains the ideal J generated by the images of P and Q, and $R/J \simeq \bar{B} \otimes_A \bar{C} =: \bar{R}$ (right exactness of $\otimes$). By Lemma 2.1.7(b), $\operatorname{Spec} \bar{R} \to \operatorname{Spec} R$ is a homeomorphism onto $V(J)$, and under it F corresponds to the set of primes $\bar{\mathfrak{r}}$ of $\bar{R}$ whose preimages in $\bar{B}$ and $\bar{C}$ are 0, i.e. which contain no $\beta \otimes 1$ ($\beta \neq 0$) and no $1 \otimes \gamma$ ($\gamma \neq 0$). Since $\bar{\mathfrak{r}}$ is prime and $\beta \otimes \gamma = (\beta \otimes 1)(1 \otimes \gamma)$, this says exactly that $\bar{\mathfrak{r}}$ is disjoint from the multiplicative set $\Sigma = \{\beta \otimes \gamma \mid \beta \neq 0, \gamma \neq 0\}$. By Lemma 2.1.7(c), $\operatorname{Spec}(\Sigma^{-1}\bar{R})$ is homeomorphic to the set of such primes. Finally, localization commutes with tensor products, so

$$\Sigma^{-1}\bar{R} = \Sigma^{-1}(\bar{B} \otimes_A \bar{C}) \simeq (\bar{B} \setminus 0)^{-1}\bar{B} \otimes_A (\bar{C} \setminus 0)^{-1}\bar{C} = k(x) \otimes_A k(y) = K.$$

(The map from left to right is defined by the universal property of localization, since $\beta \otimes \gamma \mapsto \beta \otimes \gamma$ is a unit on the right; the map in the other direction sends $(\beta/\beta') \otimes (\gamma/\gamma')$ to $(\beta \otimes \gamma)/(\beta' \otimes \gamma')$; they are inverse on generators.) The composite $\operatorname{Spec} K \to \operatorname{Spec} \bar{R} \to \operatorname{Spec} R$ is Spec of $R \to K$, i.e. u. So u is a homeomorphism onto F. □

Exercise 1.8

> **Problem.** Let $X \to S$ be a surjective morphism. Show that $X \times_S Y \to Y$ is surjective for any S-scheme Y (use Prop. 1.16).

Solution. Let $y \in Y$ and let $s \in S$ be its image. Since $X \to S$ is surjective there is $x \in X$ over s, i.e. $x \in X_s$ in the notation of Prop. 1.16. By Exercise 1.7 the set $\{z \in X \times_S Y \mid p(z) =$

$x,\ q(z) = y\}$ is homeomorphic to $\operatorname{Spec}(k(x) \otimes_{k(s)} k(y))$. This tensor product of two nonzero vector spaces over the field $k(s)$ is nonzero (a basis of $k(x)$ tensored with a basis of $k(y)$ gives a basis), so its spectrum is nonempty. Hence there is z with $q(z) = y$, and $X \times_S Y \to Y$ is surjective. Thus surjectivity is stable under base change. □

Exercise 1.9

Problem. Let k be a field, $A = k(u) \otimes_k k(v)$ the tensor product of two purely transcendental extensions of k of transcendence degree 1.

(a) We have $k(u) = T^{-1}k[u]$, T the set of nonzero elements of $k[u]$. Deduce that A is the localization of $k[u, v]$ with respect to the multiplicative set T' of the nonzero elements of the form $P(u)Q(v) \in k[u, v]$.

(b) Let $\mathfrak{m}$ be a maximal ideal of $k[u, v]$. Show that there is a $P(u) \in \mathfrak{m} \setminus \{0\}$. Deduce that $T' \cap \mathfrak{m} \neq \emptyset$.

(c) Show that the maximal ideals of A are of the form gA with $g \in k[u, v] \setminus (k[u] \cup k[v])$ irreducible in $k[u, v]$.

(d) Show that $\operatorname{Spec} A$ is an infinite set and $\dim A = 1$.

Solution. (The printed "$T' \cap \mathfrak{m} \neq \emptyset$" in (b) is read as "$\neq$"; this is what is true, and what is needed in (c).)

(a) T' is multiplicative (a domain) and consists of products $P(u)Q(v)$ with $P, Q \neq 0$. Each such element is a unit in A, with inverse $P(u)^{-1} \otimes Q(v)^{-1}$, so the canonical map $k[u, v] = k[u] \otimes k[v] \to A$ induces $\alpha : T'^{-1}k[u, v] \to A$. Conversely let $R = T'^{-1}k[u, v]$. Every nonzero $P(u)$ is a unit in R (it lies in T'), so by the universal property of localization $k[u] \to R$ extends to a k-algebra map $k(u) \to R$; likewise $k(v) \to R$. By the universal property of the tensor product of k-algebras these give $\beta : A \to R$. The composites $\beta\alpha$ and $\alpha\beta$ are the identity on the generators u, v (resp. on $a \otimes 1$, $1 \otimes b$) of the respective algebras, hence are identities (for $\alpha\beta$: A is generated by the elements $a \otimes 1, 1 \otimes b$ as a ring, since $a \otimes b = (a \otimes 1)(1 \otimes b)$). So $A = T'^{-1}k[u, v]$; in particular A is a domain.

(b) The field $k[u, v]/\mathfrak{m}$ is a finitely generated k-algebra, hence a finite extension of k (Cor. 2.1.12). So the image of u is algebraic over k: there is a nonzero $P \in k[u]$ with $P(u) \in \mathfrak{m}$. Then $P(u) = P(u) \cdot 1 \in T' \cap \mathfrak{m}$.

(c) Primes of $A = T'^{-1}k[u, v]$ correspond bijectively to primes $\mathfrak{p}$ of $k[u, v]$ with $\mathfrak{p} \cap T' = \emptyset$ (Lemma 2.1.7(c)), by $\mathfrak{p} \mapsto \mathfrak{p}A$, and inclusions are preserved both ways. Since $\mathfrak{p}$ is prime, $\mathfrak{p} \cap T' = \emptyset$ is equivalent to $\mathfrak{p} \cap k[u] = 0$ and $\mathfrak{p} \cap k[v] = 0$.

Let $\mathfrak{p} \neq 0$ be such a prime. As $k[u, v]$ is a UFD, $\mathfrak{p}$ contains an irreducible element g, and (g) is prime. If $\mathfrak{p} \neq (g)$, then $0 \subsetneq (g) \subsetneq \mathfrak{p}$ and, since $\dim k[u, v] = 2$, $\mathfrak{p}$ is a maximal ideal; but maximal ideals meet T' by (b). So $\mathfrak{p} = (g)$ with g irreducible, and $g \notin k[u] \cup k[v]$ because $g \in \mathfrak{p}$ and $\mathfrak{p} \cap k[u] = \mathfrak{p} \cap k[v] = 0$.

Conversely let g be irreducible with $g \notin k[u] \cup k[v]$. Then $(g) \cap T' = \emptyset$: if $g \mid P(u)Q(v)$ with $P, Q \neq 0$, then $g \mid P$ or $g \mid Q$ (g is prime), and a divisor of a nonzero polynomial in u alone has degree 0 in v, so $g \in k[u]$ or $g \in k[v]$, a contradiction. Thus gA is a proper prime ideal of A. It is maximal: a prime strictly containing it would be 0 or $g'A$, and $gA \subseteq g'A$ means $(g) \subseteq (g')$ in $k[u, v]$, hence $g' = gh$ with h a unit (g' irreducible), so $g'A = gA$.

Finally, A is not a field since $g = u - v \notin k[u] \cup k[v]$ is irreducible (degree 1) and gives the proper nonzero ideal gA. So 0 is not maximal, and every maximal ideal of A is a nonzero prime, hence of the form gA as described. Together with the previous paragraph, the maximal ideals of A are exactly the gA with g irreducible and $g \notin k[u] \cup k[v]$.

(d) For $n \geq 1$ the polynomials $g_n = v - u^n$ are irreducible (of degree 1 in v, monic in v) and do not lie in $k[u] \cup k[v]$. They are pairwise non-associate (the coefficient of v is 1, and the units of $k[u, v]$ are the elements of k^*), so the ideals (g_n) are pairwise distinct, and so are the $g_n A$. Hence $\operatorname{Spec} A$ is infinite. By (c), every nonzero prime of A is maximal, so there is no chain $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2$ of primes, and $0 \subsetneq gA$ is a chain of length 1. Therefore $\dim A = 1$. □

Exercise 1.10

> **Problem.** Let $\pi : X \to S$, $\rho : Y \to S$ be S-schemes. Let $\operatorname{sp}(X) \times_{\operatorname{sp}(S)} \operatorname{sp}(Y)$ be the fibered product of sets defined by π, ρ, endowed with the topology induced by the product topology on $\operatorname{sp}(X) \times \operatorname{sp}(Y)$.
>
> (a) Show that there is a canonical continuous map $f : \operatorname{sp}(X \times_S Y) \to \operatorname{sp}(X) \times_{\operatorname{sp}(S)} \operatorname{sp}(Y)$.
>
> (b) Show that f is surjective.
>
> (c) Let $X = Y = \operatorname{Spec} \mathbf{C}$, $S = \operatorname{Spec} \mathbf{R}$. Show that $X \times_S Y \simeq \operatorname{Spec}(\mathbf{C} \oplus \mathbf{C})$ and that f is not injective.
>
> (d) In the situation of Exercise 1.9, with $X = \operatorname{Spec} k(u)$, $Y = \operatorname{Spec} k(v)$, $S = \operatorname{Spec} k$, show that f has infinite fibers.
>
> (e) Let $S = \operatorname{Spec} k$ with k an arbitrary field. By studying $X = Y = \mathbb{A}^1_k$, show that the image of an open subset under f is not necessarily open.

Solution. (a) Let p, q be the projections. They are continuous, and $\pi \circ p = \rho \circ q$ (the projections are S-morphisms), so $f = (p, q)$ maps $X \times_S Y$ into the subset $\{(x, y) \mid \pi(x) = \rho(y)\}$. A map into a subspace of a product, whose components are continuous, is continuous for the induced topology.

(b) Let (x, y) be a point of the target, $s = \pi(x) = \rho(y)$. By Exercise 1.7 the fiber $\{z \mid p(z) = x, q(z) = y\}$ is homeomorphic to $\operatorname{Spec}(k(x) \otimes_{k(s)} k(y))$, which is nonempty, since a tensor product of two nonzero vector spaces over a field is nonzero. So $(x, y) \in f(X \times_S Y)$.

(c) We have $\mathbf{C} = \mathbf{R}[x]/(x^2 + 1)$, so $\mathbf{C} \otimes_{\mathbf{R}} \mathbf{C} \simeq \mathbf{C}[x]/(x^2 + 1) = \mathbf{C}[x]/((x - i)(x + i))$. Concretely, define $\theta : \mathbf{C} \otimes_{\mathbf{R}} \mathbf{C} \to \mathbf{C} \times \mathbf{C}$, $a \otimes b \mapsto (ab, a\bar{b})$. This is well defined ($\mathbf{R}$-bilinear in (a, b)), multiplicative on pure tensors, so a ring homomorphism. The images of $1 \otimes 1$, $1 \otimes i$, $i \otimes 1$, $i \otimes i$ are $(1, 1), (i, -i), (i, i), (-1, 1)$. Their $\mathbf{R}$-span contains $(1, 1) \pm (-1, 1)$, hence $(0, 1)$ and $(1, 0)$, and $(i, i) \pm (i, -i)$, hence $(i, 0)$ and $(0, i)$, so it is all of $\mathbf{C} \times \mathbf{C}$. Both sides have dimension 4 over $\mathbf{R}$, so θ is an isomorphism. By Prop. 1.2, $X \times_S Y = \operatorname{Spec}(\mathbf{C} \otimes_{\mathbf{R}} \mathbf{C}) \simeq \operatorname{Spec}(\mathbf{C} \times \mathbf{C}) = \operatorname{Spec} \mathbf{C} \sqcup \operatorname{Spec} \mathbf{C}$, which has exactly two points. The target $\operatorname{sp}(X) \times_{\operatorname{sp}(S)} \operatorname{sp}(Y)$ is a single point. So f is surjective but not injective.

(d) Here $\operatorname{sp}(X)$, $\operatorname{sp}(Y)$, $\operatorname{sp}(S)$ are single points, so the target of f is one point and the only fiber is all of $\operatorname{sp}(X \times_S Y) = \operatorname{Spec} A$ (Prop. 1.2), which is infinite by Exercise 1.9(d).

(e) We assume k is infinite; for finite k the assertion is actually false, see the remark at the end. By Example 1.3 (or Prop. 1.2), $X \times_S Y = \operatorname{Spec} k[u] \otimes_k k[v] = \operatorname{Spec} k[u, v]$, and $f(\mathfrak{p}) =$

$(\mathfrak{p} \cap k[u], \mathfrak{p} \cap k[v])$. Since $\operatorname{sp}(S)$ is a point, the target is $\mathbb{A}^1_k \times \mathbb{A}^1_k$ with the product of the Zariski topologies. The open sets of $\mathbb{A}^1_k$ are $\emptyset$ and complements of finite sets of closed points (Example 2.1.4).

Let $U = D(u-v) \subset \operatorname{Spec} k[u,v]$, which contains the generic point (0), and $f((0)) = (\xi, \xi)$ where ξ is the generic point of $\mathbb{A}^1_k$. So $(\xi, \xi) \in f(U)$. For $c \in k$ let $a_c = (u-c) \in \mathbb{A}^1_k$ (a closed point, $a_c \neq a_{c'}$ for $c \neq c'$) and $b_c = (v-c)$. If $\mathfrak{p}$ is a prime with $f(\mathfrak{p}) = (a_c, b_c)$, then $u-c, v-c \in \mathfrak{p}$, hence $u - v \in \mathfrak{p}$ and $\mathfrak{p} \notin U$. Thus $(a_c, b_c) \notin f(U)$ for all $c \in k$.

Suppose $f(U)$ were open. It would contain a basic neighborhood $V_1 \times V_2$ of (ξ, ξ) with $V_i = \mathbb{A}^1_k \setminus F_i$ for finite sets F_i of closed points. As k is infinite there is $c \in k$ with $a_c \notin F_1$ and $b_c \notin F_2$ (only finitely many c are excluded). Then $(a_c, b_c) \in V_1 \times V_2 \subseteq f(U)$, contradicting the above. So $f(U)$ is not open, although U is.

Remark (finite k): the printed statement is false. Let k be finite; we show that then $f(U)$ is open for every open $U \subseteq \operatorname{Spec} k[u,v]$, so no example exists. Since $f(\bigcup U_i) = \bigcup f(U_i)$ it suffices to treat $U = D(h)$, $h \neq 0$. Pull out of h all irreducible factors lying in $k[u] \cup k[v]$: $h = P(u)Q(v)h'$, where h' has no irreducible factor in $k[u] \cup k[v]$. Let A_P (resp. B_Q) be the finite set of closed points of $\mathbb{A}^1_k$ given by the irreducible factors of P (resp. Q).

By Exercise 1.7 (with $S = \operatorname{Spec} k$), the fiber of f over $t = (t_1, t_2)$ is $\operatorname{Spec} K_t$ with $K_t = k(t_1) \otimes_k k(t_2)$, so $t \notin f(U)$ iff the image h_t of h lies in every prime of K_t. Each K_t is reduced: k is perfect, so $k(t_1) \otimes_k k(t_2)$ is reduced when one factor is finite over k, and it is a domain (a localization of $k[u,v]$) when both are $k(u), k(v)$. Hence $t \notin f(U)$ iff $h_t = 0$. We determine these t.

$t = (\xi, \xi)$: K_t is a localization of $k[u,v]$ and $h \neq 0$, so $t \in f(U)$.

$t = (\xi, b)$ with $b = (g(v))$, g irreducible: $K_t = k(u)[v]/(g)$ has $k(u)$-basis $1, \dots, v^{\deg g - 1}$. Divide $h = gq + r$ in $k[u][v]$ (g is monic in v) with $\deg_v r < \deg g$; then $h_t = 0$ iff $r = 0$ iff $g \mid h$ in $k[u,v]$. As g is prime and $g \nmid h'$, $g \nmid P(u)$ (it has positive degree in v), this holds iff $g \mid Q$, i.e. iff $b \in B_Q$. Symmetrically $(a, \xi) \notin f(U)$ iff $a \in A_P$.

$t = (a, b)$, $a = (f_a(u))$, $b = (f_b(v))$ closed points: if $a \in A_P$ or $b \in B_Q$ then $P_t = 0$ or $Q_t = 0$, so $t \notin f(U)$. Suppose $a \notin A_P$, $b \notin B_Q$. Then P_t, Q_t are nonzero elements of the fields $k(a)$, $k(b)$, hence units of K_t, and $h_t = 0$ iff $h'_t = 0$. Let α be the class of u in $k(a)$; then $K_t = k(a)[v]/(f_b)$ and $h'_t = 0$ iff $f_b \mid h'(\alpha, v)$ in $k(a)[v]$. Now $h'(\alpha, v) \neq 0$, for otherwise every coefficient (in $k[u]$) of h' as a polynomial in v is divisible by f_a, so $f_a \mid h'$, contradicting the choice of h'. Hence $\deg f_b \leq \deg_v h'$, and symmetrically $\deg f_a \leq \deg_u h'$. As k is finite, there are only finitely many irreducible polynomials of bounded degree, so only a finite set Z of pairs (a, b) of this last kind can lie outside $f(U)$.

Altogether the complement of $f(U)$ is

$$(A_P \times \mathbb{A}^1_k) \ \cup \ (\mathbb{A}^1_k \times B_Q) \ \cup \ Z,$$

a union of two products of a finite set of closed points with $\mathbb{A}^1_k$ and of a finite set of points $\{a\} \times \{b\}$, all closed in the product topology (finite sets of closed points are closed in $\mathbb{A}^1_k$). So $f(U)$ is open. Hence for finite k the image of an open set is always open, and the assertion of (e) holds only for infinite k. $\square$

Exercise 1.11

Problem. Let k be a field and $z \in \mathbb{P}^n_k(k)$. Choose homogeneous coordinates such that $z = (1, 0, \ldots, 0)$.

(a) Show that there is a morphism $p : \mathbb{P}^n_k \setminus \{z\} \to \mathbb{P}^{n-1}_k$ such that over $\overline{k}$, the algebraic closure of k, we have $p_{\overline{k}}(a_0, a_1, \ldots, a_n) = (a_1, \ldots, a_n)$ for every point of $\mathbb{P}^n_{\overline{k}}(\overline{k})$. Such a morphism is called a projection with center z.

(b) Let X be a closed subset of $\mathbb{P}^n_k$ not containing z. Show that X cannot contain $p^{-1}(y)$ for any $y \in \mathbb{P}^{n-1}_k$.

Solution. Write $\mathbb{P}^n_k = \operatorname{Proj} k[T_0, \ldots, T_n]$ and $\mathbb{P}^{n-1}_k = \operatorname{Proj} k[S_1, \ldots, S_n]$. The point $z = (1 : 0 : \cdots : 0)$ is the point of the homogeneous prime $(T_1, \ldots, T_n)$, i.e. $\{z\} = V_+(T_1, \ldots, T_n)$.

(a) Let $\phi : C = k[S_1, \ldots, S_n] \to B = k[T_0, \ldots, T_n]$ be the homomorphism of graded rings with $S_j \mapsto T_j$. Then $M = \phi(C_+)B = (T_1, \ldots, T_n)$ and $V_+(M) = \{z\}$. By Lemma 2.3.40 we get a morphism of k-schemes $p : \mathbb{P}^n_k \setminus \{z\} \to \mathbb{P}^{n-1}_k$ with $p^{-1}(D_+(S_i)) = D_+(T_i)$ $(1 \le i \le n)$, whose restriction to $D_+(T_i) = \operatorname{Spec} k[T_j/T_i]_{0 \le j \le n}$ is the morphism of affine schemes given by the inclusion $C_{(S_i)} = k[S_j/S_i]_{1 \le j \le n} \subseteq B_{(T_i)}$.

Over $\overline{k}$: $p_{\overline{k}} = p \times \mathrm{Id}_{\overline{k}}$ is the restriction of p to the open subscheme $(\mathbb{P}^n_k \setminus \{z\})_{\overline{k}} = \mathbb{P}^n_{\overline{k}} \setminus \{z\}$ (Prop. 1.4(d), Prop. 1.9) and on $D_+(T_i)_{\overline{k}} = \operatorname{Spec} \overline{k}[T_j/T_i]$ it is given by the inclusion $\overline{k}[S_j/S_i]_{1 \le j \le n} \subseteq \overline{k}[T_j/T_i]_{0 \le j \le n}$ (base change of a morphism of affine schemes, Example 1.8). Let $a = (a_0, \ldots, a_n) \in \mathbb{P}^n(\overline{k})$, $a \ne z$, i.e. $(a_1, \ldots, a_n) \ne 0$; choose $i \ge 1$ with $a_i \ne 0$. The point a is the $\overline{k}$-point of $D_+(T_i)_{\overline{k}}$ given by the $\overline{k}$-algebra homomorphism $T_j/T_i \mapsto a_j/a_i$. Composing with $p_{\overline{k}}$ we get the $\overline{k}$-point of $D_+(S_i) \subseteq \mathbb{P}^{n-1}_{\overline{k}}$ given by $S_j/S_i \mapsto a_j/a_i$ $(1 \le j \le n)$, i.e. the point $(a_1 : \cdots : a_n)$. This proves (a).

(b) Let $y \in \mathbb{P}^{n-1}_k$ be any point and suppose $p^{-1}(y) \subseteq X$. We derive $z \in X$. Choose $i \in \{1, \ldots, n\}$ with $y \in D_+(S_i)$. Let $K = k(y)$ and let $b_j \in K$ be the image of S_j/S_i in K $(1 \le j \le n;\ b_i = 1)$. Then the canonical morphism $\operatorname{Spec} K \to \operatorname{Spec} k[S_j/S_i] \subseteq \mathbb{P}^{n-1}_k$ has image $\{y\}$ and corresponds to $S_j/S_i \mapsto b_j$. The vector $(b_1, \ldots, b_n) \in K^n$ is nonzero.

Let $\phi_K : K[T_0, \ldots, T_n] \to K[t, s]$ be the graded homomorphism with $T_0 \mapsto t$, $T_j \mapsto b_j s$ $(j \ge 1)$. The ideal generated by the images is $(t, b_1 s, \ldots, b_n s) = (t, s)$ because $b_i = 1$, so by Lemma 2.3.40 (with $V_+((t, s)) = \emptyset$) we get a morphism $G_0 : \mathbb{P}^1_K \to \mathbb{P}^n_K$ with $G_0^{-1}(D_+(T_j)) = D_+(\phi_K(T_j))$. Let $G : \mathbb{P}^1_K \to \mathbb{P}^n_K \to \mathbb{P}^n_k$ be its composite with the projection of the base change (Prop. 1.9).

Claim 1: $G(D_+(s)) \subseteq p^{-1}(y)$. We have $G_0^{-1}(D_+(T_i)) = D_+(b_i s) = D_+(s)$ and $D_+(T_i) \subseteq \mathbb{P}^n_k \setminus \{z\}$. On $D_+(s) = \operatorname{Spec} K[\tau]$ $(\tau = t/s)$, the morphism G_0 into $D_+(T_i)_K = \operatorname{Spec} K[T_j/T_i]$ is given by $T_j/T_i \mapsto b_j$ $(j \ge 1)$ and $T_0/T_i \mapsto \tau$. Hence $p \circ G|_{D_+(s)}$ corresponds to the ring homomorphism $k[S_j/S_i] \to K[\tau]$, $S_j/S_i \mapsto b_j \in K \subset K[\tau]$. This is the composite of $k[S_j/S_i] \to K$ $(S_j/S_i \mapsto b_j)$ and $K \subseteq K[\tau]$, so $p \circ G|_{D_+(s)}$ factors as $D_+(s) \to \operatorname{Spec} K \to \mathbb{P}^{n-1}_k$ and its image is $\{y\}$. So $G(D_+(s)) \subseteq p^{-1}(y) \subseteq X$.

Claim 2: $G(\infty) = z$, *where* $\infty = V_+(s)$ *is the point* $(1 : 0)$ *of* $\mathbb{P}^1_K$. The point ∞ lies in $D_+(t) = \operatorname{Spec} K[\sigma]$ $(\sigma = s/t)$ as the prime (σ), and $G_0^{-1}(D_+(T_0)) = D_+(t)$, where G_0 is given by $T_j/T_0 \mapsto b_j \sigma$ $(j \ge 1)$. The preimage of (σ) in $K[T_j/T_0]$ is $(T_1/T_0, \ldots, T_n/T_0)$, and its preimage in $k[T_j/T_0]$ is the maximal ideal $(T_j/T_0)_{1 \le j \le n}$, which is the point z.

Now $G^{-1}(X)$ is closed in $\mathbb{P}^1_K$ and contains $D_+(s)$ by Claim 1. The generic point η of $\mathbb{P}^1_K$ (the prime (0) of $K[\tau, \tau^{-1}] = K[\sigma, \sigma^{-1}]$) lies in $D_+(s)$, and its closure in $D_+(t) = \operatorname{Spec} K[\sigma]$ is all of $\operatorname{Spec} K[\sigma]$, which contains ∞. So $\infty \in \overline{\{\eta\}} \subseteq G^{-1}(X)$, and $z = G(\infty) \in X$ by Claim 2. This

contradicts $z \notin X$. Hence X does not contain $p^{-1}(y)$. (The proof works for every point y of the scheme $\mathbb{P}^{n-1}_k$, closed or not, and a fortiori for the $\overline{k}$-points of $\mathbb{P}^{n-1}$ through their images.) $\square$

3.2 Applications to algebraic varieties

Conventions for this section. An algebraic variety over a field k is a scheme of finite type over k (Example 2.3). A scheme is Noetherian if it is a finite union of affine opens $\operatorname{Spec} A_i$ with A_i Noetherian; then every affine open has a Noetherian ring. Results of the text are cited by their numbers ("Prop. 2.2" is in Section 3.2, "Prop. 1.16" in Section 3.1, "Prop. 2.5.10" and "Ex. 2.3.17" in Chapter 2). Honest notes: in Ex. 2.6 the map $K(Y) \to K(X)$ only makes sense when f is dominant, and this is part of the meaning of "birational". In Ex. 2.7 the varieties $\overline{X}, \overline{Y}$ are over $\overline{k}$ and X, Y over K (the bars are lost in some printings). In Ex. 2.9 the maps are $X(\overline{k}) \to Y(\overline{k})$. In Ex. 2.10(b) the deduction about $X_{\overline{k}}$ is proved by a direct argument (Galois transitivity is proved for finite Galois K, and $\overline{k}/k$ need not be separable). Several solutions use common lemmas that are proved where first needed (Lemma L in 2.4, Lemmas M and U in 2.5, Lemma A in 2.7, Lemma C in 2.9, Lemma D in 2.10).

Exercise 2.1

> **Problem.** Let Y be a Noetherian scheme. Show that any Y-scheme X of finite type is Noetherian. Moreover, if Y is of finite dimension, then so is X.

Solution. Let $f : X \to Y$ be the structure morphism and write $Y = V_1 \cup \cdots \cup V_m$ with $V_i = \operatorname{Spec} A_i$, A_i Noetherian. Since f is quasi-compact, $f^{-1}(V_i)$ is quasi-compact, hence a finite union of affine opens $U_{ij} = \operatorname{Spec} B_{ij}$. By Def. 2.1, B_{ij} is a finitely generated A_i-algebra, i.e. a quotient of a polynomial ring $A_i[T_1, \ldots, T_{n_{ij}}]$, which is Noetherian by Hilbert's basis theorem. So B_{ij} is Noetherian, and $X = \bigcup_{i,j} U_{ij}$ is a finite union of affine opens with Noetherian rings: X is Noetherian.

Dimension. *Fact:* if a topological space T is a union of open subsets T_λ, then $\dim T = \sup_\lambda \dim T_\lambda$. Indeed, $\geq$ is clear (if $Z_0 \subsetneq \cdots \subsetneq Z_n$ are irreducible closed subsets of T_λ, their closures in T are irreducible, closed and strictly increasing, since $\overline{Z_j} \cap T_\lambda = Z_j$). Conversely, given a chain $Z_0 \subsetneq \cdots \subsetneq Z_n$ of irreducible closed subsets of T, choose λ with $T_\lambda \cap Z_0 \neq \emptyset$. Each $Z_j \cap T_\lambda$ is a nonempty open subset of the irreducible Z_j, hence irreducible and dense in Z_j, and closed in T_λ. The inclusions $Z_j \cap T_\lambda \subsetneq Z_{j+1} \cap T_\lambda$ are strict because both are dense in Z_j, resp. Z_{j+1}, and $Z_j \neq Z_{j+1}$ are closed. So $\dim T_\lambda \geq n$.

Now let $d = \dim Y < \infty$. By the fact, $\dim A_i = \dim V_i \leq d$ (note $\dim V_i \leq \dim Y$ by the easy half). For a Noetherian ring A one has $\dim A[T_1, \ldots, T_n] = \dim A + n$ (Krull; e.g. Matsumura, Commutative Ring Theory, Thm. 15.4), and a quotient of a ring has dimension at most that of the ring. So $\dim B_{ij} \leq d + n_{ij}$, and by the fact, $\dim X = \max_{i,j} \dim U_{ij} \leq d + \max_{i,j} n_{ij} < \infty$. $\square$

Exercise 2.2

Problem. Show that any open immersion into a locally Noetherian scheme is a morphism of finite type.

Solution. Let $j : U \to Y$ be an open immersion, Y locally Noetherian. Identify U with the open subscheme $j(U)$ of Y. Cover Y by affine opens $V_i = \operatorname{Spec} A_i$ with A_i Noetherian. We check the hypothesis of Prop. 2.2 for this cover.

The space $V_i = \operatorname{Spec} A_i$ is a Noetherian topological space (the ascending chain condition on radical ideals gives the descending chain condition on closed subsets), so every open subset of it is quasi-compact. In particular $j^{-1}(V_i) = U \cap V_i$ is quasi-compact. Since the principal opens $D(a) \subseteq U \cap V_i$ ($a \in A_i$) form a base of the topology of $U \cap V_i$, there are $a_{i1}, \ldots, a_{ir_i} \in A_i$ with $U \cap V_i = D(a_{i1}) \cup \cdots \cup D(a_{ir_i})$. Each $D(a_{il}) \simeq \operatorname{Spec}(A_i)_{a_{il}}$ (Lemma 2.3.7) is affine, and $\mathcal{O}(D(a_{il})) = (A_i)_{a_{il}} = A_i[1/a_{il}]$ is a finitely generated A_i-algebra. By Prop. 2.2, j is of finite type. □

Exercise 2.3

Problem. An *immersion* of schemes is a morphism which is an open immersion followed by a closed immersion.

(a) Let $f : X \to Y$ be an immersion. Show that it can be decomposed into a closed immersion followed by an open immersion.

(b) Show that the converse of (a) is true if f is moreover quasi-compact (e.g., if Y is locally Noetherian). Use the scheme-theoretic closure of f (Exercise 2.3.17).

(c) Let $f : X \to Y$, $g : Y \to Z$ be two immersions with g quasi-compact. Show that $g \circ f$ is an immersion.

Solution. We use freely that compositions of open immersions, and of closed immersions, are again open, resp. closed immersions, and that a closed immersion restricted over an open subset of the target is a closed immersion (both defining conditions are local on the target).

(a) Write $f = i \circ j$ with $j : X \to Z$ an open immersion and $i : Z \to Y$ a closed immersion. Identify X with the open subscheme $j(X)$ of Z. The set $T = i(Z \setminus j(X))$ is closed in Y (i is a homeomorphism onto a closed subset), so $W = Y \setminus T$ is open. As i is injective, $i^{-1}(W) = Z \setminus (Z \setminus j(X)) = j(X) = X$. Hence i restricts to a closed immersion $i' : X = i^{-1}(W) \to W$, and f is i' followed by the open immersion $W \hookrightarrow Y$.

(b) Let $f = \iota \circ g$ with $g : X \to W$ a closed immersion and $\iota : W \to Y$ an open immersion, and assume f quasi-compact. (If Y is locally Noetherian, ι is quasi-compact by Ex. 2.3.17(b), g is quasi-compact by Ex. 2.3.17(a), and a composition of quasi-compact morphisms is quasi-compact: if $V \subseteq Y$ is affine open then $\iota^{-1}(V)$ is a finite union of affines V_l, and $g^{-1}(V_l)$ is quasi-compact.) By Ex. 2.3.17(c),(d), $\mathcal{J} = \operatorname{Ker}(f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X)$ defines a closed subscheme $Z = V(\mathcal{J})$ of Y (the scheme-theoretic closure of $f(X)$). Since $f^{-1}(W) = X$, we have $(f_*\mathcal{O}_X)|_W = g_*\mathcal{O}_X$, and so $\mathcal{J}|_W = \operatorname{Ker}(g^{\#} : \mathcal{O}_W \to g_*\mathcal{O}_X)$. As g is a closed immersion, g identifies X with the closed subscheme of W defined by this ideal sheaf, i.e. with $Z \cap W$ (an open subscheme of Z). Therefore $X \simeq Z \cap W \hookrightarrow Z$ is an open immersion, and f is this open immersion followed by the closed immersion $Z \to Y$ (indeed $X \simeq Z \cap W \to Z \to Y$ is $X \to W \to Y$, which is f). So f is an immersion.

(c) Write $f = c_f \circ o_f$ with $o_f : X \to Y'$ open and $c_f : Y' \to Y$ closed. By (a), $g = o_g \circ c_g$ with $c_g : Y \to Z'$ closed and $o_g : Z' \to Z$ open. Then $h := g \circ c_f = o_g \circ (c_g \circ c_f)$ is a closed immersion followed by an open immersion. It is quasi-compact, being the composition of the quasi-compact g with the closed immersion c_f (quasi-compact by Ex. 2.3.17(a)). By (b), $h = c'' \circ o''$ with $o'' : Y' \to Z''$ open and $c'' : Z'' \to Z$ closed. Hence $g \circ f = h \circ o_f = c'' \circ (o'' \circ o_f)$, where $o'' \circ o_f$ is an open immersion and c'' a closed immersion. So $g \circ f$ is an immersion. □

Exercise 2.4

> **Problem.** Let X, Y be schemes over a locally Noetherian scheme S, with Y of finite type over S. Let $x \in X$. Show that for any morphism of S-schemes $f_x : \operatorname{Spec} \mathcal{O}_{X,x} \to Y$, there exist an open subset $U \ni x$ of X and a morphism of S-schemes $f : U \to Y$ such that $f_x = f \circ i_x$, where $i_x : \operatorname{Spec} \mathcal{O}_{X,x} \to U$ is the canonical morphism (in other words, the morphism f_x extends to an open neighborhood of x).

Solution. *Lemma L.* Let A be a Noetherian ring, C a finitely generated A-algebra, R an A-algebra, $\Sigma \subseteq R$ a multiplicative subset, and $\bar{\psi} : C \to \Sigma^{-1}R$ a homomorphism of A-algebras. Then there are $\sigma \in \Sigma$ and an A-algebra homomorphism $\psi : C \to R_\sigma$ such that $\bar{\psi}$ is ψ followed by $R_\sigma \to \Sigma^{-1}R$.

Proof. By Hilbert's basis theorem $C \simeq A[t_1, \dots, t_n]/(P_1, \dots, P_m)$ with $P_l \in A[t_1, \dots, t_n]$. Write $\bar{\psi}(t_i) = r_i/s$ with $r_i \in R$ and a common denominator $s \in \Sigma$, and put $\rho_i = r_i/s \in R_s$. The element $P_l(\rho_1, \dots, \rho_n) \in R_s$ has image $\bar{\psi}(P_l(t)) = 0$ in $\Sigma^{-1}R$. By the definition of localization there is $w_l \in \Sigma$ killing it in R_s. Put $\sigma = s\, w_1 \cdots w_m$. In R_σ the w_l are units, so $P_l(\rho) = 0$ there. Hence $t_i \mapsto \rho_i$ defines an A-algebra homomorphism $\psi : C \to R_\sigma$, and its composite with $R_\sigma \to \Sigma^{-1}R$ agrees with $\bar{\psi}$ on the generators t_i. □

Now the exercise. Let $s \in S$ be the image of x, and choose an affine open $S_0 = \operatorname{Spec} A \ni s$ of S; A is Noetherian. The image y of the closed point of $\operatorname{Spec} \mathcal{O}_{X,x}$ under f_x lies over s (as f_x is an S-morphism). Choose an affine open $V = \operatorname{Spec} B \ni y$ of Y inside the preimage of S_0; B is a finitely generated A-algebra because Y is of finite type over S. The preimage $f_x^{-1}(V)$ is open and contains the closed point of the local scheme $\operatorname{Spec} \mathcal{O}_{X,x}$. The closed point lies in the closure of every point, so an open set containing it contains every point. Hence $f_x^{-1}(V) = \operatorname{Spec} \mathcal{O}_{X,x}$, and f_x factors through $V = \operatorname{Spec} B$, i.e. it is given by a ring homomorphism $\bar{\psi} : B \to \mathcal{O}_{X,x}$. Since f_x is an S-morphism and S_0 is affine, $\bar{\psi}$ is an A-algebra homomorphism.

Choose an affine open $U_0 = \operatorname{Spec} C \ni x$ of X lying over S_0, and let $\mathfrak{P} \subset C$ be the prime ideal of x, so that $\mathcal{O}_{X,x} = C_{\mathfrak{P}}$. Apply Lemma L with $R = C$, $\Sigma = C \setminus \mathfrak{P}$: there are $\sigma \in C \setminus \mathfrak{P}$ and an A-homomorphism $\psi : B \to C_\sigma$ with $\bar{\psi} = (C_\sigma \to C_{\mathfrak{P}}) \circ \psi$. Put $U = D(\sigma) = \operatorname{Spec} C_\sigma \ni x$ and let $f : U \to V \subseteq Y$ be the morphism $\operatorname{Spec} \psi$. It is an S-morphism, since over the affine S_0 both structure maps correspond to $A \to C_\sigma$ (ψ is A-linear). The morphism i_x corresponds to $C_\sigma \to C_{\mathfrak{P}}$, so $f \circ i_x$ corresponds to $B \to C_\sigma \to C_{\mathfrak{P}}$, which is $\bar{\psi}$. Thus $f \circ i_x = f_x$. □

Exercise 2.5

> **Problem.** Let S be a locally Noetherian scheme. Let X, Y be S-schemes of finite type. Let us fix $s \in S$. Let $\varphi : X \times_S \operatorname{Spec} \mathcal{O}_{S,s} \to Y \times_S \operatorname{Spec} \mathcal{O}_{S,s}$ be a morphism of S-schemes. Show that there exist an open set $U \ni s$ and a morphism of S-schemes $f : X \times_S U \to Y \times_S U$ such

that φ is obtained from f by the base change $\operatorname{Spec}\mathcal{O}_{S,s} \to U$. If φ is an isomorphism, show that there exists such an f which is moreover an isomorphism.

Solution. *Setup.* Choose an affine open $S_0 = \operatorname{Spec} A \ni s$ with A Noetherian, and let $\mathfrak{p} \subset A$ be the prime of s, $\Sigma = A \setminus \mathfrak{p}$, $T = \operatorname{Spec} A_{\mathfrak{p}} = \operatorname{Spec}\mathcal{O}_{S,s}$. The morphism $T \to S$ factors through S_0. If X_0, Y_0 are the preimages of S_0 in X, Y, then $X \times_S T = X_0 \times_{S_0} T$ (a pair of morphisms $Z \to X$, $Z \to T$ with the same image in S lands in X_0 since $Z \to T \to S$ factors through S_0) and similarly for Y; and $X_0 \times_S U = X_0 \times_{S_0} U$ for $U \subseteq S_0$ open. So we may replace S by S_0, X by X_0 and Y by Y_0, and assume $S = \operatorname{Spec} A$. Then X, Y are quasi-compact and $X_a := X \times_S D(a)$ is the preimage of $D(a)$, for $a \in A$. Write $X_T = X \times_S T$, $p : X_T \to X$ the projection, and similarly for Y. For $a \in \Sigma$ the open $D(a)$ contains s, and the sets $D(a)$, $a \in \Sigma$, form a fundamental system of neighbourhoods of s among principal opens.

A morphism $Z \to T = \operatorname{Spec} A_{\mathfrak{p}}$ is a ring homomorphism $A_{\mathfrak{p}} \to \mathcal{O}(Z)$, hence is determined by its restriction to A. So two morphisms into T with the same composite to S coincide. Applied to $q_Y \circ \varphi$ and q_X (q the maps to T), this shows that φ is a T-morphism. Hence φ corresponds, via the universal property of $Y_T = Y \times_S T$, to the A-morphism $\Phi = \mathrm{pr}_Y \circ \varphi : X_T \to Y$, and a T-morphism $X_T \to Y_T$ is determined by its composite with pr_Y. Moreover, for an affine open $X_i = \operatorname{Spec} B_i$ of X we have $(X_i)_T = \operatorname{Spec}(B_i \otimes_A A_{\mathfrak{p}}) = \operatorname{Spec}\Sigma^{-1}B_i$, and for $b \in B_i$ the principal open $D(b) \subseteq (X_i)_T$ is the base change of the affine open $D(b) \subseteq X_i$ (localization commutes with base change).

Lemma M. Let $G = \operatorname{Spec} F$ with F an A-algebra, and $\Omega \subseteq G$ an open set containing $G_T := p(G \times_S T) = \{P \in \operatorname{Spec} F : \ P \cap \Sigma = \emptyset\}$. Then $G_a = G \times_S D(a) \subseteq \Omega$ for some $a \in \Sigma$.

Proof. Write $G \setminus \Omega = V(J)$. If $J\Sigma^{-1}F$ were a proper ideal, it would lie in a prime of $\Sigma^{-1}F$, i.e. there would be a prime $P \supseteq J$ of F with $P \cap \Sigma = \emptyset$, i.e. a point of $V(J) \cap G_T = \emptyset$. So $1 = j/s'$ in $\Sigma^{-1}F$ with $j \in J$, $s' \in \Sigma$. There is $w \in \Sigma$ with $w(s' - j) = 0$ in F, so $a := ws' \in J \cap \Sigma$ (as an element of F). Then $V(J) \subseteq V(a)$, i.e. $D(a) = G_a \subseteq \Omega$. □

Lemma U. Let X, Y be A-schemes of finite type and $u, v : X \to Y$ two A-morphisms with $u \circ p = v \circ p$ (equivalently $u_T = v_T$). Then $u = v$ on X_a for some $a \in \Sigma$.

Proof. Cover X by finitely many affine opens; if the assertion holds on each with elements a_k, it holds on X for $a = \prod a_k$. So let $X = \operatorname{Spec} F$. Let $Y = Y_1 \cup \cdots \cup Y_t$ with $Y_j = \operatorname{Spec} C_j$ affine open. For $x \in X_T := p(X \times_S T)$ we have $u(x) = v(x)$. Hence $\Omega := \bigcup_j \big(u^{-1}(Y_j) \cap v^{-1}(Y_j)\big)$ is an open set containing X_T. By Lemma M, $X_{a_1} \subseteq \Omega$ for some $a_1 \in \Sigma$. Cover the affine X_{a_1} by finitely many principal opens $D(f_m) = \operatorname{Spec} R_m$ with $D(f_m) \subseteq u^{-1}(Y_{j(m)}) \cap v^{-1}(Y_{j(m)})$. On $D(f_m)$, u and v correspond to A-algebra homomorphisms $u^*, v^* : C_{j(m)} \to R_m$ which agree after composing with $R_m \to \Sigma^{-1}R_m$ (because $u \circ p = v \circ p$). Let $c_1, \ldots, c_n$ generate the A-algebra $C_{j(m)}$. Each $u^*(c_l) - v^*(c_l)$ is killed by some $w_{ml} \in \Sigma$. With $a = a_1 \prod_{m,l} w_{ml}$ we get $u^* = v^*$ on $D(f_m) \cap X_a$ for all m, so $u = v$ on X_a. □

Existence of f. As X_T is quasi-compact, we can cover it by finitely many sets $D(b_m) \subseteq (X_{i(m)})_T$, $b_m \in B_{i(m)}$ (so $D(b_m)$ is the base change of the affine open $X'_m := D(b_m) \subseteq X_{i(m)}$, with $\mathcal{O}(X'_m) =: B'_m$ finitely generated over A), such that $\Phi(D(b_m)) \subseteq Y_{j(m)}$ with $Y_{j(m)} = \operatorname{Spec} C_m$ a fixed finite affine cover of Y. (Take all principal $D(b)$ contained in some $\Phi^{-1}(Y_j)$; these cover X_T since principal opens form a base; then pass to a finite subcover.) Let $X' = \bigcup_m X'_m$, an open subscheme of X containing $p(X_T)$. For each of the finitely many affine opens X_i of the chosen cover of X, Lemma M (with $\Omega = X' \cap X_i$) gives $a_i \in \Sigma$ with $(X_i)_{a_i} \subseteq X'$. With $a_0 = \prod_i a_i$

we get $X_{a_0} \subseteq X'$. Replace X by X_{a_0} and X'_m by $X'_m \cap X_{a_0} = \operatorname{Spec}(B'_m)_{a_0}$ (again affine, with ring finitely generated over A). This changes nothing after base change to T, since a_0 is a unit on T (so $(X_{a_0})_T = X_T$ and φ, Φ are unchanged). So we may assume $X = \bigcup_m X'_m$ with $X'_m = \operatorname{Spec} B'_m$ affine, B'_m finitely generated over A, and $\Phi((X'_m)_T) \subseteq Y_{j(m)}$.

On $(X'_m)_T = \operatorname{Spec} \Sigma^{-1}B'_m$ the morphism Φ is given by an A-algebra homomorphism $\bar{\psi}_m : C_m \to \Sigma^{-1}B'_m$. By Lemma L (in 2.4; C_m is finitely generated over the Noetherian A) there are $a_m \in \Sigma$ and an A-homomorphism $\psi_m : C_m \to (B'_m)_{a_m}$ inducing $\bar{\psi}_m$. Let $f_m : (X'_m)_{a_m} \to Y_{j(m)} \subseteq Y$ be $\operatorname{Spec} \psi_m$; it is an A-morphism with $f_m \circ p = \Phi$ on $(X'_m)_T$. Put $a' = \prod a_m$ and restrict all f_m to $(X'_m)_{a'}$. The overlaps $(X'_m)_{a'} \cap (X'_{m'})_{a'}$ are open subschemes of the Noetherian affine X'_m (its ring is finitely generated over the Noetherian A), hence finite unions of affine opens E_ν. On each E_ν the two morphisms $f_m, f_{m'}$ agree after composing with p (both equal Φ there), so by Lemma U they agree on $(E_\nu)_{a_\nu}$ for some $a_\nu \in \Sigma$. Let $a_1 = a' \prod_\nu a_\nu$ (finite product over all overlaps). Then the $f_m|_{(X'_m)_{a_1}}$ agree on overlaps and glue to an A-morphism $f : X_{a_1} \to Y$ with $f \circ p = \Phi$.

Put $U = D(a_1) \ni s$. Since f is an A-morphism, it maps $X_{a_1} = X \times_S U$ into the preimage $Y_{a_1} = Y \times_S U$ of U, giving a morphism of S-schemes (even of U-schemes) $\tilde{f} : X \times_S U \to Y \times_S U$. Its base change $\tilde{f}_T : X_T \to Y_T$ satisfies $\mathrm{pr}_Y \circ \tilde{f}_T = f \circ p = \Phi = \mathrm{pr}_Y \circ \varphi$, and both are T-morphisms, so $\tilde{f}_T = \varphi$.

The isomorphism case. Let φ be an isomorphism. Apply the above to φ and to φ^{-1}: we get $\tilde{f} : X_{a_1} \to Y_{a_1}$ and $\tilde{g} : Y_{a_2} \to X_{a_2}$ with $\tilde{f}_T = \varphi$, $\tilde{g}_T = \varphi^{-1}$. Put $a_3 = a_1 a_2$ and restrict both to $D(a_3)$. Then $\tilde{g} \circ \tilde{f}$ and the identity are A-morphisms $X_{a_3} \to X$ whose base changes to T agree ($\varphi^{-1}\varphi = \mathrm{id}$). By Lemma U (applied to the finite type A-scheme X_{a_3}) they agree on $X_{a_3 a_4}$ for some $a_4 \in \Sigma$. Similarly $\tilde{f} \circ \tilde{g} = \mathrm{id}$ on $Y_{a_3 a_5}$. On $U' = D(a_3 a_4 a_5)$ the restrictions of $\tilde{f}, \tilde{g}$ are mutually inverse isomorphisms $X \times_S U' \simeq Y \times_S U'$, and their base change to T is still φ. $\square$

Exercise 2.6

> **Problem.** Let $f : X \to Y$ be a morphism of integral schemes. Let ξ be the generic point of X. We say that f is *birational* if $f^\#_\xi : K(Y) \to K(X)$ is an isomorphism. Let us suppose that X and Y are of finite type over a scheme S, and that f is a morphism of S-schemes. Show that f is birational if and only if there exist a non-empty open subset U of X and an open subset V of Y such that f induces an isomorphism from U onto V.

Solution. *Convention.* The homomorphism $f^\#_\xi : \mathcal{O}_{Y,f(\xi)} \to \mathcal{O}_{X,\xi} = K(X)$ is a map $K(Y) \to K(X)$ only if $f(\xi) = \eta$, the generic point of Y, i.e. if f is dominant. So "birational" means: $f(\xi) = \eta$ and $f^\#_\xi : K(Y) = \mathcal{O}_{Y,\eta} \to K(X)$ is an isomorphism.

($\Leftarrow$) Let $U \neq \emptyset$ be open in X, V open in Y and $f|_U : U \to V$ an isomorphism. Then $\xi \in U$ is the generic point of U, and V is a nonempty open set of Y, so $\eta \in V$ and η is the generic point of V. The homeomorphism $f|_U$ sends the generic point ξ of U to the generic point η of V. The stalk of the isomorphism $f|_U$ at ξ is $f^\#_\xi : \mathcal{O}_{V,\eta} = K(Y) \to \mathcal{O}_{U,\xi} = K(X)$, an isomorphism. (Finite type is not used here.)

($\Rightarrow$) Let $f(\xi) = \eta$ and $f^\#_\xi : K(Y) \to K(X)$ be an isomorphism. Choose an affine open $\operatorname{Spec} A \subseteq S$ and a nonempty affine open $V_0 = \operatorname{Spec} B \subseteq Y$ over it. Since $\eta \in V_0$ and $f(\xi) = \eta$, the preimage $f^{-1}(V_0)$ contains ξ; choose a nonempty affine open $U_0 = \operatorname{Spec} C \subseteq f^{-1}(V_0)$. As f is an S-morphism, U_0 lies over $\operatorname{Spec} A$ and C is a finitely generated A-algebra (X is of

finite type over S). The map $f^\# : B \to C$ is A-linear, so C is generated by finitely many elements $x_1, \dots, x_m$ as a B-algebra. Both B and C are integral domains with $K(Y) = \operatorname{Frac} B$, $K(X) = \operatorname{Frac} C$, and $f^\#_\xi$ is the extension of $f^\# : B \to C$. In particular $B \to C$ is injective, and we identify $B \subseteq C \subseteq K(X) = K(Y) = \operatorname{Frac} B$ (the last equality is the hypothesis that $f^\#_\xi$ is bijective). Write $x_i = b_i/d_i$ with $b_i, d_i \in B$, $d_i \neq 0$ and put $d = d_1 \cdots d_m \neq 0$. Then $C = B[x_1, \dots, x_m] \subseteq B_d$, and $B \subseteq C$ gives $B_d \subseteq C_d$. So $C_d = B_d$.

Let $V = D(d) \subseteq V_0$ and $U = D(f^\#(d)) \subseteq U_0$. Both are nonempty ($d \neq 0$ in the domains $B \subseteq C$). We have $f(U) \subseteq V$, and $f|_U : U = \operatorname{Spec} C_d \to V = \operatorname{Spec} B_d$ is Spec of the inclusion $B_d \to C_d$, which is the identity of $B_d = C_d$. So $f|_U : U \to V$ is an isomorphism. □

Exercise 2.7

> **Problem.** Let k be a field and $\overline{k}$ the algebraic closure of k. Let $\overline{X}, \overline{Y}$ be algebraic varieties over $\overline{k}$, $\overline{f}$ a morphism from $\overline{X}$ to $\overline{Y}$. Show that there exist a finite extension K of k, algebraic varieties X, Y over K, and a morphism $f : X \to Y$, such that $X_{\overline{k}} = \overline{X}$, $Y_{\overline{k}} = \overline{Y}$, and $\overline{f} = f \times_K \overline{k}$.

Solution. Here K is a finite subextension of $\overline{k}/k$ (so $\overline{k}$ is also an algebraic closure of K), $X_{\overline{k}} = X \times_K \overline{k}$, and "=" means a fixed isomorphism of $\overline{k}$-schemes, compatible with the morphisms. Write $L = \overline{k}$ throughout. If a datum is defined over K, its base change to any finite $K' \supseteq K$ (inside L) is defined over K'; so we may always enlarge K finitely many times.

Lemma A. Let $K \subseteq L$ be fields and X a K-scheme, $p : X_L \to X$ the projection. (i) p is surjective. (ii) For $z \in X_L$ over $x \in X$, $\mathcal{O}_{X,x} \to \mathcal{O}_{X_L,z}$ is injective. (iii) If $f, g : X \to Y$ are K-morphisms with $f_L = g_L$ (Y a K-scheme), then $f = g$.

Proof. (i) The fibre of p over x is $\operatorname{Spec}(k(x) \otimes_K L)$ (Prop. 1.16), and $k(x) \otimes_K L \neq 0$ since L is a nonzero free K-module. (ii) Take an affine open $\operatorname{Spec} A \ni x$ with $x \leftrightarrow \mathfrak{p}$ and $z \leftrightarrow \mathfrak{q} \subset A \otimes_K L$. The ring $A \otimes_K L$ is free over A, so $(A \otimes_K L)_\mathfrak{q}$ is a localization of a flat A-module, hence flat over A, and as an $A_\mathfrak{p}$-module it is flat over $A_\mathfrak{p}$ (the elements of $A \setminus \mathfrak{p}$ are units). So $A_\mathfrak{p} \to (A \otimes L)_\mathfrak{q}$ is a flat local homomorphism, hence faithfully flat, hence injective. (iii) We have $p_Y \circ f_L = f \circ p_X$, so $f \circ p_X = g \circ p_X$. By (i), $f(x) = g(x)$ for all x. For $t \in \mathcal{O}_Y(W)$ and $x \in f^{-1}(W)$, the germs of $f^\#(t)$ and $g^\#(t)$ at x are $f^\#_x(t_{f(x)})$ and $g^\#_x(t_{f(x)})$, and they have the same image in $\mathcal{O}_{X_L,z}$ (z over x) because $f \circ p_X = g \circ p_X$. By (ii) they coincide. So $f^\#(t) = g^\#(t)$, and $f = g$. □

Claim 2 (morphisms descend). Let X, Y be varieties over a finite subextension K and $h : X_L \to Y_L$ an L-morphism. Then there are a finite $K_2 \supseteq K$ in L and a K_2-morphism $f : X_{K_2} \to Y_{K_2}$ with $f_L = h$.

Proof. Cover Y by finitely many affine opens $Y_j = \operatorname{Spec} C_j$, and X by finitely many affine opens $X_i = \operatorname{Spec} B_i$. Then $(X_i)_L = \operatorname{Spec}(B_i \otimes_K L)$ is covered by the principal opens $D(\bar{b})$, $\bar{b} \in B_i \otimes_K L$, with $D(\bar{b}) \subseteq h^{-1}((Y_j)_L)$ for some j. By quasi-compactness finitely many suffice, say $D(\bar{b}_m)$ ($m = 1, \dots, M$) in $(X_{i(m)})_L$ with $h(D(\bar{b}_m)) \subseteq (Y_{j(m)})_L$. Choose a finite $K_1 \supseteq K$ such that all $\bar{b}_m$ lie in $B_{i(m)} \otimes_K K_1$ (finitely many elements, each a finite sum of tensors); call them b_m. Put $W_m = D(b_m) \subseteq (X_{i(m)})_{K_1}$, an affine open with $(W_m)_L = D(\bar{b}_m)$. Since the $(W_m)_L$ cover X_L and $p : X_L \to X_{K_1}$ is surjective (Lemma A), the W_m cover X_{K_1}.

On $(W_m)_L = \operatorname{Spec}(R_m \otimes_{K_1} L)$, where $R_m = \mathcal{O}(W_m)$, the morphism h is given by a K-algebra homomorphism $\bar{\psi}_m : C_{j(m)} \to R_m \otimes_{K_1} L$. Write $C_{j(m)} = K[T_1, \dots, T_n]/(Q_1, \dots, Q_r)$ and let c_l be

the classes of T_l. Choose a finite $K_2 \supseteq K_1$ such that all $\bar{\psi}_m(c_l)$ lie in the subring $R_m \otimes_{K_1} K_2$ (the inclusion $R_m \otimes K_2 \subseteq R_m \otimes L$ is injective because L is free over K_2). Since the Q_s vanish on the $\bar{\psi}_m(c_l)$ in the big ring, they vanish in the subring, so $\bar{\psi}_m$ restricts to a K-algebra homomorphism $\psi_m : C_{j(m)} \to R_m \otimes_{K_1} K_2$, i.e. to a K_2-morphism $f_m : (W_m)_{K_2} \to (Y_{j(m)})_{K_2}$ with $(f_m)_L = h|_{(W_m)_L}$ (two L-algebra homomorphisms from $C_{j(m)} \otimes_K L$ agreeing on the $c_l \otimes 1$ agree).

On $(W_m \cap W_{m'})_{K_2}$ the morphisms $f_m, f_{m'}$ have the same base change to L, namely h restricted, so they agree by Lemma A(iii). Hence they glue to $f : X_{K_2} \to Y_{K_2}$. Then f_L and h agree on the open cover $\{(W_m)_L\}$ of X_L, so $f_L = h$. □

Claim 1 (objects descend). For every variety $\overline{X}$ over L there are a finite K and a variety X over K with $X_L \simeq \overline{X}$.

Proof. Cover $\overline{X}$ by finitely many affine opens $\overline{U}_i = \operatorname{Spec} \overline{B}_i$, $\overline{B}_i = L[T_1, \ldots, T_{n_i}]/\overline{I}_i$ with $\overline{I}_i$ generated by finitely many polynomials (Hilbert). Choose a finite K_1 containing all coefficients of these generators and of the relevant data below, and let $U_i = \operatorname{Spec} K_1[T]/I_i$ with I_i generated by the same polynomials; then $(U_i)_L = \overline{U}_i$. The set $\overline{U}_{ij} = \overline{U}_i \cap \overline{U}_j$ is open in the Noetherian space $\overline{U}_i$, hence a finite union of principal opens $D(\bar{b}_{ijl})$, $\bar{b}_{ijl} \in \overline{B}_i$. Enlarging K_1, all $\bar{b}_{ijl}$ come from $b_{ijl} \in K_1[T]/I_i$. Let $U_{ij} = \bigcup_l D(b_{ijl}) \subseteq U_i$; then $(U_{ij})_L = \overline{U}_{ij}$ (Lemma A(i): p is surjective, so the open set U_{ij} is determined by its preimage). The transition isomorphisms $\varphi_{ij} : \overline{U}_{ij} \to \overline{U}_{ji}$ descend by Claim 2 (applied to the varieties U_{ij}, U_{ji}) to K-morphisms $\varphi'_{ij} : (U_{ij})_K \to (U_{ji})_K$ over some finite $K \supseteq K_1$ (a common one for the finitely many pairs). The relations $\varphi'_{ji} \circ \varphi'_{ij} = \mathrm{id}$, $\varphi'_{ij}((U_{ij} \cap U_{ik})_K) = (U_{ji} \cap U_{jk})_K$ and $\varphi'_{ik} = \varphi'_{jk} \circ \varphi'_{ij}$ on $(U_{ij} \cap U_{ik})_K$ hold after base change to L (they hold for the φ_{ij}, being gluing data of $\overline{X}$). The equalities of morphisms then hold already over K by Lemma A(iii), and the equality of open sets because an open subset of a K-scheme is determined by its preimage under the surjection p. Thus the $(U_i)_K$ glue along the φ'_{ij} to a K-scheme X, covered by finitely many affines of finite type, i.e. a variety. Its base change X_L is glued from the $(U_i)_L = \overline{U}_i$ along the φ_{ij}, so $X_L \simeq \overline{X}$. □

Conclusion. By Claim 1 there are X_1 over K_1 and Y_1 over K'_1 with isomorphisms $\alpha : (X_1)_L \simeq \overline{X}$, $\beta : (Y_1)_L \simeq \overline{Y}$. Let K be the compositum of K_1, K'_1 (finite over k), and replace X_1, Y_1 by their base changes X, Y to K. Then $h = \beta^{-1} \circ \overline{f} \circ \alpha : X_L \to Y_L$ is an L-morphism, and Claim 2 gives, over a finite $K_2 \supseteq K$, a morphism $f : X_{K_2} \to Y_{K_2}$ with $f_L = h$. Replacing K by K_2 and X, Y by X_{K_2}, Y_{K_2}, and identifying X_L with $\overline{X}$ via α and Y_L with $\overline{Y}$ via β, we get $X_L = \overline{X}$, $Y_L = \overline{Y}$ and $f_L = \overline{f}$. □

Exercise 2.8

> **Problem.** Let $f : X \to Y$ be a morphism of schemes. We say that f has *finite fibers* if $f^{-1}(y)$ is a finite set for every $y \in Y$. We say that f is *quasi-finite* if, moreover, $\mathcal{O}_{X_y,x}$ is finite over $k(y)$ for every $x \in X_y$. Show that a morphism of finite type with finite fibers is quasi-finite. Give an example of a morphism with finite fibers which is not quasi-finite.

Solution. Let f be of finite type with finite fibers. For $y \in Y$ the fibre X_y is an algebraic variety over $K := k(y)$ (Remark 2.5) whose underlying space is the finite set $f^{-1}(y)$. Let $x \in X_y$ and choose an affine open $\operatorname{Spec} B \ni x$ of X_y; B is a finitely generated K-algebra with finitely many prime ideals, and $\mathcal{O}_{X_y,x} = B_{\mathfrak{m}}$, where $\mathfrak{m}$ is the prime of x.

Every prime of B is maximal. Suppose $\mathfrak{p}$ is a non-maximal prime. Then $B/\mathfrak{p}$ is a finitely generated K-domain which is not a field, so it has dimension $d \geq 1$ (a domain integral over K is a

field). By Noether normalization, $K[T_1,\dots,T_d] \subseteq B/\mathfrak{p}$ is an integral extension, so $\operatorname{Spec}(B/\mathfrak{p}) \to \operatorname{Spec} K[T_1,\dots,T_d]$ is surjective (Prop. 2.5.10). But $K[T_1,\dots,T_d]$ has infinitely many primes: the ideals (g) with $g \in K[T_1]$ irreducible are distinct primes of $K[T_1,\dots,T_d]$, and there are infinitely many irreducible polynomials (if $g_1,\dots,g_n$ were all of them, any irreducible factor of $g_1 \cdots g_n + 1$ would be a new one). So B would have infinitely many primes, a contradiction.

Hence B is Noetherian of dimension 0, i.e. Artinian (Atiyah–Macdonald, Thm. 8.5), so $B \simeq \prod_i B_{\mathfrak{m}_i}$ over its finitely many maximal ideals (ibid., Thm. 8.7). B has finite length as a B-module with composition factors $B/\mathfrak{m}_i$, each of which is finite over K by Zariski's lemma (Nullstellensatz). So $\dim_K B < \infty$. The local ring $B_{\mathfrak{m}}$ is one of the direct factors of B, hence $\dim_K B_{\mathfrak{m}} < \infty$: $\mathcal{O}_{X_y,x}$ is finite over $k(y)$. So f is quasi-finite.

Example. Let k be a field, $R = k[T]_{(T)}$ and $f : X = \operatorname{Spec} R \to Y = \operatorname{Spec} k$. The only fibre is X itself, which has exactly two points (the generic point (0) and the closed point (T)), so f has finite fibers. For the closed point x, $\mathcal{O}_{X_y,x} = R$ is not finite over $k(y) = k$ (it contains $k[T]$, of infinite dimension). So f is not quasi-finite (consistently, f is not of finite type). □

Exercise 2.9

Problem. Let X, Y be algebraic varieties over a field k, with X geometrically reduced. Let $f, g : X \to Y$ be two morphisms that canonically induce the same map $X(\overline{k}) \to Y(\overline{k})$. We want to show that $f = g$ as morphisms.

(a) Show that $f(x) = g(x)$ for any closed point $x \in X$, and then show the same equality for any point x.

(b) Reduce to the case X affine, $Y = \mathbb{A}^n_k$, and k algebraically closed.

(c) Using Exercise 2.3.7 and Lemma 2.1.18, show that $f = g$.

Solution. The hypothesis means $f \circ u = g \circ u$ for every k-morphism $u : \operatorname{Spec} \overline{k} \to X$ (the map $X(\overline{k}) \to Y(\overline{k})$ is $u \mapsto f \circ u$).

Lemma C. Let Z be a variety over k. (i) A point $z \in Z$ is closed iff $[k(z) : k] < \infty$. (ii) Every nonempty open subset of Z contains a closed point of Z.

Proof. (i) If z is closed, it is closed in an affine open $\operatorname{Spec} B \ni z$, so it is a maximal ideal and $k(z) = B/\mathfrak{m}$ is finite over k by Zariski's lemma. Conversely let $[k(z) : k] < \infty$ and let $z' \in \overline{\{z\}}$. Take an affine open $\operatorname{Spec} B' \ni z'$; it contains z (a neighbourhood of z' meets $\{z\}$). Let $\mathfrak{p} \subset B'$ be the prime of z. Then $B'/\mathfrak{p} \subseteq k(z)$ is a domain integral over k, hence a field, so $\mathfrak{p}$ is maximal and z is closed in $\operatorname{Spec} B'$. Thus $z' \in \overline{\{z\}} \cap \operatorname{Spec} B' = \{z\}$, so $z' = z$. (ii) A nonempty open set contains a nonempty affine open $\operatorname{Spec} B'$ with $B' \neq 0$ finitely generated over k; a maximal ideal of B' gives a point with $k(z) = B'/\mathfrak{m}$ finite over k, which is closed by (i). □

(a) Let $x \in X$ be closed. Then $[k(x) : k] < \infty$ by Lemma C(i). Choose a k-embedding $\tau : k(x) \to \overline{k}$, and let u be the composition $\operatorname{Spec} \overline{k} \to \operatorname{Spec} k(x) \to X$, a $\overline{k}$-point with image x. Then $f \circ u = g \circ u$ has image $f(x)$ and also $g(x)$. So $f(x) = g(x)$. (This part does not need X to be geometrically reduced.)

Now let $x \in X$ be arbitrary and $Z = \overline{\{x\}}$, with the reduced structure; it is a variety over k with generic point x. Closed points of Z are the points of Z with finite residue extension, hence closed in X (Lemma C(i) applied to Z and to X), so $f = g$ on them. For $W \subseteq Y$ open, $x \in f^{-1}(W)$

iff $f^{-1}(W) \cap Z \neq \emptyset$ (a nonempty open subset of the irreducible Z contains its generic point) iff $f^{-1}(W) \cap Z$ contains a closed point z of Z (Lemma C(ii)) iff $g^{-1}(W) \cap Z$ contains a closed point of Z (as $f(z) = g(z)$) iff $x \in g^{-1}(W)$. So $f(x)$ and $g(x)$ lie in the same open subsets of Y. A scheme is T_0: if $y \neq y'$, take an affine open $\operatorname{Spec} C$ containing one of them. If it misses the other, it separates them. Otherwise y, y' are distinct primes $\mathfrak{p}_y, \mathfrak{p}_{y'}$ of C, one of which is not contained in the other, say $\mathfrak{p}_{y'} \not\subseteq \mathfrak{p}_y$; for $c \in \mathfrak{p}_{y'} \setminus \mathfrak{p}_y$ the set $D(c)$ contains y but not y'. So $f(x) = g(x)$. Thus f and g have the same underlying map.

(b) Reduction to $k = \overline{k}$. Put $L = \overline{k}$, $\overline{X} = X_L$, $\overline{Y} = Y_L$, $\overline{f} = f_L$, $\overline{g} = g_L$. By hypothesis $\overline{X}$ is reduced. An L-point $v : \operatorname{Spec} L \to \overline{X}$ corresponds to the k-point $u = p_X \circ v \in X(L)$ (Prop. 2.18(a)), and $p_Y \circ \overline{f} \circ v = f \circ u$. As $f \circ u = g \circ u$, we get $p_Y \circ \overline{f} \circ v = p_Y \circ \overline{g} \circ v$, so $\overline{f} \circ v = \overline{g} \circ v$ (two L-points of $\overline{Y}$ with the same image in Y coincide, again by Prop. 2.18(a)). So $\overline{f}, \overline{g}$ satisfy the hypotheses over the algebraically closed field L, with $\overline{X}$ reduced. If we know $\overline{f} = \overline{g}$, then $f = g$ by Lemma A(iii) of Ex. 2.7.

Reduction to X affine, $Y = \mathbb{A}^n$. Now assume $k = \overline{k}$ and X reduced; by (a), $f(z) = g(z)$ for all $z \in X$. Let $z \in X$, $y = f(z) = g(z)$. Choose an affine open $V = \operatorname{Spec} C \ni y$ of Y and an affine open $U = \operatorname{Spec} B \ni z$ of X with $f(U) \subseteq V$ (hence $g(U) \subseteq V$, same maps on points). We have $C = k[T_1, \ldots, T_n]/I$, so $i : V \to \mathbb{A}^n_k$ is a closed immersion, and B is a reduced finitely generated k-algebra. A morphism $U \to V$ is a ring homomorphism $C \to B$, and composing with the surjection $k[T] \to C$ is injective on such homomorphisms. So $f|_U = g|_U$ iff $i \circ f|_U = i \circ g|_U$ as morphisms $U \to \mathbb{A}^n_k$. Since the U's cover X, it suffices to show this.

(c) Conclusion. The morphisms $i \circ f|_U, i \circ g|_U : U \to \mathbb{A}^n_k$ correspond to the images $f_l, g_l \in B$ of T_l ($l = 1, \ldots, n$). We show $f_l = g_l$. Let $\mathfrak{m} \subset B$ be a maximal ideal and x the corresponding closed point of U, which is closed in X (Lemma C). As $k = \overline{k}$, $B/\mathfrak{m} = k$ (Zariski's lemma), so $u_x : \operatorname{Spec} k \to U \subseteq X$ is a k-point with $f \circ u_x = g \circ u_x$. Apply i: the point $i \circ f \circ u_x \in \mathbb{A}^n(k) = k^n$ is the homomorphism $k[T] \to B \to B/\mathfrak{m} = k$, $T_l \mapsto f_l \bmod \mathfrak{m}$, i.e. it is the point $(f_1(x), \ldots, f_n(x))$ (this is Ex. 2.3.7, and is immediate from the correspondence between morphisms into an affine scheme and ring homomorphisms). Likewise $i \circ g \circ u_x = (g_1(x), \ldots, g_n(x))$. Hence $f_l - g_l \in \mathfrak{m}$ for every maximal ideal $\mathfrak{m}$ of B. For a finitely generated algebra B over a field, the Jacobson radical equals the nilradical (Nullstellensatz; Lemma 2.1.18). Since B is reduced, $f_l - g_l \in \bigcap_{\mathfrak{m}} \mathfrak{m} = \mathfrak{N}(B) = 0$. So $i \circ f|_U = i \circ g|_U$, hence $f|_U = g|_U$, and so $\overline{f} = \overline{g}$ by gluing over the cover. By (b), $f = g$ in the original setting. □

Exercise 2.10

Problem. Let k be a field and K/k a finite Galois extension with group G. Let X be an algebraic variety over k.

(a) Let L be an extension of k. Show that G acts transitively on $\operatorname{Spec}(L \otimes_k K)$.

(b) Let us suppose X is irreducible. Show that G acts transitively on the irreducible components of X_K. Deduce from this that the irreducible components of $X_{\overline{k}}$ have the same dimension.

(c) Let us suppose X is connected. Show that G acts transitively on the connected components of X_K.

Solution. For $\sigma \in G$, G acts on $\operatorname{Spec} K$ by $\operatorname{Spec} \sigma$, on $\operatorname{Spec}(L \otimes_k K)$ by $\operatorname{Spec}(\mathrm{id} \otimes \sigma)$, and on

$X_K = X \times_k \operatorname{Spec} K$ by $\mathrm{id} \times \operatorname{Spec} \sigma$; these are automorphisms of schemes (not of K-schemes), hence homeomorphisms, and they commute with the projection $p : X_K \to X$.

(a) Let $R = L \otimes_k K$, $\sigma_R = \mathrm{id} \otimes \sigma$. Since G is a group, it is enough to show: for primes $\mathfrak{p}, \mathfrak{q}$ of R there is $\sigma \in G$ with $\sigma_R(\mathfrak{p}) = \mathfrak{q}$. R is finite over L (K/k is finite), so $R/\mathfrak{p}$ and $R/\mathfrak{q}$ are domains that are finite-dimensional over L, hence fields $\kappa(\mathfrak{p}), \kappa(\mathfrak{q})$, finite over L. We have k-homomorphisms $j_\mathfrak{p} : K \to \kappa(\mathfrak{p})$, $j_\mathfrak{q} : K \to \kappa(\mathfrak{q})$, $x \mapsto \overline{1 \otimes x}$. Let Ω be an algebraic closure of $\kappa(\mathfrak{q})$. As $\kappa(\mathfrak{p})$ is algebraic over $L \subseteq \Omega$, there is an L-embedding $\tau : \kappa(\mathfrak{p}) \to \Omega$. The two k-embeddings $\tau \circ j_\mathfrak{p}$ and $j_\mathfrak{q}$ of K in Ω have the same image, since K/k is normal (the image of any k-embedding is the set of roots in Ω of the minimal polynomials of the elements of K, all of which split in K). So $\sigma := j_\mathfrak{q}^{-1} \circ \tau \circ j_\mathfrak{p}$ is a well defined k-automorphism of K, i.e. $\sigma \in G$, with $\tau \circ j_\mathfrak{p} = j_\mathfrak{q} \circ \sigma$. Consider the ring homomorphisms $\alpha, \beta : R \to \Omega$ given by $\alpha(l \otimes x) = l \cdot \tau j_\mathfrak{p}(x)$ and $\beta(l \otimes x) = l \cdot j_\mathfrak{q}(x)$. Then $\ker \alpha = \mathfrak{p}$ (it is $R \to \kappa(\mathfrak{p})$ followed by the injective τ) and $\ker \beta = \mathfrak{q}$. Moreover $\alpha(l \otimes x) = l\, j_\mathfrak{q}(\sigma x) = \beta(l \otimes \sigma x)$, i.e. $\alpha = \beta \circ \sigma_R$, so $\mathfrak{p} = \sigma_R^{-1}(\mathfrak{q})$. Hence G acts transitively.

Lemma D. Let X be a variety over k, L/k an algebraic extension and $p : X_L \to X$ the projection. Let $c \in X_L$ be the generic point of an irreducible component C of X_L, and $x = p(c)$. Then x is the generic point of an irreducible component D of X, and $\dim C = \dim D$.

Proof. The ring $\mathcal{O}_{X_L,c}$ has a unique prime ideal (as c has no proper generization). The homomorphism $\varphi : \mathcal{O}_{X,x} \to \mathcal{O}_{X_L,c}$ is flat and local (as in Lemma A(ii) of Ex. 2.7). Suppose x had a proper generization, i.e. $\mathcal{O}_{X,x}$ had a prime $\mathcal{Q} \subsetneq \mathfrak{m}_x$. Going-down for the flat φ gives a prime $\mathfrak{q} \subseteq \mathfrak{m}_c$ with $\varphi^{-1}(\mathfrak{q}) = \mathcal{Q} \neq \mathfrak{m}_x = \varphi^{-1}(\mathfrak{m}_c)$, so $\mathfrak{q} \neq \mathfrak{m}_c$, contradicting uniqueness of the prime. So x is the generic point of $D = \overline{\{x\}}$, an irreducible component. For dimensions, let $\operatorname{Spec} A \ni x$ be an affine open; then $c \in \operatorname{Spec}(A \otimes_k L)$, corresponding to a prime $\mathfrak{q}$ with $\mathfrak{q} \cap A = \mathfrak{p}_x$, and $C \cap \operatorname{Spec}(A \otimes L) = V(\mathfrak{q})$, $D \cap \operatorname{Spec} A = V(\mathfrak{p}_x)$. The extension $A/\mathfrak{p}_x \to (A \otimes_k L)/\mathfrak{q}$ is injective and integral (L/k algebraic), so the two rings have the same dimension (Prop. 2.5.10). The affine opens $\operatorname{Spec} A \ni x$ cover D (they are those meeting D), and the corresponding $\operatorname{Spec}(A \otimes L)$ cover C (these are exactly the ones containing c, i.e. with $x \in \operatorname{Spec} A$). By the Fact in Ex. 2.1, $\dim C$ and $\dim D$ are the suprema of the equal numbers $\dim V(\mathfrak{q})$, $\dim V(\mathfrak{p}_x)$. □

(b) Let X be irreducible with generic point η. By Lemma D, every irreducible component of X_K has its generic point in $F := p^{-1}(\eta)$ (as $p(c)$ is the generic point of a component of X, hence η). Conversely, every point $c' \in F$ is the generic point of an irreducible component of X_K: F is homeomorphic to $\operatorname{Spec}(k(\eta) \otimes_k K)$ (Prop. 1.16), and this ring is finite over the field $k(\eta)$, hence Artinian, so every point of F is closed in F. A generization c'' of c' in X_K satisfies $p(c'') \rightsquigarrow \eta$, so $p(c'') = \eta$ and $c'' \in F$; hence $c'' = c'$. So c' has no proper generization, and the generic point of an irreducible component of X_K containing c' equals c'. Therefore the generic points of the irreducible components of X_K are exactly the points of F, i.e. of $\operatorname{Spec}(k(\eta) \otimes_k K)$. The action of G on X_K preserves F and restricts to the action $\operatorname{Spec}(\mathrm{id} \otimes \sigma)$ on $\operatorname{Spec}(k(\eta) \otimes_k K)$. By (a) with $L = k(\eta)$, G acts transitively on F, hence on the irreducible components of X_K (they are the closures of the points of F, and G acts by homeomorphisms).

Consequences for dimensions. Since each $\sigma \in G$ is a homeomorphism of X_K, it preserves dimension of closed subsets; by transitivity all irreducible components of X_K have the same dimension. For $\overline{k}$ we argue directly (as $\overline{k}/k$ may be inseparable, it is not a finite Galois extension; the same proof applies to any algebraic extension). Let C be an irreducible component of $X_{\overline{k}}$. By Lemma D (with $L = \overline{k}$), $\dim C = \dim D$, where D is an irreducible component of X; since

X is irreducible, $D = X$. Hence every irreducible component of $X_{\overline{k}}$ has dimension $\dim X$, in particular they all have the same dimension.

(c) Let X be connected. The space X_K is Noetherian (Ex. 2.1, K/k finite, X_K of finite type over K), so it has finitely many connected components, each open and closed. Let C be one of them and $C' = \bigcup_{\sigma\in G} \sigma(C)$, $C'' = X_K \setminus C'$. Both are open and closed in X_K (unions of connected components), and G-stable. The map p is closed: over an affine open $\operatorname{Spec} A \subseteq X$ it is $\operatorname{Spec}(A \otimes_k K) \to \operatorname{Spec} A$, induced by the injective integral homomorphism $A \to A \otimes_k K$, and such maps are closed (the image of $V(I)$ is $V(I \cap A)$, by Prop. 2.5.10 applied to $A/(I \cap A) \to (A \otimes K)/I$); closedness is local on the target. Moreover p is surjective (Lemma A(i) of Ex. 2.7). So $X = p(C') \cup p(C'')$ with both sets closed. They are disjoint: if $z = p(y') = p(y'')$ with $y' \in C'$, $y'' \in C''$, then y', y'' lie in the fibre $p^{-1}(z) \simeq \operatorname{Spec}(k(z) \otimes_k K)$, on which G acts transitively by (a). So $y'' = \sigma y'$ for some σ, and then $y'' \in C'$ by G-stability of C', a contradiction. As X is connected and $p(C') \neq \emptyset$, we get $p(C'') = \emptyset$, i.e. $C'' = \emptyset$ and $C' = X_K$. So every connected component of X_K is contained in $\bigcup_\sigma \sigma(C)$, where each $\sigma(C)$ is a connected component (images of connected components under homeomorphisms). Distinct connected components are disjoint, so every connected component equals some $\sigma(C)$, which is the claimed transitivity. □

Exercise 2.11

Problem. Let X be an algebraic variety over a field k.

(a) Show that if X is connected and if $X(k) \neq \emptyset$, then X is geometrically connected.

(b) Show through an example that (a) is false if we replace "connected" by "irreducible".

Solution. Write $p_K : X_K \to X$ for the projection, for a field extension K/k. Two general facts are used.

(F1) p_K is surjective. Indeed, for $z \in X$ the fibre $p_K^{-1}(z)$ is homeomorphic to $(X_K)_z = \operatorname{Spec} k(z) \times_k \operatorname{Spec} K = \operatorname{Spec}(k(z) \otimes_k K)$ (Prop. 1.16), which is non-empty because the tensor product of two non-zero vector spaces over a field is non-zero. If $k(z) = k$, this fibre is $\operatorname{Spec} K$, a single point.

(F2) Descent of closed sets. Let C be a closed subset of $X_{\overline{k}}$. Then there are a finite extension K'/k inside $\overline{k}$ and a closed subset $Z \subseteq X_{K'}$ with $C = q^{-1}(Z)$, where $q : X_{\overline{k}} \to X_{K'}$ is the projection; and the same holds, with Z replaced by its preimage in $X_{K''}$, for every finite $K'' \supseteq K'$. Indeed, give C its reduced structure and apply Lemma 2.6 to $K = \overline{k}$: there are a finite K' and a (reduced) closed subvariety Z of $X_{K'}$ with $Z_{\overline{k}} = C$. The underlying set of $Z_{\overline{k}} = Z \times_{X_{K'}} X_{\overline{k}}$ is $q^{-1}(Z)$. For $K'' \supseteq K'$ finite, $q = q' \circ q''$ with $q'' : X_{\overline{k}} \to X_{K''}$, $q' : X_{K''} \to X_{K'}$, so $C = q''^{-1}(q'^{-1}(Z))$.

(a) By definition we must show that $X_{\overline{k}}$ is connected. We proceed in two steps.

Step 1: $X_{K'}$ is connected for every finite extension K'/k. Let $n = [K' : k]$ and $Y = X_{K'}$, $p = p_{K'}$. Fix $x \in X(k)$. By Prop. 2.18(b), x is a point of X with $k(x) = k$, so by (F1) the fibre of p over x is a single point and $k(x) \otimes_k K' = K'$.

The morphism p is affine: over an affine open $U = \operatorname{Spec} A$ of X we have $p^{-1}(U) = \operatorname{Spec}(A \otimes_k K')$. Hence $\mathcal{B} := p_* \mathcal{O}_Y = \mathcal{O}_X \otimes_k K'$ is a locally free $\mathcal{O}_X$-module of rank n (choose a k-basis of K'), and $\mathcal{O}_Y(Y) = \mathcal{B}(X)$.

Let $e \in \mathcal{O}_Y(Y)$ be an idempotent. Multiplication by e and by $1-e$ are $\mathcal{O}_X$-linear, so $\mathcal{B} = \mathcal{M} \oplus \mathcal{N}$ with $\mathcal{M} = e\mathcal{B}$, $\mathcal{N} = (1-e)\mathcal{B}$. For an affine open $U = \operatorname{Spec} A$ the modules $\mathcal{M}(U)$, $\mathcal{N}(U)$ are direct summands of the free A-module $A \otimes_k K' \simeq A^n$, hence finitely generated projective. The rank function of a finitely generated projective module is locally constant on $\operatorname{Spec} A$; its value at z is $r_{\mathcal{M}}(z) = \dim_{k(z)} \mathcal{M} \otimes k(z)$. So $r_{\mathcal{M}} : X \to \{0, \dots, n\}$ is locally constant, hence constant, equal to some r, because X is connected. Likewise $r_{\mathcal{N}} = n - r$ is constant.

Now compute at x: $\mathcal{B} \otimes k(x) = K'$, and $\mathcal{M} \otimes k(x) = \bar{e}\,K'$ where $\bar{e} \in K'$ is the image of e, an idempotent of the field K', hence 0 or 1. So $r \in \{0, n\}$. If $r = 0$, then all stalks of $\mathcal{M}$ are free of rank 0, i.e. $\mathcal{M} = 0$, so $e = e \cdot e \in \mathcal{M}(X)$ is 0. If $r = n$, then $\mathcal{N} = 0$ and $1 - e = 0$. Thus $\mathcal{O}_Y(Y)$ has no idempotents other than $0, 1$. Since $Y \neq \emptyset$ and a decomposition $Y = U \sqcup V$ into open and closed sets would give the idempotent 1_U, Y is connected.

Step 2: $X_{\overline{k}}$ is connected. Suppose $X_{\overline{k}} = C_1 \sqcup C_2$ with C_1, C_2 non-empty, closed (hence both open and closed). By (F2), applied to C_1, C_2 and a common finite extension K' (enlarge one of the two), there are closed sets $Z_1, Z_2 \subseteq X_{K'}$ with $C_i = q^{-1}(Z_i)$. Since q is surjective (F1 for $X_{K'}$ over K', or directly), $Z_1 \cup Z_2 = q(C_1 \cup C_2) = X_{K'}$ and $Z_1 \cap Z_2 = \emptyset$ (a point of $Z_1 \cap Z_2$ would have a preimage in $C_1 \cap C_2 = \emptyset$); also $Z_i \neq \emptyset$ since $C_i \neq \emptyset$. So $X_{K'}$ is disconnected, contradicting Step 1. Hence $X_{\overline{k}}$ is connected, i.e. X is geometrically connected.

(b) Let $k = \mathbf{R}$ and $X = \operatorname{Spec} \mathbf{R}[x, y]/(x^2 + y^2)$. The polynomial $x^2 + y^2$ is irreducible in $\mathbf{R}[x, y]$: otherwise it would be a product of two real linear forms and its real zero set would contain a real line, whereas $x^2 + y^2 > 0$ for $(x, y) \neq (0, 0)$ in $\mathbf{R}^2$. As $\mathbf{R}[x, y]$ is factorial, $(x^2 + y^2)$ is prime, so X is an integral (in particular irreducible) variety, and $(0, 0) \in X(\mathbf{R})$. But over $\mathbf{C}$ we have $x^2 + y^2 = (x + iy)(x - iy)$, so $X_{\mathbf{C}} = \operatorname{Spec} \mathbf{C}[x, y]/((x + iy)(x - iy))$ is the union of the two distinct lines $V(x + iy)$ and $V(x - iy)$, hence reducible. So X is irreducible with a rational point but not geometrically irreducible. (By (a), X is nevertheless geometrically connected: the two lines meet at the origin.) □

Exercise 2.12

> **Problem.** Let X be an irreducible algebraic variety over a field k. We call the irreducible components of $X_{\overline{k}}$ the geometric irreducible components of X. Let us assume X is integral. Show that the number of geometric irreducible components of X is equal to $[K(X) \cap k^s : k]$, where k^s is the separable closure of k.

Solution. *Meaning of the statement.* Put $L = K(X)$ and let $k_1 \subseteq L$ be the set of elements of L that are algebraic and separable over k; this is a subfield (the separable closure of k in L), and it is what $K(X) \cap k^s$ denotes (compare the proof of Cor. 2.14(d)). We show that $[k_1 : k]$ is finite and that it equals the number of irreducible components of $X_{\overline{k}}$.

Step 0: $m := [k_1 : k] < \infty$. Let $\operatorname{Spec} A \subseteq X$ be a non-empty affine open set, so $L = \operatorname{Frac} A$. By Noether normalization L is finite over $F = k(T_1, \dots, T_d)$. For a finite subextension $k' \subseteq k_1$ of k, the ring $k' \otimes_k F$ is a localization of $k'[T_1, \dots, T_d]$, hence a domain, and it is finite over F of dimension $[k' : k]$, hence a field; it maps injectively to L (a ring homomorphism from a field), so $[k' : k] \leq [L : F]$. Thus $[k_1 : k] \leq [L : F]$.

Step 1: $L \otimes_k k^s$ is a product of exactly m fields. Since k_1/k is finite separable, $k_1 = k(\theta)$ with minimal polynomial f of degree m, and $f = \prod_{i=1}^{m} (T - r_i)$ with distinct $r_i \in k^s$. By the Chinese

remainder theorem

$$k_1 \otimes_k k^s \simeq k^s[T]/(f) \simeq \prod_{i=1}^{m} k^s_{(i)},$$

where $k^s_{(i)}$ is k^s regarded as a k_1-algebra via $\theta \mapsto r_i$. Hence

$$L \otimes_k k^s \simeq L \otimes_{k_1} (k_1 \otimes_k k^s) \simeq \prod_{i=1}^{m} L \otimes_{k_1} k^s_{(i)}.$$

Claim: if Ω is a field extension of k_1 which is algebraic and separable over k_1, then $L \otimes_{k_1} \Omega$ is a field. (Apply it to $\Omega = k^s_{(i)}$, which is separable algebraic over the image of k_1 since k^s/k is separable.) Proof of the claim. First let Ω be finite over k_1, say $\Omega = k_1[T]/(P)$ with P monic, irreducible and separable over k_1. Then $L \otimes_{k_1} \Omega = L[T]/(P)$. Let Q be a monic divisor of P in $L[T]$. In a splitting field of P over L we have $P = \prod(T - r_j)$ with distinct r_j, separable algebraic over k_1, and $Q = \prod_{j \in J}(T - r_j)$ for some subset J. The coefficients of Q lie in L and are polynomials in the r_j, hence are separable algebraic over k_1, hence over k. By definition of k_1 they lie in k_1. So $Q \in k_1[T]$ divides the irreducible polynomial P of $k_1[T]$, and $Q = 1$ or $Q = P$. Hence P is irreducible in $L[T]$ and $L[T]/(P)$ is a field. For general Ω, write Ω as the directed union of its finite subextensions Ω_α/k_1; then $L \otimes_{k_1} \Omega$ is the directed union of the subrings $L \otimes_{k_1} \Omega_\alpha$ (tensoring inclusions of vector spaces over a field preserves injectivity), which are fields; a directed union of fields is a field. This proves the claim, and Step 1.

Step 2: two facts on localization. Let D be a domain which is a k-algebra, $S = D \smallsetminus \{0\}$, E any k-algebra. Every $s \in S$ acts injectively on $E \otimes_k D$ (multiplication by s is an injective k-linear map of D, and $E \otimes_k -$ preserves injectivity). Hence $E \otimes_k D \subseteq S^{-1}(E \otimes_k D) = E \otimes_k \operatorname{Frac} D$. Moreover, in any ring R with a multiplicative set S of non-zero-divisors, the minimal primes of R do not meet S (an element of a minimal prime $\mathfrak{q}$ is a zero divisor, because $R_\mathfrak{q}$ has a nilpotent maximal ideal), and so $\mathfrak{q} \mapsto S^{-1}\mathfrak{q}$ is a bijection from the minimal primes of R onto the minimal primes of $S^{-1}R$ (primes of $S^{-1}R$ correspond order-preservingly to primes of R missing S, and a prime missing S which contains a smaller prime also contains it missing S).

Step 3: the components of X_{k^s}. Let $K = k^s$ and $p : X_K \to X$. Let η be the generic point of X and $\operatorname{Spec} A$ a non-empty affine open of X (A a domain with fraction field L). By Step 2 (with $D = A$, $E = K$), $A \otimes_k K \subseteq L \otimes_k K$ is the localization at $A \smallsetminus \{0\}$, and the minimal primes of $A \otimes K$ correspond to those of $L \otimes_k K$. By Step 1 the latter is a finite product of m fields, so it has exactly m primes, all minimal. Hence $\operatorname{Spec}(A \otimes K) = p^{-1}(\operatorname{Spec} A)$ has exactly m irreducible components, and their generic points are the m primes $w_1, \ldots, w_m$ of $A \otimes K$ missing $A \smallsetminus \{0\}$, i.e. the points of X_K lying over η. These m points form the set $p^{-1}(\eta)$, which does not depend on the affine open chosen. For every non-empty affine open U, the $\overline{\{w_j\}} \cap p^{-1}(U)$ are the m irreducible components of $p^{-1}(U)$, in particular they are pairwise non-comparable, and they cover $p^{-1}(U)$. Since the $p^{-1}(U)$ cover X_K, we get $X_K = \bigcup_j \overline{\{w_j\}}$ with no inclusions among the $\overline{\{w_j\}}$ (an inclusion would persist after intersecting with some $p^{-1}(U)$, where all w_j lie). Thus X_{k^s} has exactly m irreducible components.

Step 4: from k^s to $\overline{k}$. $\overline{k}/k^s$ is purely inseparable. Apply Prop. 2.7(c) to the variety X_{k^s} over k^s: the projection $X_{\overline{k}} = (X_{k^s})_{\overline{k}} \to X_{k^s}$ is a homeomorphism. Hence $X_{\overline{k}}$ also has exactly $m = [K(X) \cap k^s : k]$ irreducible components. □

Exercise 2.13

Problem. Let X be an algebraic variety over a field k. Show that X is geometrically reduced (resp. irreducible; connected) if and only if $X_{k'}$ is reduced (resp. irreducible; connected) for every finite extension k' of k.

Solution. We take all finite extensions k' inside $\overline{k}$. Let $q_{k'} : X_{\overline{k}} \to X_{k'}$ be the projection; it is surjective (fibres are $\operatorname{Spec}(\kappa \otimes_{k'} \overline{k}) \neq \emptyset$, Prop. 1.16).

"Only if". *Reduced:* for an affine open $\operatorname{Spec} A$ of X, $A \otimes_k k' \subseteq A \otimes_k \overline{k}$ (inclusion of vector spaces tensored with A), and a subring of a reduced ring is reduced. *Irreducible, connected:* $X_{k'}$ is the continuous image of $X_{\overline{k}}$ under $q_{k'}$, and a continuous image of an irreducible (resp. connected) space is irreducible (resp. connected).

"If". *Reduced.* Let $\operatorname{Spec} A$ be an affine open of X and $a \in A \otimes_k \overline{k}$ nilpotent. Then a lies in $A \otimes_k k'$ for some finite k' (it is a finite sum of tensors), and it is nilpotent there, hence 0. So $X_{\overline{k}}$ is covered by the reduced schemes $(\operatorname{Spec} A)_{\overline{k}}$.

Irreducible. Suppose $X_{\overline{k}} = C_1 \cup C_2$ with C_i closed and $C_i \neq X_{\overline{k}}$. By (F2) of Exercise 2.11 (a consequence of Lemma 2.6, proved there) there are a common finite extension k' and closed sets $Z_i \subseteq X_{k'}$ with $C_i = q_{k'}^{-1}(Z_i)$. Since $q_{k'}$ is surjective, $Z_1 \cup Z_2 = X_{k'}$, and $Z_i \neq X_{k'}$ because $C_i \neq X_{\overline{k}}$. Thus $X_{k'}$ is reducible, a contradiction. (Also $X_{\overline{k}} \neq \emptyset$ since $X \neq \emptyset$ in this case.)

Connected. Suppose $X_{\overline{k}} = C_1 \sqcup C_2$ with C_i non-empty and closed. As above get a finite k' and closed $Z_i \subseteq X_{k'}$ with $C_i = q_{k'}^{-1}(Z_i)$. Then $Z_1 \cup Z_2 = X_{k'}$, $Z_1 \cap Z_2 = \emptyset$ (a point in the intersection would have a preimage in $C_1 \cap C_2$), and $Z_i \neq \emptyset$. So $X_{k'}$ is disconnected, a contradiction. □

Exercise 2.14

Problem. Let X be an algebraic variety over a field k. Let us suppose X is geometrically reduced (resp. integral; irreducible; connected). We want to show that for any field extension K/k, X_K is reduced (resp. integral; irreducible; connected).

(a) Show that we may assume that k is algebraically closed and K finitely generated over k.

(b) Let Y be an integral algebraic variety over k such that $K(Y) = K$. Applying Proposition 2.15 to Y, show that X_K is reduced.

(c) Let $P(S)$ be a monic polynomial in $k(T_1, \ldots, T_d)[S]$. Let $P = Q_1 Q_2$ be a decomposition of P in $K(T_1, \ldots, T_d)[S]$ in monic polynomials. Show that $Q_i \in k(T_1, \ldots, T_d)[S]$ (consider the sub-k-algebra A of K generated by the elements of K which appear in the coefficients of the Q_i, and apply a homomorphism $A \to k$ to Q_i).

(d) Let us suppose X is integral. Applying Proposition 2.15 to X, show that $K(X) \otimes_k K$ is integral. Deduce from this that X_K is integral.

(e) Show assertion (d) for irreducible algebraic varieties and for connected algebraic varieties.

Solution. *Remark on the plan.* I follow the ingredients of the outline (Prop. 2.15 and the specialization argument of (c)), but in an order that avoids reducing to K finitely generated: the specialization argument is applied directly to an arbitrary K over an algebraically closed k, so the reduction (a) only needs "k algebraically closed". Throughout, "P" (the property) stands for one of: reduced, integral, irreducible, connected. Two simple facts:

(N) If N is a nil ideal of a k-algebra B and E is any k-algebra, then $E \otimes_k N \to E \otimes_k B$ has image consisting of nilpotent elements (a sum of tensors $e \otimes n$, each nilpotent, of commuting nilpotents), so $\mathrm{Spec}(E \otimes B/N) \to \mathrm{Spec}(E \otimes B)$ is a homeomorphism. Hence $(X_{\mathrm{red}})_K \to X_K$ is a homeomorphism for every K/k (apply to an affine cover).

(S) For every extension K/k, $p_K : X_K \to X$ is surjective (Prop. 1.16, as in Exercise 2.11), and $A \otimes_k K \subseteq A \otimes_k \Omega$ for $K \subseteq \Omega$.

(a) Reduction to k algebraically closed. Let K/k be arbitrary, and let Ω be an algebraically closed extension of K. The algebraic closure of k in Ω is an algebraic closure $\overline{k}$ of k, and $X_\Omega = (X_{\overline{k}})_\Omega$. The variety $Y := X_{\overline{k}}$ over $\overline{k}$ has property P by hypothesis. Suppose we know the theorem for algebraically closed base fields; then $Y_\Omega = X_\Omega$ has P. Now $X_\Omega \to X_K$ is surjective, so X_K is irreducible, resp. connected, if X_Ω is; and $A \otimes_k K \subseteq A \otimes_k \Omega$ shows that X_K is reduced if X_Ω is. Integral means reduced and irreducible. So it suffices to treat algebraically closed k and arbitrary K/k. (The reduction to K finitely generated is not needed.) *From now on k is algebraically closed.* Then "geometrically P" just means P.

(c) Key lemma. Let $F = k(T_1, \ldots, T_d)$ with k algebraically closed, K/k any extension, and $P \in F[S]$ monic and irreducible over F. Then P is irreducible in $K(T_1, \ldots, T_d)[S]$.

Proof. Suppose $P = Q_1 Q_2$ in $K(T)[S]$ with Q_i monic of degree $n_i \geq 1$. Choose $u \in K[T] \smallsetminus \{0\}$ such that all coefficients of Q_1, Q_2 lie in $K[T][1/u]$, and $d_0 \in k[T] \smallsetminus \{0\}$ such that the coefficients of P lie in $k[T][1/d_0]$. Let $A \subseteq K$ be the sub-k-algebra generated by all coefficients (in K) of u and of the numerators of the coefficients of the Q_i (written over the denominator u). It is a finitely generated k-algebra and a domain. Let $\beta \neq 0$ be one coefficient of u and replace A by $A_\beta \subseteq K$. Choose a maximal ideal $\mathfrak{m}$ of A_β. By the Nullstellensatz $A_\beta/\mathfrak{m}$ is a finite extension of the algebraically closed field k, so it is k, giving a k-algebra homomorphism $\varphi : A_\beta \to k$. Extend it coefficientwise to $\tilde{\varphi} : A_\beta[T] \to k[T]$. It is the identity on $k[T]$, and $\tilde{\varphi}(u) \neq 0$ because β is a unit of A_β, so $\varphi(\beta) \neq 0$. Hence $\tilde{\varphi}$ extends to a ring homomorphism $R := A_\beta[T][1/(ud_0)] \to k(T)$, which is the identity on $k[T][1/d_0]$, in particular fixes the coefficients of P. All coefficients of P, Q_1, Q_2 lie in R. Applying $\tilde{\varphi}$ to $P = Q_1 Q_2$ in $R[S]$ gives $P = \tilde{\varphi}(Q_1)\tilde{\varphi}(Q_2)$ in $F[S]$ with monic factors of degrees $n_1, n_2 \geq 1$, contradicting the irreducibility of P over F. □

(In the form stated in (c): if $P = Q_1 Q_2$ with P irreducible over F, then Q_1 or Q_2 equals 1, so both lie in $F[S]$. For general monic P, factor P into irreducibles P_j over F; each P_j stays irreducible over $K(T)$, so by unique factorization the monic factors Q_i are products of some of the P_j and lie in $F[S]$.)

Domain statement. Let k be algebraically closed, L a finitely generated field extension of k (the function field of an integral variety Y over k) and K/k any extension. Then $L \otimes_k K$ is a domain.

Proof. Since Y is integral and geometrically reduced ($k = \overline{k}$), Prop. 2.15 gives $L = F[S]/(P)$ with $F = k(T_1, \ldots, T_d)$ and $P \in F[S]$ monic, irreducible, separable. Then $L \otimes_k K = (F \otimes_k K)[S]/(P)$. The ring $F \otimes_k K$ is the localization of $k[T] \otimes_k K = K[T]$ at $k[T] \smallsetminus \{0\}$, so it is a domain contained in $K(T)$. Since P is monic, division with remainder by P shows $(F \otimes K)[S]/(P) \hookrightarrow K(T)[S]/(P)$ (if $g \in (F \otimes K)[S]$ is divisible by P in $K(T)[S]$, the quotient lies in $(F \otimes K)[S]$). By the key lemma P is irreducible in $K(T)[S]$, so $K(T)[S]/(P)$ is a field, and $L \otimes_k K$ is a subring of a field. □

(b) Reduced. Let X be reduced over $k = \overline{k}$, K/k arbitrary, $\mathrm{Spec}\, A$ an affine open of X. As

A is reduced and Noetherian, with minimal primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$, we have $A \hookrightarrow \prod_i A/\mathfrak{p}_i \hookrightarrow \prod_i L_i$, $L_i = \operatorname{Frac}(A/\mathfrak{p}_i)$, the function field of the integral variety $\operatorname{Spec} A/\mathfrak{p}_i$. Tensoring with K (exact over k) gives $A \otimes_k K \hookrightarrow \prod_i L_i \otimes_k K$, and each $L_i \otimes_k K$ is a domain by the domain statement. So $A \otimes_k K$ is reduced, and X_K is reduced. (In the language of the outline: this is (b) with $Y = \operatorname{Spec} A/\mathfrak{p}_i$.)

(d) Integral. Let X be integral over $k = \overline{k}$, $L = K(X)$. For each non-empty affine open $U = \operatorname{Spec} A$ of X we have $A \otimes_k K \subseteq L \otimes_k K$ and the latter is a domain, so $A \otimes_k K$ is a domain: $U_K = p_K^{-1}(U)$ is integral (non-empty by (S)). For two non-empty affine opens U, U' of X the intersection $U \cap U'$ is non-empty, hence $p_K^{-1}(U \cap U') \neq \emptyset$ by (S). A space covered by irreducible open subsets that pairwise intersect is irreducible. So X_K is irreducible and reduced, i.e. integral. By (a), this proves the integral case for arbitrary k (note that in the general case the property used is that $X_{\overline{k}}$ is integral).

(e) Irreducible. Let $k = \overline{k}$ and X irreducible. Then X_{red} is integral over k, so $(X_{\mathrm{red}})_K$ is integral by (d), in particular irreducible. By (N) $(X_{\mathrm{red}})_K \to X_K$ is a homeomorphism, so X_K is irreducible. By (a), irreducible case done.

(e) Connected. Let $k = \overline{k}$ and X connected, with irreducible components $X_1, \ldots, X_n$ (finitely many, X being Noetherian). Each X_i with its reduced structure is irreducible, so by the previous paragraph $(X_i)_K$, a closed subscheme of X_K with underlying set $p_K^{-1}(X_i)$, is irreducible, hence connected. Moreover $X_K = \bigcup_i p_K^{-1}(X_i)$ and $p_K^{-1}(X_i) \cap p_K^{-1}(X_j) = p_K^{-1}(X_i \cap X_j)$ is non-empty iff $X_i \cap X_j \neq \emptyset$, by surjectivity (S). Because X is connected, the graph on $\{1, \ldots, n\}$ with an edge $i - j$ iff $X_i \cap X_j \neq \emptyset$ is connected (otherwise X would be the disjoint union of two non-empty closed sets, each a union of components). If $X_K = U \sqcup V$ with U, V open and closed, every connected set $p_K^{-1}(X_i)$ lies in U or in V; adjacent ones lie on the same side since they meet; thus all lie on the same side and the other set is empty. So X_K is connected. By (a), done. □

Exercise 2.15

> **Problem.** Let X be a geometrically integral algebraic variety over a field k. Show that for any integral algebraic variety Y over k, the fibered product $X \times_{\operatorname{Spec} k} Y$ is integral. Show the same assertion after subsequently replacing "integral" everywhere in hypothesis for X and Y by "reduced", "irreducible", and "connected".

Solution. Precisely, we prove four statements, with $Z = X \times_k Y$ and $p : Z \to Y$ the second projection: (i) X geometrically reduced, Y reduced $\Rightarrow$ Z reduced; (ii) X geometrically integral, Y integral $\Rightarrow$ Z integral; (iii) X geometrically irreducible, Y irreducible $\Rightarrow$ Z irreducible; (iv) X geometrically connected, Y connected $\Rightarrow$ Z connected. (Varieties are non-empty in (iii), (iv).) We use Exercise 2.14: for every extension K/k, X_K is reduced (resp. integral, irreducible, connected) if X is geometrically so.

Preliminaries. Let $U = \operatorname{Spec} A \subseteq X$, $V = \operatorname{Spec} B \subseteq Y$ be affine opens, so $U \times_k V = \operatorname{Spec}(A \otimes_k B)$; these cover Z. (1) If B is a domain, $K = \operatorname{Frac} B$, $S = B \setminus \{0\}$, then every $s \in S$ is a non-zero-divisor on $A \otimes_k B$ (multiplication by s is injective on B, and $A \otimes_k -$ preserves injectivity). Hence $A \otimes_k B \subseteq S^{-1}(A \otimes_k B) = A \otimes_k K$. (2) If N is the nilradical of Y, then as in (N) of Exercise 2.14, $X \times_k Y_{\mathrm{red}} \to X \times_k Y$ is a homeomorphism. (3) p is surjective (the fibre over y is $X_{k(y)} \neq \emptyset$, by Prop. 1.16), and so are all projections of fibre products over k of non-empty

varieties; moreover, for non-empty k-algebras A, B we have $A \otimes_k B \neq 0$, so the product of two non-empty varieties is non-empty.

(i) Let B be reduced, with minimal primes $\mathfrak{q}_i$, and $K_i = \operatorname{Frac}(B/\mathfrak{q}_i)$. Then $B \hookrightarrow \prod_i K_i$ and $A \otimes_k B \hookrightarrow \prod_i A \otimes_k K_i$. Now $A \otimes_k K_i$ is the ring of the affine open U_{K_i} of X_{K_i}, which is reduced by Exercise 2.14. So $A \otimes B$ is reduced for all affine U, V, and Z is reduced.

(ii) Let B be a domain with $K = \operatorname{Frac} B = K(Y)$. By (1), $A \otimes_k B \subseteq A \otimes_k K$, the ring of $U_K \subseteq X_K$, and X_K is integral by Exercise 2.14, so $A \otimes_k K$ is a domain (it is non-zero). Hence $U \times V$ is integral for all non-empty affine opens U, V. Two non-empty opens $U \times V$, $U' \times V'$ meet, since $(U \cap U') \times (V \cap V') \neq \emptyset$ by (3) (X, Y irreducible). A space covered by irreducible open sets which pairwise meet is irreducible; so Z is irreducible, and reduced by (i), hence integral.

(iii), (iv): the generic fibre is dense. By (2) we may replace Y by Y_{red} in (iii) (same underlying space for Z); so let Y be integral with generic point ξ, $K = k(\xi) = K(Y)$. By Prop. 1.16 and the transitivity of fibre products, $G := p^{-1}(\xi)$ is homeomorphic to $Z \times_Y \operatorname{Spec} K = X_K$. *Claim: G is dense in Z.* Let $O \subseteq Z$ be non-empty open; it contains a non-empty basic open $D(f) \subseteq U \times V = \operatorname{Spec}(A \otimes B)$ with $f \in A \otimes B$ not nilpotent. By (1), f is not nilpotent in $A \otimes K$ either, so there is a prime $\mathfrak{P}$ of $A \otimes K = S^{-1}(A \otimes B)$ with $f \notin \mathfrak{P}$. It corresponds to a prime of $A \otimes B$ missing S, i.e. whose contraction to B is 0; this is a point of $D(f) \cap G$. So G meets every non-empty open set.

(iii) Let X be geometrically irreducible and Y irreducible, w.l.o.g. integral. By Exercise 2.14, X_K is irreducible, so G is irreducible. As G is dense in Z, $Z = \overline{G}$ is irreducible.

(iv) Let X be geometrically connected and Y connected, with irreducible components $Y_1, \ldots, Y_n$ (reduced structures; Y is Noetherian). Put $Z_i = X \times_k Y_i \subseteq Z$, a closed subscheme with underlying set $p^{-1}(Y_i)$. By the density argument applied to Y_i (integral), Z_i contains the dense connected subset $G_i \simeq X_{K(Y_i)}$ (connected by Exercise 2.14), hence Z_i is connected (the closure of a connected set is connected). Further $Z_i \cap Z_j = p^{-1}(Y_i \cap Y_j)$ is non-empty iff $Y_i \cap Y_j \neq \emptyset$ (surjectivity of p). Since Y is connected, the graph on the components is connected, so, exactly as at the end of Exercise 2.14, $Z = \bigcup_i Z_i$ is connected. □

Exercise 2.16

> **Problem.** Let A be an integral domain with field of fractions K, and let B be a finitely generated A-algebra. We suppose that $B \otimes_A K$ is finite over K.
>
> (a) For any $b \in B$, show that there exists a $g \in A \smallsetminus \{0\}$ such that gb is integral over A.
> (b) Show that there exists an $h \in A \smallsetminus \{0\}$ such that B_h is finite over A_h.

Solution. Let $S = A \smallsetminus \{0\}$, so $B \otimes_A K = S^{-1}B$.

(a) $S^{-1}B$ is a finite K-algebra, so $b/1$ is integral over K: there are $c_1, \ldots, c_n \in K$ with $(b/1)^n + c_1(b/1)^{n-1} + \cdots + c_n = 0$ in $S^{-1}B$. Write $c_i = a_i/s$ with $a_i \in A$ and a common $s \in S$. Then $sb^n + a_1 b^{n-1} + \cdots + a_n = 0$ in $S^{-1}B$, so there is $t \in S$ with $t(sb^n + a_1 b^{n-1} + \cdots + a_n) = 0$ in B. Put $g = ts \in S$. Then

$$g\,b^n + ta_1 b^{n-1} + \cdots + ta_n = 0 \quad \text{in } B.$$

Multiplying by g^{n-1} gives

$$(gb)^n + ta_1(gb)^{n-1} + ta_2 g\,(gb)^{n-2} + \cdots + ta_n g^{n-1} = 0,$$

a monic equation for gb with coefficients $ta_i g^{i-1} \in A$. So gb is integral over A.

(b) Let $b_1, \dots, b_m$ generate B as an A-algebra. By (a) choose $g_i \in S$ with $g_i b_i$ integral over A, and put $h = g_1 \cdots g_m \in S$. In B_h the element $b_i/1 = (g_i b_i)/g_i$ is integral over A_h because $g_i b_i$ is integral over A and $1/g_i \in A_h$. Now B_h is generated as an A_h-algebra by $b_1/1, \dots, b_m/1$, finitely many integral elements, so it is a finite A_h-module. □

Exercise 2.17

> **Problem.** Let Y be an irreducible scheme with generic point ξ, and let $f : X \to Y$ be a dominant morphism of finite type.
>
> (a) Let x be a closed point of X_ξ. Let Z be the reduced subscheme $\overline{\{x\}}$ of X. Show that there exists a dense open subset V of Y such that $f^{-1}(V) \cap Z \to V$ is surjective (use Exercises 2.16 and 2.1.8).
>
> (b) Deduce from this that $f(X)$ contains a dense open subset of Y.
>
> (c) Show that $f(X)$ is a constructible subset of Y, that is $f(X)$ is a finite disjoint union of sets Z_i, with each Z_i the intersection of an open subset and a closed subset.

Solution. *Remark on hypotheses.* Parts (a), (b) hold as stated for any irreducible scheme Y. For (c) I assume that Y is Noetherian (or at least locally Noetherian and quasi-compact), the setting of the cited Hartshorne exercise; the proof is a Noetherian induction and I do not claim (c) for non-Noetherian Y. In (c) I also drop the assumptions "Y irreducible" and "f dominant" (they cannot be kept in a Noetherian induction): I prove that for every finite type morphism $f : X \to Y$ with Y Noetherian, $f(X)$ is constructible.

Preliminary: $\xi \in f(X)$, i.e. $X_\xi \neq \emptyset$. Let $V_0 = \operatorname{Spec} R$ be a non-empty affine open of Y (it contains ξ). Then $f^{-1}(V_0) = \bigcup_{i=1}^N \operatorname{Spec} C_i$ is a finite union of affine opens (f is quasi-compact). The set $f(X) \cap V_0 = f(f^{-1}V_0)$ is dense in V_0, being the trace on the open V_0 of the dense set $f(X)$. Since $\operatorname{Spec} R$ is irreducible and $f(f^{-1}V_0)$ is the finite union of the images of $\operatorname{Spec} C_i$, one of these images, say that of $\operatorname{Spec} C = \operatorname{Spec} C_1$, is dense. Then $\ker(R \to C)$ is contained in every prime of R, hence in the nilradical $\mathfrak{p}_0$ of R, which is the prime corresponding to ξ. Let $T = R \setminus \mathfrak{p}_0$. The ring $T^{-1}C$ is non-zero (otherwise $t \cdot 1 = 0$ in C for some $t \in T$, so $t \in \ker \subseteq \mathfrak{p}_0$). The ideal $\mathfrak{p}_0 T^{-1}C$ consists of nilpotents, so it is proper, and $T^{-1}C/\mathfrak{p}_0 T^{-1}C \neq 0$ has a prime ideal; it gives a prime of C whose contraction to R is $\mathfrak{p}_0$. So $\xi \in f(X)$ and $X_\xi \simeq f^{-1}(\xi) \neq \emptyset$ (Prop. 1.16). Moreover X_ξ is of finite type over $\kappa = k(\xi)$ (Prop. 2.4(c)), so it has closed points, and for a closed point x (closed in an affine open $\operatorname{Spec} C'$ of X_ξ, i.e. a maximal ideal of a finitely generated κ-algebra) $k(x)$ is a finite extension of κ (Nullstellensatz); also $k(x)$ is the residue field of x as a point of X.

(a) Keep $V_0 = \operatorname{Spec} R$ and $\mathfrak{p}_0$ as above; $\bar{R} = R/\mathfrak{p}_0$ is a domain with $\operatorname{Frac} \bar{R} = \kappa$. The point x lies in $f^{-1}(V_0)$; choose an affine open $\operatorname{Spec} C \subseteq f^{-1}(V_0)$ containing x. C is a finitely generated R-algebra (finite type). Let $\mathfrak{q} \subset C$ be the prime of x. The closure of x in $\operatorname{Spec} C$ is $V(\mathfrak{q})$, so $Z \cap \operatorname{Spec} C = \operatorname{Spec} D$ with $D = C/\mathfrak{q}$ a domain with $\operatorname{Frac} D = k(x)$. Since $f(x) = \xi$, the kernel of $R \to D$ is $\mathfrak{q} \cap R = \mathfrak{p}_0$; so $\bar{R} \hookrightarrow D$ and D is a finitely generated $\bar{R}$-algebra. Then $D \otimes_{\bar{R}} \kappa$ is the localization of D at $\bar{R} \setminus \{0\}$, a κ-subalgebra of $k(x)$, which is finite-dimensional over κ. So $D \otimes_{\bar{R}} \kappa$ is finite over κ, and Exercise 2.16(b) (with $A = \bar{R}$, $B = D$) gives $h \in \bar{R} \setminus \{0\}$ with D_h finite, i.e. integral, over $\bar{R}_h$. The map $\bar{R}_h \to D_h$ is injective (D is a domain containing $\bar{R}$). By the lying-over theorem for integral extensions (Exercise 2.1.8), $\operatorname{Spec} D_h \to \operatorname{Spec} \bar{R}_h$ is surjective.

Lift h to $\tilde{h} \in R \smallsetminus \mathfrak{p}_0$ and let $V = D(\tilde{h}) \subseteq V_0$. It is a non-empty open subset of the irreducible scheme Y (it contains ξ), hence dense. Under the homeomorphism $\operatorname{Spec} \bar{R} \to \operatorname{Spec} R$ we have $\operatorname{Spec} \bar{R}_h = V$ as sets. Let $y \in V$. By surjectivity there is a prime of D_h over y, i.e. a point w of $\operatorname{Spec} D \subseteq Z$ with $f(w) = y \in V$. Thus $w \in f^{-1}(V) \cap Z$ and $f^{-1}(V) \cap Z \to V$ is surjective.

(b) By the preliminary, X_ξ has a closed point x; with Z and V as in (a), $V = f(f^{-1}(V) \cap Z) \subseteq f(X)$. So $f(X)$ contains the dense open set V.

(c) Let Y be Noetherian. We prove by Noetherian induction on the closed subsets $T \subseteq Y$ the statement $\mathcal{S}(T)$: *for every finite type morphism $f : X \to Y$ with $f(X) \subseteq T$, $f(X)$ is constructible.* (Finite unions of locally closed sets are the same as finite disjoint unions of locally closed sets, since complements and finite intersections of such sets are again finite unions of locally closed sets.) Suppose $\mathcal{S}(T')$ holds for all closed $T' \subsetneq T$ (vacuous if $T = \emptyset$, where the statement is trivial).

Reduction to T irreducible. Let $T = T_1 \cup \cdots \cup T_r$ be the irreducible components. If $r \geq 2$, put $X_i = f^{-1}(T_i)$ with its reduced structure, $f_i : X_i \to Y$ the restriction. $X_i \to X$ is a closed immersion, hence of finite type (Prop. 2.4(a)), so f_i is of finite type (Prop. 2.4(b)), $f_i(X_i) \subseteq T_i \subsetneq T$, and $f(X) = \bigcup f_i(X_i)$ is constructible by the induction hypothesis.

The irreducible case. Let T be irreducible, with the reduced structure; it is an integral Noetherian scheme and $T \to Y$ is a closed immersion. Let $X' = f^{-1}(T)_{\mathrm{red}}$. Then X' is reduced, so $X' \to Y$ factors through T (universal property of the reduced closed subscheme); X' is of finite type over Y (closed immersion followed by f), hence Noetherian (Exercise 2.1), hence $X' \to T$ is quasi-compact, and by Prop. 2.4(e) $f' : X' \to T$ is of finite type. Also $f'(X') = f(X)$. Let $T' = \overline{f(X)} \subseteq T$. If $T' \neq T$, then $f(X) \subseteq T' \subsetneq T$ and $\mathcal{S}(T')$ gives constructibility. If $T' = T$, then f' is dominant, and by (b) there is a dense open $V \subseteq T$ contained in $f(X)$. Let $T'' = T \smallsetminus V \subsetneq T$ (closed) and $X'' = f^{-1}(T'')_{\mathrm{red}}$, $f'' : X'' \to Y$ of finite type as above with $f''(X'') = f(X) \cap T'' \subseteq T''$. Then

$$f(X) = V \ \cup \ (f(X) \cap T''),$$

where V is open in the closed subset T of Y, hence locally closed in Y, and $f(X) \cap T'' = f''(X'')$ is constructible by $\mathcal{S}(T'')$. So $f(X)$ is constructible. This proves $\mathcal{S}(T)$ for all closed T, and in particular, for $T = Y$, the claim (and the case of the exercise). □

Exercise 2.18

Problem. Give an analogous version of Corollary 2.26(b) for projective varieties.

Solution. The exercise is open-ended; here is the statement I prove. Let $k = \mathbf{F}_p$, $\overline{k}$ an algebraic closure, and $X = V_+(I)$ a closed subvariety of $\mathbf{P}^n_k = \operatorname{Proj} k[T_0, \ldots, T_n]$, I a homogeneous ideal. By Prop. 2.18(e), $X(\overline{k}) = \{[t_0 : \cdots : t_n] \in \mathbf{P}^n(\overline{k}) \mid P(t) = 0 \ \forall P \in I\}$. Let $\overline{F}_X : X(\overline{k}) \to X(\overline{k})$, $s \mapsto F_X \circ s$.

Claim. (b′) $\overline{F}_X([t_0 : \cdots : t_n]) = [t_0^p : \cdots : t_n^p]$. (c′) $X(k)$ is identified with the fixed points of $\overline{F}_X$; explicitly, $[t]$ is fixed iff $[t] \in \mathbf{P}^n(\mathbf{F}_p)$.

Proof of (b′). Let $s : \operatorname{Spec} \overline{k} \to \mathbf{P}^n_k$ correspond to $[t]$. Choose i with $t_i \neq 0$; then s factors through $D_+(T_i) = \operatorname{Spec} k[T_0/T_i, \ldots, T_n/T_i]$ and corresponds to the homomorphism $T_j/T_i \mapsto t_j/t_i$ (this is

the identification of Prop. 2.18(e), Example 2.3.32). By Lemma 2.22(a) applied to the morphism s, $F_{\mathbf{P}^n} \circ s = s \circ F_{\operatorname{Spec} \overline{k}}$, and $F_{\operatorname{Spec} \overline{k}}$ corresponds to the Frobenius $\varphi : a \mapsto a^p$ of $\overline{k}$. So $F_{\mathbf{P}^n} \circ s$ corresponds to $T_j/T_i \mapsto \varphi(t_j/t_i) = t_j^p/t_i^p$, i.e. to the point $[t_0^p : \cdots : t_n^p]$ (note $t_i^p \neq 0$). For the closed immersion $\iota : X \to \mathbf{P}^n_k$, Lemma 2.22(a) gives $\iota \circ F_X = F_{\mathbf{P}^n} \circ \iota$, so $\iota \circ \overline{F}_X(s) = F_{\mathbf{P}^n} \circ (\iota \circ s)$. Since $s \mapsto \iota \circ s$ is the identification $X(\overline{k}) \subseteq \mathbf{P}^n(\overline{k})$ (injective, as ι is a monomorphism), (b′) follows. As a check, $[t^p]$ is again in $X(\overline{k})$: every $P \in I$ has coefficients in $\mathbf{F}_p$, so $P(t^p) = P(t)^p = 0$.

Proof of (c′). $[t]$ is fixed iff $[t^p] = [t]$, i.e. there is $\lambda \in \overline{k}^*$ with $t_j^p = \lambda t_j$ for all j. Take i with $t_i \neq 0$ and rescale t so that $t_i = 1$ (replacing t by ct replaces λ by $c^{p-1}\lambda$); then $\lambda = 1$, i.e. $t_j^p = t_j$ for all j, i.e. all $t_j \in \mathbf{F}_p$. Conversely such a point is fixed. So the fixed points in $X(\overline{k})$ are those $[t] \in X(\overline{k}) \cap \mathbf{P}^n(\mathbf{F}_p)$, which by Prop. 2.18(e) (with $K = k = \mathbf{F}_p$) and 2.18(c) is exactly $X(k) \subseteq X(\overline{k})$. □

Exercise 2.19

> **Problem.** Show that assertions (b) and (c) of Corollary 2.26 remain true after replacing k by the field of $q = p^f$ elements and F_X by F_X^f.

Solution. Let $k = \mathbf{F}_q$, $q = p^f$, X an algebraic variety over k (in particular an $\mathbf{F}_p$-scheme), $\overline{k}$ an algebraic closure. First, F_X^f is a morphism of k-schemes: for the structural morphism $\pi : X \to \operatorname{Spec} k$ we have $\pi \circ F_X^f = F^f_{\operatorname{Spec} k} \circ \pi$ (Lemma 2.22(a)) and $F^f_{\operatorname{Spec} k}$ corresponds to $a \mapsto a^q$, which is the identity of $\mathbf{F}_q$. So $s \mapsto F_X^f \circ s$ defines a map $\overline{F} : X(\overline{k}) \to X(\overline{k})$. Let $\varphi : \overline{k} \to \overline{k}$, $a \mapsto a^p$.

For $s \in X(\overline{k})$, Lemma 2.22(a) (applied to the morphism s of $\mathbf{F}_p$-schemes) gives

$$F_X^f \circ s = s \circ F^f_{\operatorname{Spec} \overline{k}},$$

and $F^f_{\operatorname{Spec} \overline{k}}$ corresponds to $\varphi^f : a \mapsto a^q$.

(b) Let $X = V(I) \subseteq \operatorname{Spec} k[T_1, \ldots, T_n]$, so $X = \operatorname{Spec} k[T]/I$ is affine and s corresponds to a k-homomorphism $s^\# : k[T]/I \to \overline{k}$, with $t_i = s^\#(T_i)$. The morphism $s \circ F^f_{\operatorname{Spec} \overline{k}}$ corresponds to $\varphi^f \circ s^\#$, which sends T_i to t_i^q. Hence

$$\overline{F}(t_1, \ldots, t_n) = (t_1^q, \ldots, t_n^q)$$

under the identification of Prop. 2.18(d).

(c) (For arbitrary X, not only affine.) By Prop. 2.18(b), s corresponds to a pair (x, θ) with $x \in X$ and $\theta : k(x) \to \overline{k}$ a k-homomorphism, x being the image point of s and θ induced by $s_x^\# : \mathcal{O}_{X,x} \to \overline{k}$. The point $\overline{F}(s) = s \circ F^f_{\operatorname{Spec} \overline{k}}$ has the same image point x, and its map $\mathcal{O}_{X,x} \to \overline{k}$ is $\varphi^f \circ s_x^\#$, so it corresponds to $(x, \varphi^f \circ \theta)$. Therefore s is fixed by $\overline{F}$ iff $\theta(a)^q = \theta(a)$ for all $a \in k(x)$, iff $\theta(k(x)) \subseteq \mathbf{F}_q = k$ (the roots of $T^q - T$ in $\overline{k}$ form the subfield $\mathbf{F}_q$). In that case θ is an injective k-linear map from $k(x)$ into the one-dimensional space k, so $k(x) = k$ and $\theta = \mathrm{id}$; conversely $k(x) = k$ gives $\theta = \mathrm{id}$ (a k-homomorphism $k \to \overline{k}$). By Prop. 2.18(b),(c), the pairs (x, θ) with $k(x) = k$ are exactly the image of $X(k) \hookrightarrow X(\overline{k})$. So the fixed points of $\overline{F}$ are exactly $X(k)$. (The same argument with $f = 1$ proves Cor. 2.26(c) for all varieties over $\mathbf{F}_p$.) □

Exercise 2.20

Problem. Let S be a scheme over $\mathbf{F}_p$. Let X be an S-scheme.

(a) Show that the relative Frobenius $F_{X/S} : X \to X^{(p)}$ commutes with base change $T \to S$ (i.e. $F_{X/S} \times \mathrm{Id}_T = F_{X_T/T}$).

(b) Let $f : X \to Y$ be a morphism of S-schemes. Show that f canonically induces a morphism of S-schemes $f^{(p)} : X^{(p)} \to Y^{(p)}$ such that $f^{(p)} \circ F_{X/S} = F_{Y/S} \circ f$. Note that in the special case when $Y = S$, $f^{(p)}$ is just the structural morphism of $X^{(p)}$. Let us suppose that f satisfies a property (P) which is stable by base change (e.g., closed immersion). Show that $f^{(p)}$ also satisfies (P).

(c) Let us suppose that S is the spectrum of a field. Show that the morphism $\phi : X^{(p)} \to X$ induces an injective homomorphism $\mathcal{O}_X \to \phi_* \mathcal{O}_{X^{(p)}}$. Deduce from this that if $X^{(p)}$ is integral, then so is X. Show that the converse is false in general.

Solution. Notation of the text: X has structural morphism π_X; $X^{(p)} = X \times_S S$ where the second factor is the S-scheme $F_S : S \to S$; $\phi_X : X^{(p)} \to X$ and $q_X : X^{(p)} \to S$ are the projections; $F_{X/S}$ is the unique S-morphism with $\phi_X \circ F_{X/S} = F_X$ and $q_X \circ F_{X/S} = \pi_X$. All schemes below are $\mathbf{F}_p$-schemes (e.g. T is, via $T \to S$), so Lemma 2.22 applies. We argue with Z-valued points for an arbitrary scheme Z (Yoneda).

(a) Let $g : T \to S$ be the given morphism and $X_T = X \times_S T$ with projections $a : X_T \to X$, $b : X_T \to T$. We have two schemes:

$$(X_T)^{(p)}(Z) = \{(u, t') \mid u : Z \to X_T,\ t' : Z \to T,\ b \circ u = F_T \circ t'\},$$
$$(X^{(p)})_T(Z) = \{(w, t) \mid w : Z \to X^{(p)},\ t : Z \to T,\ q_X \circ w = g \circ t\}.$$

Write $u = (x, t_1)$ with $x : Z \to X$, $t_1 : Z \to T$, $\pi_X x = g t_1$; then the condition $bu = F_T t'$ says $t_1 = F_T t'$, so $(X_T)^{(p)}(Z) = \{(x, t') \mid \pi_X \circ x = g \circ F_T \circ t' = F_S \circ g \circ t'\}$ (using $g F_T = F_S g$, Lemma 2.22(a)). Write $w = (x, s)$ with $x : Z \to X$, $s : Z \to S$, $\pi_X x = F_S s$; then $q_X w = s$, so $(X^{(p)})_T(Z) = \{(x, t) \mid \pi_X \circ x = F_S \circ g \circ t\}$. These are the same functor of Z, giving a canonical isomorphism $\theta : (X_T)^{(p)} \simeq (X^{(p)})_T$, under which $(u, t') = ((x, F_T t'), t')$ corresponds to (x, t'). This is the identification with respect to which the statement is meant.

Now take a Z-point (x, t) of X_T (so $\pi_X x = g t$). On one hand $(F_{X/S} \times \mathrm{Id}_T)(x, t) = (F_{X/S} \circ x, t)$, which as a point of $(X^{(p)})_T$ corresponds to the pair $(\phi_X \circ F_{X/S} \circ x, t) = (F_X \circ x, t)$. On the other hand, $F_{X_T/T}$ is characterized by $\phi_{X_T} \circ F_{X_T/T} = F_{X_T}$ and $q_{X_T} \circ F_{X_T/T} = b$. So $F_{X_T/T}(x, t) = (u', t')$ with $u' = F_{X_T} \circ (x, t)$ and $t' = b(x, t) = t$. Since $a \circ F_{X_T} = F_X \circ a$ (Lemma 2.22(a)), $u' = (F_X \circ x, F_T \circ t)$, and under θ the point (u', t) corresponds to the pair $(F_X \circ x, t)$. The two points agree, so $F_{X/S} \times \mathrm{Id}_T = F_{X_T/T}$.

(b) Define $f^{(p)} = f \times \mathrm{Id}_S : X \times_{S, F_S} S \to Y \times_{S, F_S} S$ (this makes sense since $\pi_Y f = \pi_X$); then $q_Y \circ f^{(p)} = q_X$, so $f^{(p)}$ is an S-morphism, and $\phi_Y \circ f^{(p)} = f \circ \phi_X$. To prove $f^{(p)} \circ F_{X/S} = F_{Y/S} \circ f$ we compare projections to $Y^{(p)} = Y \times_{S, F_S} S$:

$$\phi_Y f^{(p)} F_{X/S} = f \phi_X F_{X/S} = f F_X, \qquad \phi_Y F_{Y/S} f = F_Y f = f F_X,$$
$$q_Y f^{(p)} F_{X/S} = q_X F_{X/S} = \pi_X, \qquad q_Y F_{Y/S} f = \pi_Y f = \pi_X,$$

using Lemma 2.22(a). By the uniqueness in the universal property of the fibered product, they are equal. If $Y = S$ (with Id_S), then $Y^{(p)} = S \times_{S,F_S} S \simeq S$ and $f^{(p)} = \pi_X \times \mathrm{Id}_S : X^{(p)} \to S$ is q_X, the structural morphism of $X^{(p)}$.

Now let (P) be stable under base change and satisfied by f. We have canonical isomorphisms

$$X \times_Y Y^{(p)} = X \times_Y (Y \times_{S,F_S} S) \simeq X \times_{S,F_S} S = X^{(p)}$$

(transitivity of fibered products, using $\pi_Y f = \pi_X$), under which the second projection $X \times_Y Y^{(p)} \to Y^{(p)}$ becomes $f \times \mathrm{Id}_S = f^{(p)}$. Thus $f^{(p)}$ is the base change of f by $\phi_Y : Y^{(p)} \to Y$, and has (P).

(c) Now $S = \operatorname{Spec} k$, $\operatorname{char} k = p$. Then $X^{(p)} = X \times_k k'$, where $k' = \operatorname{Spec} k$ is a k-scheme via F_S, i.e. $k \to k$, $a \mapsto a^p$, and $\phi = \phi_X$ is the base change of $\operatorname{Spec} k' \to \operatorname{Spec} k$ (writing k' for k with the k-algebra structure $a \mapsto a^p$). For an affine open $U = \operatorname{Spec} A$ of X, $\phi^{-1}(U) = \operatorname{Spec}(A \otimes_k k')$ and $\phi^{\#} : A \to A \otimes_k k'$ is $a \mapsto a \otimes 1$. As k' is a non-zero k-vector space, $1 \in k'$ is part of a k-basis, so $A \to A \otimes_k k'$ is injective (indeed split). A section of $\mathcal{O}_X$ over an arbitrary open set which maps to 0 vanishes on every affine open subset, so it is 0: $\mathcal{O}_X \to \phi_* \mathcal{O}_{X^{(p)}}$ is injective.

Suppose $X^{(p)}$ integral. By Lemma 2.24, ϕ is a homeomorphism, so $X \neq \emptyset$ is irreducible. For U open in X, $\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X^{(p)}}(\phi^{-1}U)$, and the latter is a domain when $U \neq \emptyset$; so every $\mathcal{O}_X(U)$ is a domain, and X is integral.

The converse is false. Let p be a prime, $k = \mathbf{F}_p(t)$ and $K = k[x]/(x^p - t)$, a field since $x^p - t$ is Eisenstein at the prime t of $\mathbf{F}_p[t]$, hence irreducible over $\mathbf{F}_p(t)$ by Gauss' lemma. Then $X = \operatorname{Spec} K$ is an integral variety over k. Its Frobenius twist is $X^{(p)} = \operatorname{Spec}(K \otimes_k k')$ where $k \to k' = k$ is $a \mapsto a^p$. Now

$$K \otimes_k k' = k'[x]/(x^p - t^p) = k[x]/((x - t)^p),$$

because the image of t in k' is t^p, and $x^p - t^p = (x - t)^p$ in characteristic p. This ring has the non-zero nilpotent element $x - t$, so $X^{(p)}$ is not reduced, hence not integral. (This is consistent with Cor. 2.27: X is not geometrically reduced.) □

Exercise 2.21

> **Problem.** Let B be a finitely generated algebra over a ring A. Show that there exist a subring A_0 of A that is finitely generated over $\mathbf{Z}$ and a finitely generated A_0-algebra B_0 such that $B = B_0 \otimes_{A_0} A$. If B is homogeneous, we can take B_0 homogeneous.

Solution. *The statement is false as printed unless B is of finite presentation over A (automatic if A is Noetherian, which is the situation of the Remark following the exercise).* Here is a counterexample. Let $A = \mathbf{F}_p[x_1, x_2, \dots]$ (infinitely many variables), $J = (x_1, x_2, \dots)$ and $B = A/J \simeq \mathbf{F}_p$, which is generated by 1 over A. Suppose $A_0 \subseteq A$ is finitely generated over $\mathbf{Z}$ and B_0 is a finitely generated A_0-algebra with $B_0 \otimes_{A_0} A \simeq B$ as A-algebras. Then A_0 is Noetherian (Hilbert basis theorem), $B_0 = A_0[T_1, \dots, T_n]/I_0$ with $I_0 = (g_1, \dots, g_m)$ finitely generated, and $B \simeq A[T_1, \dots, T_n]/(g_1, \dots, g_m)$ is a finitely presented A-algebra. But then J would be finitely generated: choose $a_i \in A$ with $a_i \mapsto$ the class of T_i in B (possible as $A \to B$ is onto) and let $\mathrm{ev} : A[T] \to A$ be the substitution $T_i \mapsto a_i$, which is onto. The composite $A[T] \xrightarrow{\mathrm{ev}} A \to B$ agrees

with the quotient map $A[T] \to B$ (both are A-algebra maps with the same values on the T_i), so $\mathrm{ev}^{-1}(J) = (g_1, \ldots, g_m)$ and $J = \mathrm{ev}((g_1, \ldots, g_m)) = (g_1(a), \ldots, g_m(a))$ is finitely generated, which is false. So no such A_0, B_0 exist.

Correct statement and proof. Let B be a finitely presented A-algebra, i.e. $B = A[T_1, \ldots, T_n]/(g_1, \ldots, g_m)$ (this holds whenever A is Noetherian, since then every ideal of $A[T_1, \ldots, T_n]$ is finitely generated). Let A_0 be the subring of A generated by the (finitely many) coefficients of $g_1, \ldots, g_m$; it is finitely generated over $\mathbf{Z}$. Then $g_j \in A_0[T]$; put $B_0 = A_0[T_1, \ldots, T_n]/(g_1, \ldots, g_m)$, a finitely generated A_0-algebra. By right exactness of $- \otimes_{A_0} A$,

$$B_0 \otimes_{A_0} A = A[T_1, \ldots, T_n]/(g_1, \ldots, g_m)A[T_1, \ldots, T_n] = B.$$

Homogeneous case. Here "B homogeneous" means $B = A[T_0, \ldots, T_n]/I$ with I a homogeneous ideal, graded by degree. If I is finitely generated (finite presentation, e.g. A Noetherian), then it is generated by finitely many homogeneous elements: the homogeneous components of finitely many generators lie in I and generate I. Choose $g_1, \ldots, g_m$ homogeneous generators and define A_0, $B_0 = A_0[T_0, \ldots, T_n]/(g_1, \ldots, g_m)$ as above. The g_j are homogeneous with coefficients in A_0, so B_0 is a homogeneous A_0-algebra (graded quotient), and the isomorphism $B_0 \otimes_{A_0} A \simeq B$ is an isomorphism of graded A-algebras (the grading on $B_0 \otimes_{A_0} A$ being the one induced by $(B_0)_d \otimes A$, as in the text). □

3.3 Some global properties of morphisms

Conventions. A closed immersion is a morphism $f : X \to Y$ which is a homeomorphism onto a closed subset of Y and such that $f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective. "Algebraic variety over k" means a scheme of finite type over k; dimension and irreducible components only depend on the underlying topological space, so in Ex. 3.4 closed subvarieties are given their reduced structure. Honest notes. In Ex. 3.5 one needs $X \neq \varnothing$. The criterion in Ex. 3.13(b) is false as printed when S is not separated (a counterexample is given) and is proved with the extra condition that y lies over the same point of S as x, which is automatic if S is separated. In Ex. 3.14 the hypotheses "birational" and "S Noetherian" are not needed.

Exercise 3.1

> **Problem.** Let $f : X \to Y$ be a morphism of schemes. We suppose that there exist open subsets Y_i of Y such that $X = \bigcup_i f^{-1}(Y_i)$, and that the restrictions $f : f^{-1}(Y_i) \to Y_i$ are closed immersions. Show that if $f(X)$ is closed in Y, then f is a closed immersion.

Solution. Put $Y' = \bigcup_i Y_i$; then $f(X) \subseteq Y'$.

f is injective. If $f(x) = f(x') = y$, then $y \in Y_i$ for some i, so $x, x' \in f^{-1}(Y_i)$, where f is injective; thus $x = x'$.

f is a closed map. Let $F \subseteq X$ be closed. For each i, $F \cap f^{-1}(Y_i)$ is closed in $f^{-1}(Y_i)$, and $f : f^{-1}(Y_i) \to Y_i$ is a closed immersion, hence a closed map; so $f(F) \cap Y_i = f(F \cap f^{-1}(Y_i))$ is closed in Y_i. Since the Y_i form an open cover of Y', $f(F)$ is closed in Y'. Let C be the closure of $f(F)$ in Y. Since $f(X)$ is closed in Y, $C \subseteq f(X) \subseteq Y'$, so C is also the closure of $f(F)$ in Y', which is $f(F)$. Thus $f(F)$ is closed in Y.

An injective continuous closed map is a homeomorphism onto its image, and $f(X)$ is closed by hypothesis.

$f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$ *is surjective.* It suffices to check this on stalks at every $y \in Y$. If $y \notin f(X)$, then $Y \setminus f(X)$ is an open neighbourhood of y with $f^{-1}(Y \setminus f(X)) = \varnothing$, so $(f_*\mathcal{O}_X)_y = 0$ and there is nothing to prove. If $y \in f(X)$, then $y \in Y_i$ for some i, and on Y_i the sheaf $f_*\mathcal{O}_X$ is the sheaf $(f|_{f^{-1}(Y_i)})_*\mathcal{O}_{f^{-1}(Y_i)}$, for which $\mathcal{O}_{Y_i} \to (f|)_*\mathcal{O}_{f^{-1}(Y_i)}$ is surjective by hypothesis. Hence f is a closed immersion. □

Exercise 3.2

> **Problem.** Let X be a scheme. Show that the following properties are equivalent: (i) X is separated; (ii) X is separated over an affine scheme; (iii) for any scheme Y, every morphism of schemes $X \to Y$ is separated.

Solution. (i)⇒(iii). Let $u : X \to Y$ be any morphism. The structure morphism $X \to \operatorname{Spec} \mathbf{Z}$ equals $X \xrightarrow{u} Y \to \operatorname{Spec} \mathbf{Z}$ and is separated by (i). By Prop. 3.9(e) (if $g \circ u$ is separated, then u is separated; no hypothesis on g), u is separated.

(iii)⇒(ii). Take $Y = \operatorname{Spec} \mathbf{Z}$, which is affine.

(ii)⇒(i). Let $X \to \operatorname{Spec} A$ be separated. The morphism $\operatorname{Spec} A \to \operatorname{Spec} \mathbf{Z}$ is separated (Prop. 3.4), so the composite $X \to \operatorname{Spec} \mathbf{Z}$ is separated by Prop. 3.9(b). That is, X is separated. □

Exercise 3.3

> **Problem.** Let $f : X \to Y$ be a surjective closed morphism between two Noetherian schemes (or, more generally, topological spaces). Show that we have the inequality $\dim X \geq \dim Y$ (reduce to the case when X is irreducible and use induction on $\dim X$).

Solution. We only use that f is a continuous, surjective, closed map of topological spaces with X Noetherian. Dimension is the supremum of the lengths n of chains $Z_0 \subsetneq \cdots \subsetneq Z_n$ of irreducible closed subsets. We prove by induction on $n \geq 0$:

Claim $P(n)$. If $f : X \to Y$ is a continuous, closed, surjective map with X Noetherian and Y contains a chain $Y_0 \subsetneq \cdots \subsetneq Y_n$ of irreducible closed subsets, then X contains such a chain of length n.

Applying $P(n)$ to chains of Y of every length gives $\dim X \geq \dim Y$ (also when $\dim Y = \infty$).

$P(0)$: $Y_0 \neq \varnothing$ and f is surjective, so $X \neq \varnothing$ and contains an irreducible closed subset (a point closure).

$P(n-1) \Rightarrow P(n)$. Let $Z = f^{-1}(Y_n)$, a closed subset of X, hence Noetherian, with finitely many irreducible components $Z_1, \ldots, Z_m$. The restriction $Z \to Y_n$ is surjective and closed (a closed subset of Z is closed in X, its image is closed in Y and contained in Y_n). Hence $Y_n = \bigcup_j f(Z_j)$ is a finite union of closed subsets; as Y_n is irreducible, $Y_n = f(Z_j)$ for some j. Put $W = Z_j \cap f^{-1}(Y_{n-1})$. It is closed in X, hence Noetherian, and $f(W) = Y_{n-1}$ because

$Y_{n-1} \subseteq Y_n = f(Z_j)$. So $f|_W : W \to Y_{n-1}$ is surjective and closed, and $Y_0 \subsetneq \cdots \subsetneq Y_{n-1}$ is a chain in Y_{n-1}. By $P(n-1)$, W contains a chain $T_0 \subsetneq \cdots \subsetneq T_{n-1}$ of irreducible closed subsets (closed in W, hence in X). Since $f(W) = Y_{n-1} \neq Y_n = f(Z_j)$, we have $W \subsetneq Z_j$, so $T_0 \subsetneq \cdots \subsetneq T_{n-1} \subsetneq Z_j$ is a chain of length n in X (Z_j is irreducible and closed). □

Exercise 3.4

Problem. Let k be a field. Let $Z = \mathbb{A}^n_k$ or $\mathbb{P}^n_k$. Let us consider two closed subvarieties X, Y of Z, pure of respective dimensions q, r.

(a) Show that the irreducible components of $X \times_k Y$ are of dimension $q + r$.

(b) Show that $\dim(X \cap Y) \geq q + r - n$ if $X \cap Y \neq \varnothing$ (identify $X \cap Y$ with the subset $\Delta(Z) \cap (X \times_k Y)$ of $Z \times_k Z$ and use Corollary 2.5.26).

(c) If $Z = \mathbb{P}^n_k$ and $q + r \geq n$, show that $X \cap Y \neq \varnothing$ (if $X = V_+(I)$, $Y = V_+(J)$, consider $V(I) \cap V(J)$ in $\mathbb{A}^{n+1}_k$ and use Corollary 2.5.21).

(d) Show that every projective plane curve X (i.e. X is a closed subvariety of $\mathbb{P}^2_k$, pure of dimension 1) is connected.

Solution. Varieties are nonempty. We use Prop. 2.5.19: for an integral variety, every nonempty open subset has the same dimension, equal to the transcendence degree of the function field.

(a) Let $X_1, \ldots, X_a$ and $Y_1, \ldots, Y_b$ be the irreducible components of X, Y (reduced structures). As sets $X \times_k Y = \bigcup_{i,j} X_i \times_k Y_j$ (the $X_i \times_k Y_j \to X \times_k Y$ are closed immersions), so every irreducible component of $X \times_k Y$ is an irreducible component of some $X_i \times_k Y_j$. So it suffices to treat X, Y integral of dimensions q, r. Let C be an irreducible component of $X \times_k Y$. It meets an affine open subset $\operatorname{Spec}(A \otimes_k B)$, with $\operatorname{Spec} A \subseteq X$, $\operatorname{Spec} B \subseteq Y$ nonempty affine opens (A, B domains of finitely generated type over k, of dimensions q, r by Prop. 2.5.19). Then $C \cap \operatorname{Spec}(A \otimes B) = V(P)$ for a minimal prime P of $R := A \otimes_k B$, and $\dim C = \dim R/P = \operatorname{trdeg}_k \operatorname{Frac}(R/P)$ (Prop. 2.5.19 applied to C_{red} and its open subset). We show $\operatorname{trdeg}_k \operatorname{Frac}(R/P) = q + r$.

R is a free B-module (as A is a free k-module), so $B \to R$ is flat and injective. Flat extensions satisfy going-down, so $P \cap B$ is a minimal prime of B, i.e. $P \cap B = 0$. Let $L = \operatorname{Frac}(B)$ and invert $1 \otimes (B \setminus 0)$: then R is replaced by $R' = A \otimes_k L$ and P by a minimal prime P' of R' with $\operatorname{Frac}(R/P) = \operatorname{Frac}(R'/P')$. Choose a transcendence basis $y_1, \ldots, y_r$ of L over k and put $F = k(y_1, \ldots, y_r)$; L is finite over F. The ring $A_1 := A \otimes_k F$ is a localization of the polynomial ring $A[y_1, \ldots, y_r]$, hence a domain with fraction field $\operatorname{Frac}(A)(y_1, \ldots, y_r)$ of transcendence degree $q + r$ over k. Now $R' = A_1 \otimes_F L$ is free over A_1, so again by going-down $P' \cap A_1 = 0$ and $A_1 \hookrightarrow R'/P'$. Moreover R'/P' is generated over A_1 by elements of L, which are algebraic over F, so R'/P' is integral over A_1. Hence $\operatorname{Frac}(R'/P')$ is algebraic over $\operatorname{Frac}(A_1)$, and has transcendence degree $q + r$. This proves (a).

(b) *Lemma.* Let T be an irreducible closed subset (reduced structure) of dimension d of an affine variety V and $f_1, \ldots, f_m \in \mathcal{O}(V)$. Then every irreducible component of $T \cap V(f_1, \ldots, f_m)$ has dimension $\geq d - m$.

Induction on m, $m = 0$ being trivial. Let E be an irreducible component of $T \cap V(f_1, \ldots, f_m)$. It lies in an irreducible component E' of $T \cap V(f_1, \ldots, f_{m-1})$, with $\dim E' \geq d - m + 1$, and E is a component of $E' \cap V(f_m)$ (it is maximal irreducible in the bigger set). If $f_m|_{E'} = 0$

then $E = E'$. Otherwise $f_m|_{E'} \in \mathcal{O}(E')$ is non-nilpotent (E' is reduced), and Cor. 2.5.26 gives $\dim E = \dim E' - 1 \geq d - m$.

Case $Z = \mathbb{A}^n_k = \operatorname{Spec} k[x_1, \ldots, x_n]$. Then $Z \times_k Z = \operatorname{Spec} k[x, y]$ and $\Delta(Z) = V(x_1 - y_1, \ldots, x_n - y_n)$. Let $P = X \times_k Y$, a closed subset of $Z \times Z$ with image $p_1^{-1}(X) \cap p_2^{-1}(Y)$. The diagonal Δ is a closed immersion, and $\Delta(z) \in p_1^{-1}(X) \cap p_2^{-1}(Y)$ iff $z \in X \cap Y$. Hence Δ is a homeomorphism from $X \cap Y$ onto $\Delta(Z) \cap P = P \cap V(f_1, \ldots, f_n)$ with $f_i = x_i - y_i$. Let C be an irreducible component of $X \cap Y$ and $C' \subseteq P \cap V(f_1, \ldots, f_n)$ the corresponding component, so $\dim C = \dim C'$. Choose an irreducible component D of P containing C'; by (a), $\dim D = q + r$. Then C' is a component of $D \cap V(f_1, \ldots, f_n)$, so by the Lemma $\dim C \geq q + r - n$.

Case $Z = \mathbb{P}^n_k$. Let C be an irreducible component of $X \cap Y$ and $z \in C$; say $z \in U_i = D_+(T_i) \simeq \mathbb{A}^n_k$. Then $X \cap U_i$, $Y \cap U_i$ are nonempty closed subvarieties of U_i which are pure of dimensions q, r (Prop. 2.5.19 applied to each component), and $C \cap U_i$ is an irreducible component of $(X \cap U_i) \cap (Y \cap U_i)$ with $\dim(C \cap U_i) = \dim C$. The affine case gives $\dim C \geq q + r - n$.

(c) Let $B = k[T_0, \ldots, T_n]$ and replace I, J by their radicals, which are homogeneous. As $X, Y \neq \varnothing$, $I, J \subseteq \mathfrak{m}_0 := (T_0, \ldots, T_n)$ (a proper homogeneous prime is contained in B_+). Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$ be the minimal primes of I; they are homogeneous (Lemma 2.3.35(a)) and different from $\mathfrak{m}_0$ (as $X \neq \varnothing$, some prime $\mathfrak{q} \in \operatorname{Proj} B$ contains I, and $\mathfrak{q} \subsetneq \mathfrak{m}_0$, so $\mathfrak{m}_0$ is not minimal over I), and $V_+(\mathfrak{p}_i)$ are the irreducible components of X, of dimension q. By Cor. 2.5.21, $\dim V(\mathfrak{p}_i) = \dim B/\mathfrak{p}_i = q + 1$. So $V(I) \subseteq \mathbb{A}^{n+1}_k$ is a closed subvariety pure of dimension $q + 1$, and likewise $V(J)$ is pure of dimension $r + 1$. Both contain the origin $\mathfrak{m}_0$. Apply (b) in $\mathbb{A}^{n+1}_k$: every irreducible component of $V(I) \cap V(J) = V(I+J)$ has dimension $\geq (q+1)+(r+1)-(n+1) = q + r - n + 1 \geq 1$. Let $\mathfrak{P}$ be a minimal prime of $I + J$; it is homogeneous (Lemma 2.3.35(a)) and $\dim B/\mathfrak{P} \geq 1$, so $\mathfrak{P} \neq \mathfrak{m}_0$. A homogeneous prime containing $B_+ = \mathfrak{m}_0$ equals $\mathfrak{m}_0$, so $\mathfrak{P} \in \operatorname{Proj} B$ and $\mathfrak{P} \in V_+(I) \cap V_+(J) = X \cap Y$.

(d) Suppose $X = X_1 \sqcup X_2$ with X_1, X_2 nonempty, disjoint and closed. Each X_i is a union of irreducible components of X, hence a closed subvariety of $\mathbb{P}^2_k$ pure of dimension 1. Apply (c) with $n = 2$, $q = r = 1$ (so $q + r \geq n$): $X_1 \cap X_2 \neq \varnothing$, a contradiction. □

Exercise 3.5

> **Problem.** Let $X \to S$ be a proper morphism. Let $f : X \to Y$ be an open immersion of S-schemes. Let us suppose that Y is separated over S and connected. Show that f is an isomorphism.

Solution. The statement needs $X \neq \varnothing$ (if $X = \varnothing$ the empty map is an open immersion into any $Y \neq \varnothing$ and is not an isomorphism); we assume it. The composite $X \xrightarrow{f} Y \to S$ is proper and $Y \to S$ is separated, so f is proper (Prop. 3.16(e)), in particular a closed morphism. Thus $f(X)$ is open (open immersion) and closed (closed morphism) and nonempty in the connected space Y, so $f(X) = Y$. A surjective open immersion is an isomorphism onto the open subscheme $f(X) = Y$. □

Exercise 3.6

> **Problem.** Let $f : X \to Y$ be a separated morphism. Show that any section of f is a closed immersion (apply Proposition 3.9(f) to $X \times_X Y \to X \times_Y Y$).

Solution. Let $s : Y \to X$ with $f \circ s = \mathrm{Id}_Y$. Regard X as a scheme over Y via f (separated by hypothesis), and consider the X-scheme Y (structure morphism s) and the X-scheme X (structure morphism Id_X). Prop. 3.9(f) (with base Y, separated scheme X over it, $X_1 = Y$ and $X_2 = X$ over X) says that the canonical morphism

$$\varphi : Y \times_X X \to Y \times_Y X$$

is a closed immersion; φ is characterized by $\mathrm{pr}_1 \circ \varphi = p_1$ and $\mathrm{pr}_2 \circ \varphi = p_2$, where p_1, p_2 are the projections of $Y \times_X X$ and $\mathrm{pr}_1, \mathrm{pr}_2$ those of $Y \times_Y X$. Now $p_1 : Y \times_X X \to Y$ is an isomorphism (the base change of Id_X) with $p_2 = s \circ p_1$, and $\mathrm{pr}_2 : Y \times_Y X \to X$ is an isomorphism (base change of Id_Y). Therefore

$$\mathrm{pr}_2 \circ \varphi \circ p_1^{-1} = p_2 \circ p_1^{-1} = s,$$

so s is the composite of an isomorphism, the closed immersion φ and an isomorphism, hence a closed immersion. □

Exercise 3.7

> **Problem.** Let X be a Y-scheme. Show that X is separated over Y if and only if for every Y-scheme Y', the sections of $X \times_Y Y' \to Y'$ are closed immersions (hint: $\Delta_{X/Y}$ is a section of the second projection $X \times_Y X \to X$).

Solution. ($\Rightarrow$) If $X \to Y$ is separated, then $X \times_Y Y' \to Y'$ is separated for every Y' (Prop. 3.9(c)), and its sections are closed immersions by Ex. 3.6.

($\Leftarrow$) Take $Y' = X$ (a Y-scheme via the structure morphism). Then $X \times_Y Y' = X \times_Y X \to Y' = X$ is the second projection, and $\Delta_{X/Y}$ is a section of it since $\mathrm{pr}_2 \circ \Delta = \mathrm{Id}_X$. By hypothesis $\Delta_{X/Y}$ is a closed immersion, i.e. $X \to Y$ is separated. □

Exercise 3.8

> **Problem.** Let $f : X \to Y$ be a morphism with X affine and Y separated. Show that $f^{-1}(V)$ is affine for every affine open subset V of Y (use Proposition 3.9(f)).

Solution. Here "Y separated" means separated over $\operatorname{Spec} \mathbf{Z}$. Let $V \subseteq Y$ be an affine open subset; X and V are Y-schemes (via f and the inclusion). Prop. 3.9(f) (base $\operatorname{Spec} \mathbf{Z}$, separated scheme Y) says that the canonical morphism

$$X \times_Y V \to X \times_{\mathbf{Z}} V$$

is a closed immersion. The scheme $X \times_{\mathbf{Z}} V$ is affine, being a fibred product of affine schemes, and a closed subscheme of an affine scheme is affine. Finally $X \times_Y V$ is identified with the open subscheme $f^{-1}(V)$ of X. Hence $f^{-1}(V)$ is affine. □

Exercise 3.9

Problem. Let S be a locally Noetherian scheme, $s \in S$. Let Y be a scheme of finite type over $\operatorname{Spec} \mathcal{O}_{S,s}$. Show that there exist an open neighborhood U of s and a scheme of finite type $X \to U$ such that $Y = X \times_U \operatorname{Spec} \mathcal{O}_{S,s}$. Moreover, if Y is separated, we can take $X \to U$ separated.

Solution. Choose an affine open neighbourhood $\operatorname{Spec} A$ of s; A is Noetherian, s corresponds to a prime $\mathfrak{p}$ and $R := \mathcal{O}_{S,s} = A_{\mathfrak{p}} = \varinjlim_{h \notin \mathfrak{p}} A_h$. All neighbourhoods U below are of the form $D(g) = \operatorname{Spec} A_g$, $g \in A \setminus \mathfrak{p}$; the rings A_g are Noetherian. For an A_g-algebra C we write $C_h = C \otimes_{A_g} A_{gh}$ and $C_R = C \otimes_{A_g} R = \varinjlim_h C_h$. Finite type algebras over A_g are finitely presented.

Step 0: an algebra lemma. Let D, C be A_g-algebras of finite type.

(i) Every R-algebra homomorphism $\varphi : D_R \to C_R$ is $\varphi_0 \otimes R$ for some A_{gh}-algebra homomorphism $\varphi_0 : D_h \to C_h$, $h \notin \mathfrak{p}$.

(ii) If $\varphi_1, \varphi_2 : D \to C$ become equal over R, they become equal over A_{gh} for some $h \notin \mathfrak{p}$.

(iii) If $\varphi : D \to C$ is such that φ_R is an isomorphism, then φ_h is an isomorphism for some $h \notin \mathfrak{p}$.

(iv) If O_1, O_2 are open subsets of $\operatorname{Spec} C$ with $(O_1)_R \subseteq (O_2)_R$ (preimages in $\operatorname{Spec} C_R$), then $O_1 \cap \operatorname{Spec} C_h \subseteq O_2 \cap \operatorname{Spec} C_h$ for some $h \notin \mathfrak{p}$ (in particular equal if equal over R).

Proof. Write $D = A_g[x_1, \dots, x_m]/(P_1, \dots, P_l)$. (i) Write $\varphi(x_i) = c_i/h$ with $c_i \in C$ and a common h. The elements $P_j(c_1/h, \dots, c_m/h) \in C_h$ vanish in $C_R = \varinjlim C_h$, hence in $C_{hh'}$ for some h'. Then $x_i \mapsto c_i/h$ defines φ_0 on $D_{hh'}$, and $\varphi_0 \otimes R = \varphi$ as both agree on generators. (ii) The finitely many elements $\varphi_1(x_i) - \varphi_2(x_i)$ vanish in C_R, hence in some C_{h_i}; take $h = \prod h_i$. (iii) By (i) the inverse of φ_R is $\psi_0 \otimes R$ with $\psi_0 : C_h \to D_h$; then $\psi_0 \varphi_h$ and $\varphi_h \psi_0$ become the identities over R, hence over some $A_{ghh'}$ by (ii). (iv) We may assume $O_1 = D(c)$ (a finite union of such, $\operatorname{Spec} C$ being Noetherian). Then $D(c)_R \subseteq (O_2)_R = \bigcup_j D(c'_j)_R$ means that c'_j generate the unit ideal in $C_c \otimes R$, say $1 = \sum u_j c'_j$; this identity already holds in $C_c \otimes A_{gh}$ for some h.

Step 1: bi-principal opens. If $U_1 = \operatorname{Spec} B_1$, $U_2 = \operatorname{Spec} B_2$ are affine opens of a scheme and $y \in U_1 \cap U_2$, there is an open neighbourhood W of y principal in both. Indeed, choose $a \in B_1$ with $y \in D_{U_1}(a) \subseteq U_1 \cap U_2$; $D_{U_1}(a)$ is an open subset of U_2, so there is $b \in B_2$ with $y \in D_{U_2}(b) \subseteq D_{U_1}(a)$. Now b restricts to an element $b| \in \mathcal{O}(D_{U_1}(a)) = (B_1)_a$, say $b| = b'/a^N$ with $b' \in B_1$. Then $D_{U_2}(b) = D_{U_1}(a) \cap D(b|) = D_{U_1}(ab')$, which is principal in both.

Y is quasi-compact; cover it by finitely many affine opens $Y_\alpha = \operatorname{Spec} B_\alpha$, with B_α finitely generated R-algebras (hence Noetherian). Each $Y_\alpha \cap Y_\beta$ is an open subset of the Noetherian space Y_α, so quasi-compact; by the preceding paragraph it is a finite union of opens $W_{\alpha\beta\gamma} = D_{Y_\alpha}(b_{\alpha\beta\gamma}) = D_{Y_\beta}(b_{\beta\alpha\gamma})$ $(\gamma \in \Gamma_{\alpha\beta})$. The identity of $W_{\alpha\beta\gamma}$ induces isomorphisms of R-algebras $\rho_{\alpha\beta\gamma} : (B_\beta)_{b_{\beta\alpha\gamma}} \to (B_\alpha)_{b_{\alpha\beta\gamma}}$, inverse to the corresponding $\rho_{\beta\alpha\gamma}$.

Step 2: descent. Choose $g \notin \mathfrak{p}$ and finitely presented A_g-algebras B^0_α with $B^0_\alpha \otimes R = B_\alpha$ and lifts $b^0_{\alpha\beta\gamma} \in B^0_\alpha$ of the $b_{\alpha\beta\gamma}$ (possible since each is a finite set of finitely many elements, with finitely many relations). Put $X_\alpha = \operatorname{Spec} B^0_\alpha$ and $X_{\alpha\beta} = \bigcup_\gamma D(b^0_{\alpha\beta\gamma}) \subseteq X_\alpha$. Using Lemma (i),(iii) (the rings $(B^0_\alpha)_{b^0}$ are of finite type over A_g and their R-base change is $(B_\alpha)_b$) and shrinking g finitely often, we find A_g-isomorphisms $\rho^0_{\alpha\beta\gamma}$ with $\rho^0 \otimes R = \rho$ and $\rho^0_{\beta\alpha\gamma} = (\rho^0_{\alpha\beta\gamma})^{-1}$; let

$\theta^0_{\alpha\beta\gamma} : D(b^0_{\alpha\beta\gamma}) \to D(b^0_{\beta\alpha\gamma}) \subseteq X_\beta$ be the isomorphism of schemes they define. Two such maps for γ, γ' agree on $D(b^0_{\alpha\beta\gamma} b^0_{\alpha\beta\gamma'})$ after $\otimes R$ (both are induced by the identity of Y), these are morphisms from an affine scheme to the affine scheme X_β, i.e. ring homomorphisms, so they agree after shrinking g (Lemma (ii)). Thus they glue to $\theta_{\alpha\beta} : X_{\alpha\beta} \to X_{\beta\alpha}$, an isomorphism with inverse $\theta_{\beta\alpha}$ and $\theta_{\alpha\beta} \otimes R = \mathrm{Id}$ on $Y_\alpha \cap Y_\beta$. The cocycle conditions $\theta_{\alpha\beta}(X_{\alpha\beta} \cap X_{\alpha\delta}) = X_{\beta\alpha} \cap X_{\beta\delta}$ and $\theta_{\beta\delta}\theta_{\alpha\beta} = \theta_{\alpha\delta}$ on $X_{\alpha\beta} \cap X_{\alpha\delta}$ are finitely many equalities between open subsets of the X_β and between morphisms from affine pieces (finitely many principal opens of X_α) to affine schemes; all hold after $\otimes R$, hence after shrinking g by Lemma (iv) and (ii). Replacing U by $D(g)$ so that all this holds, the X_α glue to a scheme $X \to U := \operatorname{Spec} A_g$ which is of finite type (finitely many X_α of finite type), and $X \times_U \operatorname{Spec} R$ is obtained by gluing the Y_α along $Y_\alpha \cap Y_\beta$ by the identity, i.e. it is Y.

Step 3: separatedness. Suppose Y separated over R. Let $G = \{\mathfrak{q} \in \operatorname{Spec} A_g : \mathfrak{q} \subseteq \mathfrak{p}\}$ be the set of generizations of s, closed under generization. For a U-scheme T, the map $T_R \to T$ is a homeomorphism onto $\pi_T^{-1}(G)$ with the subspace topology (it is $\operatorname{Spec} S^{-1}C \to \operatorname{Spec} C$ on affine opens). Put $P = X \times_U X$, so $P_R = Y \times_R Y$. Since $\Delta_{Y/R} = \Delta_{X/U} \times_U R$, the set $D := \Delta(X) \cap \pi_P^{-1}(G)$ is the image of $\Delta_{Y/R}$, which is closed in $\pi_P^{-1}(G)$ by hypothesis.

Let Z be the closure of $\Delta(X)$ in P. The open set $O = \bigcup_\alpha X_\alpha \times_U X_\alpha$ contains $\Delta(X)$ and $\Delta(X) \cap (X_\alpha \times_U X_\alpha) = \Delta(X_\alpha)$ is closed in it by Prop. 3.4; so $\Delta(X) = F \cap O$ with F closed, hence $Z \subseteq F$ and $Z \cap O = \Delta(X)$. Therefore $T := Z \setminus \Delta(X) = Z \setminus O$ is closed in P.

$T \cap \pi_P^{-1}(G) = \varnothing$. The space X is Noetherian with irreducible components having generic points $x_1, \ldots, x_m$, and $Z = \bigcup_j \overline{\{\Delta(x_j)\}}$. If $z \in Z \cap \pi_P^{-1}(G)$, then $z \in \overline{\{\Delta(x_j)\}}$ for some j; so $\Delta(x_j)$ is a generization of z and lies in $\pi_P^{-1}(G)$, thus in D. As D is closed in $\pi_P^{-1}(G)$ and $z \in \overline{\{\Delta(x_j)\}} \cap \pi_P^{-1}(G)$, we get $z \in D \subseteq \Delta(X)$, so $z \notin T$.

The Noetherian space T has finitely many irreducible components T_i with generic points τ_i, and $\pi_P(\tau_i) = \mathfrak{q}_i \notin G$, i.e. $\mathfrak{q}_i \not\subseteq \mathfrak{p}$. Pick $h_i \in \mathfrak{q}_i \setminus \mathfrak{p}$ and $h = \prod h_i$. Then $\pi_P(T_i) \subseteq V(\mathfrak{q}_i)$ misses $D(h)$, so $T \cap \pi_P^{-1}(D(h)) = \varnothing$. Over $D(h)$ we get that $\Delta(X) \cap \pi_P^{-1}(D(h)) = Z \cap \pi_P^{-1}(D(h))$ is closed, i.e. the diagonal of $X \times_U D(h) \to D(h)$ has closed image, so this morphism is separated by Cor. 3.5. Replace U by $D(h)$; still $X_R = Y$. □

Exercise 3.10

Problem. (Graph of a morphism) Let $f : X \to Y$ be a morphism of S-schemes. We define the graph Γ_f of f as being the image of $(\mathrm{Id}_X, f) : X \to X \times_S Y$.

(a) Show that (Id_X, f) is a section of the projection morphism $X \times_S Y \to X$. Deduce from this that Γ_f is a closed subset of $X \times_S Y$ if Y is separated over S.

(b) Let us suppose X is reduced and let us endow Γ_f with the structure of a reduced closed subscheme. Show that if Y is separated over S, then the projection $X \times_S Y \to X$ induces an isomorphism from Γ_f onto X.

Solution. (a) Let $p : X \times_S Y \to X$ be the first projection. By definition $p \circ (\mathrm{Id}_X, f) = \mathrm{Id}_X$, so $\gamma := (\mathrm{Id}_X, f)$ is a section of p. If $Y \to S$ is separated, so is its base change $p : X \times_S Y \to X$ (Prop. 3.9(c)), and by Ex. 3.6 the section γ is a closed immersion. In particular $\Gamma_f = \gamma(X)$ is closed.

(b) Let X be reduced and Y/S separated; $\gamma : X \to P := X \times_S Y$ is a closed immersion, defining a closed subscheme Γ' of P with $\Gamma' \simeq X$ via γ. This subscheme is reduced and has underlying set Γ_f. The reduced closed subscheme structure on a closed subset is unique (its ideal sheaf is the radical ideal of the set). Hence $\Gamma' = \Gamma_f$ and γ induces an isomorphism $j : X \to \Gamma_f$. Then $p|_{\Gamma_f} \circ j = p \circ \gamma = \mathrm{Id}_X$, so $p|_{\Gamma_f} = j^{-1}$ is an isomorphism $\Gamma_f \to X$. □

Exercise 3.11

Problem. Let X be a Y-scheme.

(a) Show that for any Y-scheme Z, $X_{\mathrm{red}} \times_Y Z \to X \times_Y Z$ is a closed immersion and a homeomorphism on the underlying topological spaces. Deduce from this that $X \to Y$ separated $\Leftrightarrow X_{\mathrm{red}} \to Y$ separated $\Leftrightarrow X_{\mathrm{red}} \to Y_{\mathrm{red}}$ separated.

(b) Let $\{F_i\}_i$ be a finite number of closed subsets of X such that $X = \bigcup_i F_i$. Let us endow the F_i with the structure of reduced closed subschemes. Show that $X \to Y$ is separated if and only if $F_i \to Y$ is separated for every i.

(c) Let us suppose X is of finite type over Y. Show that $X \to Y$ proper $\Leftrightarrow X_{\mathrm{red}} \to Y$ proper $\Leftrightarrow X_{\mathrm{red}} \to Y_{\mathrm{red}}$ proper. Also show that (b) is true for proper morphisms (by supposing $X \to Y$ of finite type).

Solution. (a) Let $i : X_{\mathrm{red}} \to X$ be the canonical closed immersion. We have $X_{\mathrm{red}} \times_Y Z = X_{\mathrm{red}} \times_X (X \times_Y Z)$, so $i \times \mathrm{Id}_Z$ is a base change of i, hence a closed immersion (Prop. 1.23 and Rem. 1.20). To see it is a homeomorphism, work on affine opens: with $X = \operatorname{Spec} B$, $Z = \operatorname{Spec} C$ over $\operatorname{Spec} A$, the morphism is given by $B \otimes_A C \to (B/\mathfrak{N}) \otimes_A C$ ($\mathfrak{N}$ the nilradical), which is surjective with kernel the image of $\mathfrak{N} \otimes C$, an ideal generated by nilpotent elements and so contained in the nilradical. A closed immersion defined by an ideal inside the nilradical is a homeomorphism ($V(I) = \operatorname{Spec}$ when $I \subseteq \mathrm{Nil}$).

Equivalences. If $X \to Y$ is separated, then $X_{\mathrm{red}} \to X \to Y$ is a composite of separated morphisms (closed immersions are separated, Prop. 3.9(a),(b)). Conversely, let $X_{\mathrm{red}} \to Y$ be separated. The morphism $i \times i : X_{\mathrm{red}} \times_Y X_{\mathrm{red}} \to X \times_Y X$ is a composite of two morphisms as in the first part, so a closed immersion, and $\Delta_X \circ i = (i \times i) \circ \Delta_{X_{\mathrm{red}}}$. As i is surjective, $\Delta_X(X) = (i \times i)(\Delta_{X_{\mathrm{red}}}(X_{\mathrm{red}}))$ is closed in $X \times_Y X$ (image of a closed subset by a closed immersion). By Cor. 3.5, $X \to Y$ is separated. Next, $X_{\mathrm{red}} \to Y$ factors through Y_{red}, and $Y_{\mathrm{red}} \to Y$ is a closed immersion, hence separated. If $X_{\mathrm{red}} \to Y_{\mathrm{red}}$ is separated then so is the composite $X_{\mathrm{red}} \to Y$ (Prop. 3.9(b)); if $X_{\mathrm{red}} \to Y$ is separated, so is $X_{\mathrm{red}} \to Y_{\mathrm{red}}$ (Prop. 3.9(e)).

(b) Let $j_i : F_i \to X$ be the closed immersions. If $X \to Y$ is separated, so is $F_i \to X \to Y$. Conversely suppose all $F_i \to Y$ separated. The morphism $j_i \times j_i : F_i \times_Y F_i \to X \times_Y X$ is a closed immersion (composite of base changes of j_i), and $\Delta_X \circ j_i = (j_i \times j_i) \circ \Delta_{F_i}$. Hence $\Delta_X(F_i) = (j_i \times j_i)(\Delta_{F_i}(F_i))$ is closed. Then $\Delta_X(X) = \bigcup_i \Delta_X(F_i)$ is a finite union of closed sets, hence closed. By Cor. 3.5, $X \to Y$ is separated.

(c) Recall proper = finite type + separated + universally closed. Let $X \to Y$ be of finite type.

First equivalence. If $X \to Y$ is proper, so is $X_{\mathrm{red}} \to X \to Y$ (closed immersions are proper, Prop. 3.16(a),(b)). Conversely, let $X_{\mathrm{red}} \to Y$ be proper. Then $X \to Y$ is of finite type by hypothesis and separated by (a). For any Y-scheme Z, let $h : X_{\mathrm{red}} \times_Y Z \to X \times_Y Z$ be the homeomorphism above. The projection of $X_{\mathrm{red}} \times_Y Z$ to Z is closed and equals $\mathrm{pr}_Z \circ h$; so for a

closed $C \subseteq X \times_Y Z$, $\mathrm{pr}_Z(C) = \mathrm{pr}_Z(h(h^{-1}C))$ is closed. So $X \to Y$ is universally closed, hence proper.

Second equivalence. If $X_{\mathrm{red}} \to Y$ is proper, then since $Y_{\mathrm{red}} \to Y$ is separated, $X_{\mathrm{red}} \to Y_{\mathrm{red}}$ is proper by Prop. 3.16(e). If $X_{\mathrm{red}} \to Y_{\mathrm{red}}$ is proper, then composing with the closed immersion $Y_{\mathrm{red}} \to Y$ (proper), $X_{\mathrm{red}} \to Y$ is proper (Prop. 3.16(b)).

Finite closed covers. Let $X = \bigcup_i F_i$, F_i reduced closed subschemes. If $X \to Y$ is proper, each $F_i \to X \to Y$ is proper. Conversely let all $F_i \to Y$ be proper. Then $X \to Y$ is of finite type (hypothesis) and separated by (b). For a Y-scheme Z let $C \subseteq X \times_Y Z$ be closed. The morphisms $F_i \times_Y Z \to X \times_Y Z$ are closed immersions whose images E_i (preimages of F_i under the projection to X) cover $X \times_Y Z$. Since $F_i \times_Y Z \to Z$ is closed and $C \cap E_i$ corresponds to a closed subset of $F_i \times_Y Z$, the set $\mathrm{pr}_Z(C \cap E_i)$ is closed. Hence $\mathrm{pr}_Z(C) = \bigcup_i \mathrm{pr}_Z(C \cap E_i)$ is closed. So $X \to Y$ is universally closed and thus proper. □

Exercise 3.12

Problem. Let $f : X \to Y$ be a surjective morphism. We suppose that Y is connected and that all of the fibers X_y are connected.

(a) Show that if X is proper over Y, then X is connected.

(b) Study the example $\operatorname{Spec} k[T_1, T_2]/T_1(T_1T_2 - 1) \to \operatorname{Spec} k[T_1]$, where k is a field. Deduce from this that (a) is false if f is not proper.

Solution. (a) Suppose $X = X_1 \sqcup X_2$ with X_1, X_2 open and closed (hence both closed). Since f is proper it is a closed map, so $f(X_1)$ and $f(X_2)$ are closed in Y. For $y \in Y$ the fibre X_y is nonempty (surjectivity); its underlying space is $f^{-1}(y)$ with the subspace topology, so $f^{-1}(y) = (f^{-1}(y) \cap X_1) \sqcup (f^{-1}(y) \cap X_2)$ is a decomposition into disjoint open and closed subsets of the connected set $f^{-1}(y)$. Hence $f^{-1}(y)$ lies entirely in X_1 or entirely in X_2, so y belongs to exactly one of $f(X_1)$, $f(X_2)$. Thus $Y = f(X_1) \sqcup f(X_2)$ is a disjoint union of two closed sets. As Y is connected, one of them is empty, say $f(X_2) = \varnothing$, so $X_2 = \varnothing$. Therefore X is connected.

(b) Put $X = \operatorname{Spec} k[T_1, T_2]/(T_1(T_1T_2 - 1))$ and $Y = \operatorname{Spec} k[T_1] = \mathbb{A}^1_k$, which is connected. Fibres: for the point $y = (T_1)$ the residue field is k and $X_y = \operatorname{Spec} k[T_2]$, a line, connected. For every other point y of $\mathbb{A}^1_k$ (generic point or closed point (π), $\pi \neq T_1$ irreducible), T_1 is a unit in $\kappa = k(y)$ and $X_y = \operatorname{Spec} \kappa[T_2]/(T_1(T_1T_2 - 1)) = \operatorname{Spec} \kappa[T_2]/(T_2 - T_1^{-1}) = \operatorname{Spec} \kappa$, a point. So f is surjective with connected fibres. But X is not connected: the ideals (T_1) and $(T_1T_2 - 1)$ of $k[T_1, T_2]$ are comaximal ($T_1 \cdot T_2 - (T_1T_2 - 1) = 1$), so by the Chinese remainder theorem

$$k[T_1, T_2]/(T_1(T_1T_2 - 1)) \simeq k[T_2] \times k[T_1, T_1^{-1}],$$

and X is the disjoint union of two nonempty open and closed pieces. Hence the conclusion of (a) fails; by (a), f is therefore not proper (indeed the image of the closed piece $V(T_1T_2 - 1)$ is $D(T_1)$, which is not closed in Y). □

Exercise 3.13

Problem. (Rational maps) Let X, Y be schemes over S which are integral, with $Y \to S$ separated. A rational map from X to Y, denoted by $X \dashrightarrow Y$, is an equivalence class of

morphisms of S-schemes from a non-empty open subscheme of X to Y. Two such morphisms $U \to Y$, $V \to Y$ are called equivalent if they coincide on $U \cap V$. Let us fix a rational map $X \dashrightarrow Y$.

(a) Show that in every equivalence class, there exists a unique element $f : U \to Y$ such that U is maximal for the inclusion relation, and that every element $g : V \to Y$ of the class verifies $g = f|_V$. We call U the domain of definition of $X \dashrightarrow Y$. We then denote the rational map associated to f by $f : X \dashrightarrow Y$.

(b) Let us suppose that f is dominant, that X, Y are of finite type over S, and that S is locally Noetherian. Let $x \in X$. Show that f is defined at x (i.e., $x \in U$) if and only if there exists a $y \in Y$ such that the image of $\mathcal{O}_{Y,y}$ under $K(Y) \to K(X)$ is a local ring dominated by $\mathcal{O}_{X,x}$.

(c) Let Γ_f be the closure in $X \times_S Y$ of the graph of $f : U \to Y$. We call it the graph of the rational map f. We endow Γ_f with the structure of a reduced closed subscheme. Show that Γ_f is integral and that the projection $p : X \times_S Y \to X$ induces a birational morphism from Γ_f to X.

(d) Show that there exist a birational morphism $g : Z \to X$ and a morphism $\tilde{f} : Z \to Y$ such that $\tilde{f} = f \circ g$ on $g^{-1}(U)$ (we say that we have eliminated the indetermination of $X \dashrightarrow Y$). Moreover, if Y is proper (resp. projective) over S, we can choose g proper (resp. projective).

Solution. Preliminary: equivalence is an equivalence relation. Reflexivity and symmetry are clear. Let $f : U \to Y$, $g : V \to Y$, $h : W \to Y$ with $f = g$ on $U \cap V$ and $g = h$ on $V \cap W$. Then $f = h$ on $U \cap V \cap W$, which is a nonempty open (as X is irreducible) and thus dense subset of $U \cap W$. Since $U \cap W$ is reduced and Y/S is separated, Prop. 3.11 gives $f = h$ on $U \cap W$.

(a) Let $\mathcal{C}$ be the class and U_0 the union of the domains of its members. Any two members agree on the intersection of their domains (by definition of equivalence, as the class is an equivalence class), so the members glue to a morphism of S-schemes $f : U_0 \to Y$. Then $f|_V = g$ for each member $g : V \to Y$, so $f \in \mathcal{C}$ and f has maximal domain. If $f' : U' \to Y$ is another member with maximal domain, then $U' \subseteq U_0$ and $f' = f|_{U'}$; maximality of U' forces $U' = U_0$ and $f' = f$.

(c) Let U be the domain of f, nonempty, dense and irreducible. Since Y/S is separated, $\gamma = (\mathrm{Id}_U, f) : U \to U \times_S Y$ is a closed immersion (Ex. 3.10(a)); its image Γ_U is closed in the open subset $p^{-1}(U) = U \times_S Y$ of $X \times_S Y$. Let $\Gamma_f = \overline{\Gamma_U}$. It is irreducible (closure of the image of an irreducible space), and reduced by definition, so it is integral. Moreover $\Gamma_f \cap p^{-1}(U) = \Gamma_U$ (closure in an open subset $= \overline{\Gamma_U} \cap p^{-1}(U)$, and Γ_U is closed there). Hence the open subscheme $\Gamma_f \cap p^{-1}(U)$ of Γ_f is the reduced closed subscheme of $U \times_S Y$ supported on Γ_U, and by Ex. 3.10(b) (applied to $f|_U$, U reduced) p restricts to an isomorphism $\Gamma_f \cap p^{-1}(U) \to U$, with inverse γ. As $p^{-1}(U) \cap \Gamma_f$ and U are dense open in the integral schemes Γ_f, X, the morphism $p|_{\Gamma_f} : \Gamma_f \to X$ is birational (an isomorphism above the dense open subset U, so also an isomorphism of function fields).

(d) Take $Z = \Gamma_f$, $g = p|_{\Gamma_f}$ and $\tilde{f} = q|_{\Gamma_f}$, $q : X \times_S Y \to Y$ the second projection. By (c), g is birational. Over U, $g^{-1}(U) = \Gamma_U$ with $g^{-1} = \gamma$ (identified with U), so $\tilde{f} \circ \gamma = q \circ \gamma = f$, i.e. $\tilde{f} = f \circ g$ on $g^{-1}(U)$. If $Y \to S$ is proper (resp. projective), then so is its base change $p : X \times_S Y \to X$ (Prop. 3.16(c), Cor. 3.32(c)), and g is the composite of the closed immersion $\Gamma_f \to X \times_S Y$ and p; closed immersions are proper and projective (Prop. 3.16(a), Cor. 3.32(a)) and compositions preserve both (Prop. 3.16(b), Cor. 3.32(b)). So g is proper (resp. projective).

(b): the statement, a counterexample, and a corrected version. Let $f : U \to Y$ be dominant, so $f(\eta_X) = \eta_Y$ (Exercise 2.4.11) and f induces $\varphi : K(Y) = \mathcal{O}_{Y,\eta_Y} \to \mathcal{O}_{X,\eta_X} = K(X)$, independent of the representative. "The image of $\mathcal{O}_{Y,y}$ is a local ring dominated by $\mathcal{O}_{X,x}$" means $\varphi(\mathcal{O}_{Y,y}) \subseteq \mathcal{O}_{X,x}$ and $\varphi(\mathfrak{m}_y) \subseteq \mathfrak{m}_x$.

As printed, the "if" direction is false when S is not separated. Let S be the affine line over k with doubled origin $0_1, 0_2$ (locally Noetherian), $X = S_1$ and $Y = S_2$ the two open copies of $\mathbb{A}^1_k$ in S, with the inclusions as structure morphisms. Both are integral and of finite type over S, and $Y \to S$ is separated (an open immersion). The inclusion $U = S_1 \setminus \{0_1\} = S_2 \setminus \{0_2\} \to Y$ is an S-morphism, dominant, with $K(Y) = K(X) = k(T)$ and $\varphi = \mathrm{id}$. For $x = 0_1$ and $y = 0_2$ we have $\mathcal{O}_{Y,y} = k[T]_{(T)} = \mathcal{O}_{X,x}$ and φ is a local isomorphism. But f is not defined at x: an S-morphism $V \to Y$ with $V \ni 0_1$ open in X would have to satisfy that $V \to Y \to S$ equals the inclusion $V \subseteq S_1 \subseteq S$, whose image contains 0_1, while the image of $Y \to S$ does not contain 0_1.

Corrected statement. f is defined at x if and only if there is $y \in Y$ lying over the same point of S as x such that $\varphi(\mathcal{O}_{Y,y}) \subseteq \mathcal{O}_{X,x}$ and $\varphi(\mathfrak{m}_y) \subseteq \mathfrak{m}_x$. If S is separated, the condition on the image in S is automatic: let $T = \operatorname{Spec} \mathcal{O}_{X,x}$, $a : T \to X \to S$ and $b : T \to \operatorname{Spec} \mathcal{O}_{Y,y} \to Y \to S$ (the latter using the local homomorphism $\mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ induced by φ). They coincide on the generic point of T because φ is compatible with S; the set $(a, b)^{-1}(\Delta_S(S))$ is closed in T and contains the generic point, so it is T; evaluating at the closed point gives a point $\Delta_S(s)$ whose two projections are the images of x and of y in S, and these projections coincide (Rem. 3.3), so x and y lie over the same point of S. The hypotheses "finite type" and "locally Noetherian" are not needed below beyond finite generation of affine algebras.

($\Rightarrow$) If $x \in U$ and $y = f(x)$, then y lies over the same point of S as x since f is an S-morphism. The local homomorphism $f_x^{\#} : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is compatible with the maps to the fraction fields (the generic points specialize to x, y), so it is the restriction of φ; local means $\varphi(\mathfrak{m}_y) \subseteq \mathfrak{m}_x$.

($\Leftarrow$) Let $t \in S$ be the common image of x and y. Choose an affine open $S' = \operatorname{Spec} \Lambda \ni t$, Λ Noetherian, an affine open $Y' = \operatorname{Spec} B \ni y$ over S' and an affine open $\operatorname{Spec} A \ni x$ over S', so $B = \Lambda[b_1, \ldots, b_m]$ is a finitely generated Λ-algebra with $B \subseteq K(Y)$, $A \subseteq K(X)$ (integrality). By hypothesis $\varphi(b_i) \in \mathcal{O}_{X,x} = A_{\mathfrak{p}_x}$, and since φ is S-compatible, φ maps Λ into the image of Λ in A. So there is $g \in A \setminus \mathfrak{p}_x$ with $\varphi(B) \subseteq A_g$, giving a Λ-algebra homomorphism $B \to A_g$ and an S-morphism $F : D(g) \to Y' \subseteq Y$, with $F(\eta_X) = \eta_Y$. We show F is equivalent to f. The open set $W_0 = D(g) \cap U \cap f^{-1}(Y')$ contains η_X; choose a nonempty affine open $W_1 = \operatorname{Spec} A_1 \subseteq W_0$. Then $F|_{W_1}$ and $f|_{W_1}$ correspond to Λ-algebra homomorphisms $B \to A_1$ whose composites with $A_1 \hookrightarrow K(X)$ both equal $\varphi|_B$. So they are equal: $F = f$ on W_1. As W_1 is dense in $D(g) \cap U$, X is reduced and Y/S separated, Prop. 3.11 yields $F = f$ on $D(g) \cap U$. Thus F belongs to the class of f and by (a) $D(g) \subseteq U$, so $x \in U$. $\square$

Exercise 3.14

> **Problem.** Let X, Y be integral separated schemes over a Noetherian scheme S. Let $f : X \dashrightarrow Y$ be a birational map (i.e., f comes from an isomorphism from a non-empty open subset $U \subseteq X$ onto an open subset of Y). Show that f is defined everywhere if and only if $\Gamma_f \to X$ is an isomorphism.

Solution. Let $U_{\max}$ be the domain of definition of f, and $p : \Gamma_f \to X$ the projection (Ex. 3.13(c)).

Only the separatedness of Y over S is used (it is what Ex. 3.10 and 3.13 need); the facts that f is birational and that S is Noetherian are not used.

($\Rightarrow$) Suppose $U_{\max} = X$, so $f : X \to Y$ is a morphism. The graph $\gamma(X)$ of f, $\gamma = (\mathrm{Id}_X, f)$, is closed in $X \times_S Y$ (Ex. 3.10(a)), contains the graph $\Gamma_{U'}$ of the restriction to any dense open U', and $\Gamma_{U'} = \gamma(U')$ is dense in $\gamma(X) \simeq X$ (as γ is a homeomorphism onto its image and U' is dense in X). So $\Gamma_f = \overline{\Gamma_{U'}} = \gamma(X)$, the reduced closed subscheme on it. By Ex. 3.10(b) (X reduced, Y/S separated), $p : \Gamma_f \to X$ is an isomorphism.

($\Leftarrow$) Suppose $p : \Gamma_f \to X$ is an isomorphism. Put $\tilde{f} = q \circ p^{-1} : X \to Y$, a morphism of S-schemes. By Ex. 3.13(d) (with $Z = \Gamma_f$, $g = p$), $\tilde{f} \circ p = q|_{\Gamma_f} = f \circ p$ on $p^{-1}(U_{\max})$; hence $\tilde{f}|_{U_{\max}} = f$. So $\tilde{f}$ is a member of the class of f defined on all of X, and by Ex. 3.13(a) the domain of definition is X. □

Exercise 3.15

> **Problem.** We say that a morphism of schemes $f : X \to Y$ is finite (resp. integral) if for every affine open subset V of Y, $f^{-1}(V)$ is affine and $\mathcal{O}_X(f^{-1}(V))$ is finite (resp. integral) over $\mathcal{O}_Y(V)$. Let $f : X \to Y$ be a morphism such that there exists an affine open covering $Y = \bigcup_i Y_i$ with $f^{-1}(Y_i)$ affine and $\mathcal{O}_X(f^{-1}(Y_i))$ finite over $\mathcal{O}_Y(Y_i)$. We want to show that f is finite. (a) Show that we can reduce to the case when Y is affine and that $Y_i = D(h_i)$ is a principal open subset. (b) Show that X then verifies condition (3.2), Subsection 2.3.1. Deduce from this that the canonical morphism $X \to \operatorname{Spec} \mathcal{O}_X(X)$ is an isomorphism (hence X is affine). (c) Show that $\mathcal{O}_X(X)$ is finite over $\mathcal{O}_Y(Y)$.

Solution. Conventions: results of the same chapter are cited by their number ("Prop. 3.9", "Prop. 1.2" for Section 3.1, "Prop. 2.4" for Section 3.2), results of earlier chapters with the chapter prefix ("Prop. 2.3.25"); "Ex. 2.5" means Exercise 2.5 of Section 3.2, "Ex. 1.8" is an exercise of Section 3.1, "Ex. 2.3.17" is Exercise 3.17 of Section 2.3 (Chapter 2), and "Ex. 3.17" is an exercise of the present section. The $\mathcal{O}_Y(W)$-module structure on $\mathcal{O}_X(f^{-1}(W))$ is always given by $f^{\#}(W)$. For a ring homomorphism $\psi : A \to B$ and $g \in A$, the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ pulls $D(g)$ back to $D(\psi(g))$, and $\mathcal{O}(D(\psi(g))) = B_{\psi(g)}$ is the localization of the A-algebra B at (the image of) g (Prop. 2.3.14, Lemma 2.3.7).

(a) Reduction. Let V be an affine open subset of Y. We must show that $f^{-1}(V)$ is affine and $\mathcal{O}_X(f^{-1}(V))$ is finite over $\mathcal{O}_Y(V)$.

Claim: V is a finite union of open sets D such that $D = D_V(h)$ for some $h \in \mathcal{O}_Y(V)$ and also $D = D_{Y_i}(g)$ for some i and some $g \in \mathcal{O}_Y(Y_i)$. Let $y \in V$ and choose i with $y \in Y_i$. The principal open sets form a base of the affine scheme Y_i, so there is $g \in \mathcal{O}_Y(Y_i)$ with $y \in D_{Y_i}(g) \subseteq V \cap Y_i$. Now $D_{Y_i}(g)$ is open in the affine scheme V, so there is $h \in \mathcal{O}_Y(V)$ with $y \in D_V(h) \subseteq D_{Y_i}(g)$. Write $h|_{D_{Y_i}(g)} = a/g^m \in \mathcal{O}_Y(Y_i)_g$ with $a \in \mathcal{O}_Y(Y_i)$. Then $D_V(h)$ is the set of points of $D_{Y_i}(g) = \operatorname{Spec} \mathcal{O}_Y(Y_i)_g$ where a/g^m is a unit, that is $D_V(h) = D_{Y_i}(ag)$. Since V is quasi-compact (Ex. 2.1.6), finitely many such sets cover V. This proves the claim.

Let $D = D_V(h) = D_{Y_i}(g)$ be one of these sets and $X_i = f^{-1}(Y_i) = \operatorname{Spec} B_i$, $A_i = \mathcal{O}_Y(Y_i)$. By the preliminary remark, $f^{-1}(D) = D_{X_i}(g)$ is affine, with ring $(B_i)_g$, which is finite over $(A_i)_g = \mathcal{O}_Y(D)$ (a localization of a finite module is finite: generators stay generators). So the hypothesis of the exercise holds for the morphism $f^{-1}(V) \to V$ with the covering $\{D_V(h_k)\}_{k=1}^n$ of V. This is the required reduction: it suffices to prove the following statement.

(S) Let $Y = \operatorname{Spec} A$ be affine and $h_1, \dots, h_n \in A$ with $D(h_1) \cup \dots \cup D(h_n) = Y$ (equivalently $(h_1, \dots, h_n) = A$). Let $f : X \to Y$ be a morphism such that $X_k := f^{-1}(D(h_k))$ is affine and $\mathcal{O}_X(X_k)$ is finite over A_{h_k} for every k. Then X is affine and $B := \mathcal{O}_X(X)$ is finite over A.

(b) Write h_k also for $f^{\#}(Y)(h_k) \in B$. Then $X_k = X_{h_k}$ in the notation of Def. 2.3.11. Indeed, let $x \in X$ and $y = f(x)$. The homomorphism $f_x^{\#} : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ is local, so the preimage of $\mathfrak{m}_x$ is $\mathfrak{m}_y$, and therefore $(h_k)_x$ is a unit of $\mathcal{O}_{X,x}$ iff $(h_k)_y \notin \mathfrak{m}_y$ iff $y \in D(h_k)$.

Condition (3.2) holds: the X_k form a finite covering of X by affine open sets, and $X_k \cap X_l = f^{-1}(D(h_k h_l))$ is the principal open subset of the affine scheme X_k defined by (the restriction of) h_l, hence is affine (Lemma 2.3.7) and has the trivial finite affine covering by itself.

By Prop. 2.3.12, restriction induces isomorphisms of A-algebras $B_{h_k} \xrightarrow{\sim} \mathcal{O}_X(X_{h_k}) = \mathcal{O}_X(X_k)$.

Let $\pi : X \to \operatorname{Spec} B$ be the canonical morphism, corresponding to id_B under Prop. 2.3.25. By the construction in Prop. 2.3.25, $\pi(x)$ is the prime ideal of B which is the preimage of $\mathfrak{m}_x$ under $B \to \mathcal{O}_{X,x}$. So for $b \in B$ we have $\pi(x) \in D(b)$ iff $b_x \notin \mathfrak{m}_x$, i.e. $\pi^{-1}(D(b)) = X_b$. In particular $\pi^{-1}(D(h_k)) = X_k$. The restriction $\pi_k : X_k \to D(h_k) = \operatorname{Spec} B_{h_k}$ is a morphism of affine schemes. The ring homomorphism $\pi^{\#}(D(h_k)) : B_{h_k} \to \mathcal{O}_X(X_k)$ compatible with restrictions satisfies $\pi^{\#}(D(h_k)) \circ (B \to B_{h_k}) = (\text{restriction } B \to \mathcal{O}_X(X_k))$, because $\pi^{\#}(Y') = \mathrm{id}_B$ for $Y' = \operatorname{Spec} B$. By the universal property of localization, it is the isomorphism of Prop. 2.3.12. A morphism between affine schemes inducing an isomorphism on rings of global sections is an isomorphism (Prop. 2.3.25). So π_k is an isomorphism for every k.

The elements $h_k \in B$ generate the unit ideal of B (they do already in A), so the sets $D(h_k)$ cover $\operatorname{Spec} B$. As π is an isomorphism over each member of an open covering of the target, it is an isomorphism. Hence $X \simeq \operatorname{Spec} B$ is affine.

(c) We use the following lemma, which we also use in Ex. 3.16.

Lemma L. Let A be a ring, $h_1, \dots, h_n \in A$ with $(h_1, \dots, h_n) = A$, and M an A-module such that M_{h_k} is a finite A_{h_k}-module for each k. Then M is a finite A-module.

Proof. A generator of M_{h_k} has the form m/h_k^e with $m \in M$; since $h_k^{-e} \in A_{h_k}$, the elements $m/1$ also generate. So there are finitely many $m_{kj} \in M$ whose images generate M_{h_k} for each k. Let M' be the submodule generated by all m_{kj}, and $Q = M/M'$. By exactness of localization $Q_{h_k} = M_{h_k}/M'_{h_k} = 0$. For $q \in Q$ there is e (the same for all k) with $h_k^e q = 0$ for all k. The elements h_k^e again generate the unit ideal (a high power of $(h_1, \dots, h_n)$ lies in $(h_1^e, \dots, h_n^e)$), so $1 = \sum_k c_k h_k^e$ and $q = \sum c_k h_k^e q = 0$. So $M = M'$ is finite. □

Apply this to $M = B$ viewed as A-module: $B_{h_k} \simeq \mathcal{O}_X(X_k)$ is finite over A_{h_k}. So B is finite over A. This proves (S). Together with (a), f is finite. □

Exercise 3.16

> **Problem.** Present a statement analogous to that of Exercise 3.15 by replacing finite morphism with integral morphism.

Solution. Statement. Let $f : X \to Y$ be a morphism of schemes such that there is an affine open covering $Y = \bigcup_i Y_i$ with $f^{-1}(Y_i)$ affine and $\mathcal{O}_X(f^{-1}(Y_i))$ integral over $\mathcal{O}_Y(Y_i)$. Then f is integral, i.e. for every affine open subset V of Y, $f^{-1}(V)$ is affine and $\mathcal{O}_X(f^{-1}(V))$ is integral over $\mathcal{O}_Y(V)$.

Proof. We follow Ex. 3.15. In (a) of that proof, the only place where finiteness is used is that a localization $(B_i)_g$ of a finite $(A_i)_g$-module is finite; here $(B_i)_g$ is integral over $(A_i)_g$, because b/g^m satisfies the equation obtained from an integral equation of b over A_i by dividing the coefficients by powers of g. So we reduce to the situation (S) with "integral" instead of "finite". Part (b) is unchanged (it never uses finiteness). So $X \simeq \operatorname{Spec} B$, $B = \mathcal{O}_X(X)$, and $B_{h_k} \simeq \mathcal{O}_X(X_k)$ is integral over A_{h_k} for all k.

It remains to show that B is integral over A. Let $b \in B$ and $M = A[b] \subseteq B$ the A-subalgebra generated by b. Since localization is exact, M_{h_k} is the subalgebra $A_{h_k}[b/1]$ of B_{h_k}. As $b/1$ is integral over A_{h_k}, this subalgebra is a finite A_{h_k}-module. By Lemma L, $M = A[b]$ is a finite A-module, so b is integral over A (a finitely generated A-module $A[b]$ which is a ring containing b gives an integral equation by the determinant trick). Hence B is integral over A. □

Exercise 3.17

> **Problem.** Show that the following properties are true. (a) Any finite morphism is of finite type and quasi-finite. (b) The class of finite morphisms verifies hypotheses (1)–(3) of Lemma 3.15. (c) Let $\rho : A \to B$ be a finite ring homomorphism. Then for any ideal I of B, we have $(\operatorname{Spec}\rho)(V(I)) = V(\rho^{-1}(I))$. (d) Any finite morphism is proper (see also Exercise 3.22).

Solution. (a) Let $f : X \to Y$ be finite. Cover Y by affine opens $Y_j = \operatorname{Spec} A_j$. Then $f^{-1}(Y_j) = \operatorname{Spec} B_j$ with B_j finite, hence finitely generated, over A_j. By Prop. 2.2 (the criterion for finite type), f is of finite type.

Quasi-finiteness (Ex. 2.8): let $y \in Y$ and choose $Y_j \ni y$. The fibre is $X_y = f^{-1}(Y_j) \times_{Y_j} \operatorname{Spec} k(y) = \operatorname{Spec} D$ with $D = B_j \otimes_{A_j} k(y)$ (Prop. 1.2). This is a finite-dimensional $k(y)$-algebra, so it is Artinian, it has only finitely many prime ideals, and $D \simeq \prod_{\mathfrak{q}} D_{\mathfrak{q}}$ is the product of its (local) localizations. Thus X_y is a finite set, and for $x \in X_y$ the local ring $\mathcal{O}_{X_y,x} = D_{\mathfrak{q}}$ is a direct factor of D, hence finite over $k(y)$. So f is quasi-finite.

(b) *(1) Closed immersions are finite.* Let $i : Z \to Y$ be a closed immersion and $V = \operatorname{Spec} A$ an affine open of Y. The restriction $i^{-1}(V) \to V$ is a closed immersion, so by Prop. 2.3.20 $i^{-1}(V) \simeq \operatorname{Spec} A/J$ is affine, the ring map $A \to \mathcal{O}_Z(i^{-1}V)$ being the surjection $A \to A/J$. Thus $\mathcal{O}_Z(i^{-1}V)$ is generated by 1 as an A-module.

(2) Composition. Let $f : X \to Y$ and $g : Y \to Z$ be finite and $W = \operatorname{Spec} C$ an affine open of Z. Then $g^{-1}(W) = \operatorname{Spec} B$ is affine with B finite over C, and $f^{-1}(g^{-1}(W)) = \operatorname{Spec} D$ is affine since $\operatorname{Spec} B$ is an affine open of Y, with D finite over B. So D is a finite C-module (a product of generators is a system of generators).

(3) Base change. Let $f : X \to Y$ be finite, $g : Y' \to Y$ a morphism, $X' = X \times_Y Y'$ and $f' : X' \to Y'$. Let V' be an affine open subset of Y'. Cover V' by affine opens $V'_k = \operatorname{Spec} A'_k$ such that $g(V'_k) \subseteq V_k$ for an affine open $V_k = \operatorname{Spec} A_k$ of Y. Then $f'^{-1}(V'_k) = f^{-1}(V_k) \times_{V_k} V'_k$ (Ex. 1.6(b)) $= \operatorname{Spec}(B_k \otimes_{A_k} A'_k)$, where $B_k = \mathcal{O}_X(f^{-1}(V_k))$ is finite over A_k. Hence $B_k \otimes_{A_k} A'_k$ is finite over $A'_k = \mathcal{O}_{Y'}(V'_k)$. Ex. 3.15, applied to $f'^{-1}(V') \to V'$ and the covering $\{V'_k\}$, shows that $f'^{-1}(V')$ is affine with ring finite over $\mathcal{O}_{Y'}(V')$. As V' was arbitrary, f' is finite.

(c) The inclusion $\subseteq$ is clear: if $\mathfrak{q} \supseteq I$ then $\rho^{-1}(\mathfrak{q}) \supseteq \rho^{-1}(I)$. Conversely let $\mathfrak{p}$ be a prime of A containing $J := \rho^{-1}(I)$. The induced map $\bar\rho : A/J \to B/I$ is injective and finite. Put $R = A/J$ localized at $\mathfrak{p}/J$, which is a local ring with maximal ideal $\mathfrak{m} = \mathfrak{p}R$, and $B' = S^{-1}(B/I)$ with S

the image of $A \setminus \mathfrak{p}$. Since localization is exact, $R \to B'$ is injective, so $B' \neq 0$ (as $R \neq 0$), and B' is a finite R-module. By Nakayama, $\mathfrak{m}B' \neq B'$. Let $\mathfrak{n}$ be a maximal ideal of B' containing $\mathfrak{m}B'$. Then $\mathfrak{n} \cap R$ is a proper ideal containing $\mathfrak{m}$, so equals $\mathfrak{m}$. Let $\mathfrak{q}$ be the preimage of $\mathfrak{n}$ in B (a prime containing I). Then $\rho^{-1}(\mathfrak{q})$ is the preimage of $\mathfrak{n} \cap R = \mathfrak{m}$ in A, namely $\mathfrak{p}$. So $\mathfrak{p} \in (\operatorname{Spec} \rho)(V(I))$.

(d) Let $f : X \to Y$ be finite. It is of finite type by (a). *Separated:* this is local on Y (Def. 3.2), so we may assume $Y = \operatorname{Spec} A$ affine; then X is affine, and f is separated by Prop. 3.4. *Universally closed:* finite morphisms are stable under base change by (b), so it is enough to show that every finite morphism f is closed. Being closed is local on Y (if $F \subseteq X$ is closed, then $f(F) \cap V = f(F \cap f^{-1}V)$ is closed in V for all V in an open covering of Y, so $f(F)$ is closed). So let $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, B finite over A. A closed subset of X is $V(I)$, and $f(V(I)) = V(\rho^{-1}(I))$ is closed by (c). Hence f is proper. □

Exercise 3.18

> **Problem.** Let K be a number field. We know that the ring of integers $\mathcal{O}_K$ of K is finite over $\mathbb{Z}$. Show that if X is a separated connected scheme containing $\operatorname{Spec} \mathcal{O}_K$ as an open subscheme, then $X = \operatorname{Spec} \mathcal{O}_K$ (use Exercises 3.17(d) and 3.5).

Solution. Let $U = \operatorname{Spec} \mathcal{O}_K$ and $j : U \to X$ the open immersion; we regard X as a scheme over $S = \operatorname{Spec} \mathbb{Z}$ ("separated" means separated over $\mathbb{Z}$). The structural morphism $U \to S$ is finite (the preimage of an affine open $D(n) \subseteq S$ is $\operatorname{Spec}(\mathcal{O}_K)_n$, finite over $\mathbb{Z}_n$; or use Ex. 3.15 with the covering $\{S\}$), hence proper (Ex. 3.17(d)). We recall the argument of Ex. 3.5. The composite $U \xrightarrow{j} X \to S$ is proper and $X \to S$ is separated, so j is proper (Prop. 3.16(e)). In particular j is closed, so $j(U)$ is open (an open immersion) and closed in X. It is non-empty ($\mathcal{O}_K \neq 0$), and X is connected, hence $j(U) = X$. An open immersion which is surjective is an isomorphism. So $U = X$, i.e. $X = \operatorname{Spec} \mathcal{O}_K$. □

Exercise 3.19

> **Problem.** Let X be a proper algebraic variety over a field k. Let $f : X \to Y$ be a morphism of k-schemes with Y affine. Show that $f(X)$ is a finite set of closed points (hint: f factors into $X \to \operatorname{Spec} \mathcal{O}_X(X) \to Y$).

Solution. Write $Y = \operatorname{Spec} C$, where C is a k-algebra via the structure morphism. As sets $f(X) = f(i(X_{\mathrm{red}}))$ where $i : X_{\mathrm{red}} \to X$ is the canonical closed immersion, which is a homeomorphism (Prop. 2.4.2 and its proof). The reduced scheme X_{red} is proper over k (Prop. 3.16(a),(b)). We may therefore replace X by X_{red} and f by $f \circ i$, and assume X reduced. (We do this because Cor. 3.19 is available only for reduced varieties; for general X one can use Remark 3.20 instead.)

Put $B = \mathcal{O}_X(X)$. By Cor. 3.19, B is a finite-dimensional k-vector space. Let $\pi : X \to \operatorname{Spec} B$ be the canonical morphism ($\rho(\pi) = \mathrm{id}_B$) and $\varphi = \rho(f) : C \to B$ (Prop. 2.3.25). Since f is a morphism of k-schemes, φ is a k-algebra homomorphism (apply the naturality of ρ to the structure morphisms). Then $\operatorname{Spec}(\varphi) \circ \pi$ and f both correspond to φ (for the first one: $\rho(\operatorname{Spec} \varphi \circ \pi) = \rho(\pi) \circ \rho(\operatorname{Spec} \varphi) = \mathrm{id}_B \circ \varphi$), so by injectivity of ρ (Prop. 2.3.25)

$$f = \operatorname{Spec}(\varphi) \circ \pi.$$

Hence $f(X) \subseteq \mathrm{Spec}(\varphi)(\mathrm{Spec}\, B)$.

Now $\mathrm{Spec}\, B$ is a finite set: every prime $\mathfrak{q}$ of B is maximal ($B/\mathfrak{q}$ is a domain of finite dimension over k, hence a field), distinct maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ are pairwise comaximal, and by the Chinese remainder theorem $B \to \prod B/\mathfrak{m}_l$ is surjective, so $r \leq \dim_k B$. Therefore $f(X)$ is finite. Finally, if $\mathfrak{q} \in \mathrm{Spec}\, B$ and $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$, then $C/\mathfrak{p}$ embeds as a k-subalgebra of the finite extension field $B/\mathfrak{q}$ of k. A domain of finite dimension over k is a field, so $\mathfrak{p}$ is maximal, i.e. a closed point of Y. Every point of $f(X)$ is of this form. □

Exercise 3.20

> **Problem.** (a) Show that Corollary 3.32 is true for quasi-projective morphisms (use Exercise 2.3(c)). (b) Show that a morphism of finite type from an affine scheme to an affine scheme is quasi-projective.

Solution. Recall (Def. 3.35) that $h : X \to W$ is quasi-projective if it is the composite of an open immersion $a : X \to Z$ of finite type and a projective morphism $Z \to W$. Cor. 3.32 consists of the statements: (3.32a) closed immersions are quasi-projective; (3.32b) compositions of quasi-projective morphisms are quasi-projective; (3.32c) quasi-projective morphisms are stable under base change; (3.32d) if $X \to S$ and $Y \to S$ are quasi-projective, so is $X \times_S Y \to S$; (3.32e) if $X \to Y \to Z$ has quasi-projective composite and $Y \to Z$ is separated, then $X \to Y$ is quasi-projective.

The proof of Cor. 3.32 for projective morphisms deduces (d) and (e) from (a)–(c) by Lemma 3.15 (note that (b) of that lemma has to be read, as in its proof and in Prop. 3.16(e), as: if $g \circ f$ has P and g is separated, then f has P). Hence it is enough to prove (3.32a)–(3.32c) for quasi-projective morphisms: then Lemma 3.15(a),(b) gives (3.32d) and (3.32e), since the hypotheses (1)–(3) of the lemma are exactly (3.32a)–(3.32c).

A lemma on immersions (this is the content of Ex. 2.3 which we need). Recall that an immersion is an open immersion followed by a closed immersion.

Lemma R. Let Q be a scheme, $\Omega \subseteq Q$ an open subscheme, $\kappa : X \to \Omega$ a closed immersion and $e = \omega \circ \kappa : X \to Q$, where $\omega : \Omega \to Q$ is the inclusion. Assume that e is of finite type. Then $e = j \circ g$ where $j : \overline{X} \to Q$ is a closed immersion and $g : X \to \overline{X}$ is an open immersion of finite type.

Proof. The morphism e is quasi-compact (it is of finite type). Let $\overline{X}$ be its scheme-theoretic closure (Ex. 2.3.17): $\overline{X} = V(\mathcal{J})$ with $\mathcal{J} = \mathrm{Ker}(\mathcal{O}_Q \to e_*\mathcal{O}_X)$ is a scheme, $j : \overline{X} \to Q$ is a closed immersion, and $e = j \circ g$ for a morphism $g : X \to \overline{X}$. For $V \subseteq \Omega$ open, $e_*\mathcal{O}_X(V) = \mathcal{O}_X(\kappa^{-1}V)$, so $\mathcal{J}|_\Omega = \mathrm{Ker}(\kappa^\#) =: \mathcal{J}_\kappa$. Since κ is a closed immersion, κ identifies X with the closed subscheme $(V(\mathcal{J}_\kappa), \mathcal{O}_\Omega/\mathcal{J}_\kappa)$ of Ω (Prop. 2.2.24). This closed subscheme of Ω is the open subscheme $\overline{X} \cap \Omega = j^{-1}(\Omega)$ of $\overline{X}$. So κ factors as $X \xrightarrow{\kappa'} j^{-1}(\Omega) \to \Omega$ with κ' an isomorphism. By Ex. 1.6(a) (a closed immersion is a monomorphism), $g = \kappa'$ followed by the inclusion $j^{-1}(\Omega) \subseteq \overline{X}$. Hence g is an open immersion.

It remains to see that g is quasi-compact; then g is of finite type by Prop. 2.4(e), since $j \circ g = e$ is of finite type. Let Q_l be an affine open covering of Q. The sets $\overline{X}_l = j^{-1}(Q_l)$ are affine (Prop. 2.3.20) and cover $\overline{X}$, and $g^{-1}(\overline{X}_l) = e^{-1}(Q_l)$ is quasi-compact. Let $V \subseteq \overline{X}$ be an affine

open subset. As in the proof of Prop. 2.2, $V \cap \overline{X}_l$ is a union of open sets which are principal in V and in $\overline{X}_l$; by quasi-compactness of V, V is a finite union of such sets $W = D_{\overline{X}_l}(s)$. Then $g^{-1}(W)$ is the principal open subset of the quasi-compact scheme $T = g^{-1}(\overline{X}_l)$ defined by $g^{\#}(s)$. Write $T = T_1 \cup \cdots \cup T_m$ as a finite union of affine opens (Ex. 2.3.14). Then $g^{-1}(W) \cap T_c = D_{T_c}(g^{\#}(s)|_{T_c})$ is affine, so $g^{-1}(W)$ is a finite union of affine opens, hence quasi-compact. Therefore $g^{-1}(V)$ is quasi-compact. $\square$

Part (a), step 1: closed immersions (3.32a). A closed immersion $i : X \to Y$ is $i \circ \mathrm{id}_X$, where id_X is an open immersion of finite type and i is projective (Cor. 3.32(a)).

Part (a), step 2: base change (3.32c). Let $h = b \circ a$ with $a : X \to Z$ an open immersion of finite type and $b : Z \to Y$ projective, and let $Y' \to Y$ be a morphism. Put $Z' = Z \times_Y Y'$ and $X' = X \times_Y Y' = X \times_Z Z'$. The projection $X' \to Z'$ is the base change of the open immersion a, hence an open immersion (it is the inclusion of the open subscheme which is the preimage of $a(X)$, Ex. 1.6(b)); it is of finite type by Prop. 2.4(c). The morphism $Z' \to Y'$ is projective by Cor. 3.32(c). So $X' \to Y'$ is quasi-projective.

Part (a), step 3: composition (3.32b). Let $X \xrightarrow{a} Z \xrightarrow{b} Y$ and $Y \xrightarrow{c} Z' \xrightarrow{d} W$ with a, c open immersions of finite type and b, d projective. We must show that $d \circ c \circ b \circ a$ is quasi-projective. Write $b = \pi \circ i$ with $i : Z \to \mathbb{P}^n_Y$ a closed immersion and $\pi : \mathbb{P}^n_Y \to Y$ the projection. Put $Q = \mathbb{P}^n_{Z'}$ with projection $\pi_Q : Q \to Z'$. Since $\mathbb{P}^n_Y = \mathbb{P}^n_{Z'} \times_{Z'} Y$ (Prop. 1.4(b)), the canonical morphism $c' : \mathbb{P}^n_Y \to Q$ is the base change of c, hence an open immersion, of finite type by Prop. 2.4(c), and $\pi_Q \circ c' = c \circ \pi$. Define $e = c' \circ i \circ a : X \to Q$. It is of finite type, as a composite of a, of the closed immersion i (Prop. 2.4(a)) and of c' (Prop. 2.4(b)).

The set $a(X)$ is open in Z, and i is a homeomorphism onto a closed subset, so $a(X) = i^{-1}(\Omega_0)$ for some open $\Omega_0 \subseteq \mathbb{P}^n_Y$. The restriction of i over Ω_0 is a closed immersion $i^{-1}(\Omega_0) \to \Omega_0$, and $i^{-1}(\Omega_0)$ is the open subscheme $a(X) \simeq X$ of Z. Thus $i \circ a = (\text{inclusion of } \Omega_0) \circ \kappa$ with $\kappa : X \to \Omega_0$ a closed immersion, and $e = c'|_{\Omega_0} \circ \kappa$ where $\Omega = c'(\Omega_0)$ is open in Q and $c'|_{\Omega_0} : \Omega_0 \to \Omega$ is an isomorphism. So Lemma R applies to e (with Ω and the closed immersion $c'|_{\Omega_0} \circ \kappa : X \to \Omega$): $e = j \circ g$, with $j : \overline{X} \to Q$ a closed immersion and $g : X \to \overline{X}$ an open immersion of finite type.

Now $\pi_Q \circ j : \overline{X} \to Z'$ is projective by definition, so $d \circ \pi_Q \circ j : \overline{X} \to W$ is projective (Cor. 3.32(b)). And

$$d \circ c \circ b \circ a = d \circ c \circ \pi \circ i \circ a = d \circ \pi_Q \circ c' \circ i \circ a = d \circ \pi_Q \circ j \circ g.$$

So $d \circ c \circ b \circ a$ is the open immersion g of finite type followed by a projective morphism, i.e. quasi-projective. This proves (3.32a)–(3.32c), and hence (3.32d), (3.32e) by Lemma 3.15 as explained. This completes part (a) of the exercise.

Part (b). Let $X = \operatorname{Spec} B \to Y = \operatorname{Spec} A$ be of finite type. Then $B = A[t_1, \ldots, t_n]/I$ for some n (Def. 2.1 with $U = X$). The surjection $A[T_1, \ldots, T_n] \to B$ gives a closed immersion $\kappa : X \to \mathbb{A}^n_A = \operatorname{Spec} A[T_1, \ldots, T_n] = D_+(T_0) \subseteq Q := \mathbb{P}^n_A$. Put $e = \omega \circ \kappa : X \to Q$, with ω the open immersion $D_+(T_0) \subseteq Q$. We check that e is of finite type. The scheme Q is separated over A (Cor. 3.10), hence separated, since $\operatorname{Spec} A \to \operatorname{Spec} \mathbb{Z}$ is separated (Prop. 3.4, 3.9(b)). So the intersection of $D_+(T_0)$ with any affine open V of Q is affine (Prop. 3.6), and $e^{-1}(V) = \kappa^{-1}(V \cap D_+(T_0))$ is affine (Prop. 2.3.20). Thus e is quasi-compact. The composite of e with the structure morphism $Q \to Y$ is $X \to Y$, which is of finite type, and $Q \to Y$ is of finite type (Cor. 3.10). By Prop. 2.4(e), e is of finite type. By Lemma R, $e = j \circ g$ with $g : X \to \overline{X}$ an open immersion of finite type and $j : \overline{X} \to Q$ a closed immersion. Then $\overline{X} \to Q \to Y$

is projective and $X \to Y$ is the composite of g and this projective morphism. So $X \to Y$ is quasi-projective. □

Exercise 3.21

> **Problem.** Let k be a field. Show that two closed subsets of dimension 1 in $\mathbb{P}^2_k$ always have a non-empty intersection (use Exercise 3.4). Deduce from this that $\mathbb{P}^1_k \times_k \mathbb{P}^1_k \not\simeq \mathbb{P}^2_k$.

Solution. Let $S = k[T_0, T_1, T_2]$, so $\mathbb{P}^2_k = \operatorname{Proj} S$, and let S_+ be the irrelevant ideal. We prove directly the case $n = 2$, $q = r = 1$ of Ex. 3.4(c), without the diagonal argument.

Step 1: an irreducible closed subset X_0 of dimension 1 of $\mathbb{P}^2_k$ equals $V_+(F)$ for a homogeneous prime element F of S. Let P be the homogeneous prime corresponding to the generic point of X_0; then $X_0 = V_+(P) \simeq \operatorname{Proj}(S/P)$ and $P \not\supseteq S_+$. By Cor. 2.5.21, $\dim \operatorname{Spec}(S/P) = \dim X_0 + 1 = 2$. As S is a finitely generated integral domain over k of dimension 3 (Cor. 2.5.17), Prop. 2.5.23(a) gives $\operatorname{ht} P = 3 - 2 = 1$. In particular $P \neq 0$. As P is homogeneous, it contains a non-zero homogeneous element, and a prime factor F of it (which is homogeneous, being a factor of a homogeneous polynomial; S is factorial) lies in P. Then $0 \subsetneq (F) \subseteq P$ with (F) prime, and $\operatorname{ht} P = 1$ forces $(F) = P$. So $X_0 = V_+(F)$ with $\deg F \geq 1$.

Step 2. Let $X, Y \subseteq \mathbb{P}^2_k$ be closed subsets of dimension 1. The space $\mathbb{P}^2_k$ is Noetherian, so X and Y have finitely many irreducible components, and each has an irreducible component X_0 (resp. Y_0) of dimension 1. By Step 1, $X_0 = V_+(F)$ and $Y_0 = V_+(G)$ with F, G homogeneous of positive degree. The ideal $(F, G) \subseteq S_+$ is proper homogeneous, so it has a minimal prime $\mathfrak{Q}$ over it. The prime $\mathfrak{Q}$ is homogeneous: the ideal $\mathfrak{Q}^h$ generated by the homogeneous elements of $\mathfrak{Q}$ is prime by Lemma 2.3.35(a), and $(F, G) \subseteq \mathfrak{Q}^h \subseteq \mathfrak{Q}$ because (F, G) is generated by homogeneous elements of $\mathfrak{Q}$, so $\mathfrak{Q}^h = \mathfrak{Q}$ by minimality. By Krull's theorem (Cor. 2.5.14(a)), $\operatorname{ht} \mathfrak{Q} \leq 2$. If $\mathfrak{Q} \supseteq S_+$, then $\mathfrak{Q} = S_+$ ($S/S_+ = k$ is a field), but $\operatorname{ht} S_+ = 3$ (the chain $0 \subset (T_0) \subset (T_0, T_1) \subset S_+$ of primes), a contradiction. So $\mathfrak{Q} \in \operatorname{Proj} S$, and $\mathfrak{Q} \in V_+(F) \cap V_+(G) = X_0 \cap Y_0 \subseteq X \cap Y$. Hence $X \cap Y \neq \emptyset$.

Step 3: $\mathbb{P}^1_k \times_k \mathbb{P}^1_k \not\simeq \mathbb{P}^2_k$. Let $a, b \in \mathbb{P}^1_k = \operatorname{Proj} k[U_0, U_1]$ be the closed points given by the homogeneous primes (U_1) and (U_0); they are distinct and have residue field k. Let $p_1 : Z = \mathbb{P}^1_k \times_k \mathbb{P}^1_k \to \mathbb{P}^1_k$ be the first projection, so $Z = \mathbb{P}^1_{\mathbb{P}^1_k}$ over $\mathbb{P}^1_k$. The sets $C_a = p_1^{-1}(a)$ and $C_b = p_1^{-1}(b)$ are closed in Z and disjoint. By Prop. 1.16 and Example 1.14, C_a is homeomorphic to the fibre $Z_a = \mathbb{P}^1_{k(a)} = \mathbb{P}^1_k$, so $\dim C_a = 1$ (Example 2.5.28), and likewise $\dim C_b = 1$. If there were an isomorphism $Z \simeq \mathbb{P}^2_k$, its image of C_a, C_b would be two disjoint closed subsets of dimension 1 of $\mathbb{P}^2_k$, contradicting Step 2. □

Exercise 3.22

> **Problem.** Let $f : \operatorname{Spec} B \to \operatorname{Spec} A$ be a finite morphism. We want to show that f is projective. (a) Let us first suppose that $B = A[T]/(P(T))$, where $P(T) = \sum_{0 \leq i \leq n} a_i T^i$ with $a_n = 1$. Let $\tilde{P}(T, S) = \sum_{0 \leq i \leq n} a_i T^i S^{n-i}$. Show that the natural morphism $\operatorname{Spec} B \to \operatorname{Proj} A[T, S]/(\tilde{P}(T, S))$ is an isomorphism. (b) Conclude in the general case.

Solution. (a) We may assume $n \geq 1$ (if $n = 0$ then $P = 1$, $B = 0$, and both sides are empty). Let $R = A[T, S]/(\tilde{P})$, graded by $\deg T = \deg S = 1$ ($\tilde{P}$ is homogeneous of degree n), and $R_+ = (T, S)R$.

$D_+(S) = \operatorname{Proj} R$. Let $\mathfrak{P} \in \operatorname{Proj} R$ be a homogeneous prime with $S \in \mathfrak{P}$. In R we have $0 = \tilde{P} = T^n + \sum_{i<n} a_i T^i S^{n-i} \equiv T^n \pmod{S}$ (all other terms contain S, and $a_n = 1$). So $T^n \in \mathfrak{P}$, hence $T \in \mathfrak{P}$, so $\mathfrak{P} \supseteq R_+$, which is excluded in $\operatorname{Proj} R$. Thus $\operatorname{Proj} R = D_+(S)$.

$R_{(S)} \simeq B$. By Prop. 2.3.38, $D_+(S) \simeq \operatorname{Spec} R_{(S)}$. Put $t = T/S$. Then $L := A[T, S, S^{-1}] = A[t][S, S^{-1}]$, graded by powers of S, with degree-0 part $A[t]$. The ideal $(\tilde{P})L$ is generated by $\tilde{P}/S^n = \sum a_i t^i = P(t) \in A[t]$, a homogeneous element of degree 0. So $R_S = L/P(t)L = (A[t]/(P(t)))[S, S^{-1}]$, whose degree-0 part is $R_{(S)} = A[t]/(P(t)) \simeq A[T]/(P(T)) = B$, with A-algebra structure the obvious one.

Therefore the natural morphism $\operatorname{Spec} B \simeq \operatorname{Spec} R_{(S)} \simeq D_+(S) = \operatorname{Proj} R$ is an isomorphism. Moreover $\operatorname{Proj} R = V_+(\tilde{P})$ is a closed subscheme of $\mathbb{P}^1_A = \operatorname{Proj} A[T, S]$ (Lemma 2.3.41). So $\operatorname{Spec} B \to \operatorname{Spec} A$ factors as a closed immersion into $\mathbb{P}^1_A$ followed by the projection: it is projective.

(b) Let B be finite over A, generated as an A-module by $b_1, \ldots, b_m$; these generate B as A-algebra. Each b_l is integral over A (determinant trick), so there is a monic polynomial $P_l \in A[T_l]$ of degree $n_l \geq 1$ with $P_l(b_l) = 0$ (multiply by T_l if necessary to have $n_l \geq 1$). Put $B_l = A[T_l]/(P_l)$ and $B' = A[T_1, \ldots, T_m]/(P_1(T_1), \ldots, P_m(T_m)) \simeq B_1 \otimes_A \cdots \otimes_A B_m$. The A-algebra homomorphism $B' \to B$, $T_l \mapsto b_l$, is well-defined and surjective. Hence $\operatorname{Spec} B \to \operatorname{Spec} B'$ is a closed immersion, and $\operatorname{Spec} B' = \operatorname{Spec} B_1 \times_{\operatorname{Spec} A} \cdots \times_{\operatorname{Spec} A} \operatorname{Spec} B_m$ (Prop. 1.2). By (a) each $\operatorname{Spec} B_l \to \operatorname{Spec} A$ is projective, hence so is the fibre product $\operatorname{Spec} B' \to \operatorname{Spec} A$ (Cor. 3.32(d)). Finally $\operatorname{Spec} B \to \operatorname{Spec} B'$ is a closed immersion, hence projective, and the composite $\operatorname{Spec} B \to \operatorname{Spec} A$ is projective by Cor. 3.32(b). □

Exercise 3.23

> **Problem.** Let X be a quasi-projective scheme over a scheme S, and let G be a finite group acting on the S-scheme X. Show that the quotient scheme X/G exists (use Exercise 2.3.21), and that G acts transitively on the fibers of $X \to X/G$. Moreover, the canonical morphism $X \to X/G$ is a finite morphism if S is locally Noetherian.

Solution. Let $\pi : X \to S$ be the structural morphism; "G acts on the S-scheme X" means that each $\sigma \in G$ acts by an automorphism φ_σ of X with $\pi \circ \varphi_\sigma = \pi$. We recall from Ex. 2.3.20 and Ex. 2.3.21 what we need. Let A be a ring and G a finite group acting on A by ring automorphisms.

(i) A is integral over A^G: for $a \in A$ the monic polynomial $\prod_{\sigma \in G}(T - \sigma a)$ has coefficients in A^G and vanishes at a. Consequently $p : \operatorname{Spec} A \to \operatorname{Spec} A^G$ is surjective, and (incomparability for integral extensions) two primes $\mathfrak{q} \subseteq \mathfrak{q}'$ of A with the same contraction to A^G are equal.

(ii) The fibres of p are the G-orbits (Ex. 2.3.20(a)). Indeed, $p \circ \varphi_\sigma = p$ since σ fixes A^G. Conversely let $\mathfrak{q}_1 \cap A^G = \mathfrak{q}_2 \cap A^G =: P_0$. For $b \in \mathfrak{q}_2$, the product $\prod_\sigma \sigma(b)$ lies in $\mathfrak{q}_2 \cap A^G = P_0 \subseteq \mathfrak{q}_1$, so some $\sigma(b) \in \mathfrak{q}_1$. Thus $\mathfrak{q}_2 \subseteq \bigcup_\sigma \sigma^{-1}(\mathfrak{q}_1)$ and by prime avoidance $\mathfrak{q}_2 \subseteq \tau(\mathfrak{q}_1)$ for some $\tau \in G$. Both primes contract to P_0, so $\mathfrak{q}_2 = \tau(\mathfrak{q}_1)$ by (i).

(iii) (Ex. 2.3.21(c) and its proof) If every point of X has a G-stable affine open neighbourhood, then X/G exists. Concretely: $q : X \to X/G$ is the map to the orbit space, X/G is covered by the open sets $V_i = q(U_i) \simeq \operatorname{Spec} A_i^G$, where $\{U_i = \operatorname{Spec} A_i\}$ is a covering of X by G-stable affine open sets, $q^{-1}(V_i) = U_i$, and $q|_{U_i} : U_i \to V_i$ is Spec of the inclusion $A_i^G \subseteq A_i$.

Existence. Let $x \in X$ and $s = \pi(x)$. Choose an affine open $S_0 = \operatorname{Spec} R \ni s$ and put $X_0 = \pi^{-1}(S_0)$. It is G-stable because $\pi\varphi_\sigma = \pi$, and $X_0 \to S_0$ is quasi-projective: it is the base change of $X \to S$ by $S_0 \to S$ (this is Cor. 3.32(c) for quasi-projective morphisms, proved in Ex. 3.20(a)). The orbit $F = Gx$ is a finite subset of X_0 (all its points lie over s). By Prop. 3.36(b), F is contained in an affine open subset U of X_0. Put $U' = \bigcap_{\sigma\in G} \varphi_\sigma(U)$. It is open, G-stable, and contains F (for $\tau \in G$, $\tau x \in \sigma(U)$ since $\sigma^{-1}\tau x \in F \subseteq U$). It is affine: X_0 is quasi-projective over $\operatorname{Spec} R$, hence separated over R (an open immersion is separated by Prop. 3.9(a), a projective morphism is separated by Cor. 3.10, and compositions are separated by Prop. 3.9(b)), hence separated (Prop. 3.4, 3.9(b)), so a finite intersection of affine opens of X_0 is affine (Prop. 3.6(ii), by induction). Thus every point of X has a G-stable affine open neighbourhood, and $q : X \to Y := X/G$ exists by (iii). Since $\pi\varphi_\sigma = \pi$, the universal property of q gives a morphism $Y \to S$, so Y is an S-scheme.

Transitivity on fibres. Let $y \in Y$, say $y \in V_i$. Then $q^{-1}(y) \subseteq q^{-1}(V_i) = U_i$ and $q^{-1}(y) = p_i^{-1}(y)$ for $p_i = q|_{U_i}$, which is a single G-orbit by (ii).

Finiteness. (We do not use that S is locally Noetherian: with Def. 3.35 the open immersion $X \to Z$ is of finite type, which is what is needed.) The morphism $X \to S$ is of finite type: it is the composite of an open immersion of finite type and a projective morphism, which is of finite type by Cor. 3.10; so Prop. 2.4(b) applies. Take $U_i = \operatorname{Spec} A_i$ as in the existence part, so $U_i \subseteq \pi^{-1}(S_0)$ with $S_0 = \operatorname{Spec} R$ affine. By definition of finite type, A_i is a finitely generated R-algebra, say generated by $a_1, \dots, a_r$. Because $\pi\varphi_\sigma = \pi$, the image of $R \to A_i$ is contained in A_i^G. Hence $A_i = A_i^G[a_1, \dots, a_r]$, and every a_l is integral over A_i^G by (i), so A_i is a finite A_i^G-module. Thus the covering $Y = \bigcup V_i$ satisfies: $q^{-1}(V_i) = U_i$ is affine, with $\mathcal{O}(U_i) = A_i$ finite over $\mathcal{O}_Y(V_i) = A_i^G$. By Ex. 3.15, q is finite. □

Exercise 3.24

> **Problem.** Let S be a locally Noetherian integral scheme with generic point ξ. Let X be a scheme of finite type over S. Show that if X_ξ is projective (resp. finite) over the function field $K(S) = k(\xi)$, then there exists a non-empty open subset U of S such that $X \times_S U \to U$ is projective (resp. finite). (Use Exercises 2.5 and 2.16.)

Solution. *Reduction.* Choose a non-empty affine open $S_0 = \operatorname{Spec} R$ of S. Then R is a Noetherian domain, ξ is the generic point of S_0, $K = \operatorname{Frac} R = k(\xi)$, and $X_\xi = \pi^{-1}(S_0) \times_{S_0} \operatorname{Spec} K$. An open subset of S_0 is open in S. So we may assume $S = \operatorname{Spec} R$.

I will not use Ex. 2.16, but I prove below (Lemma Sp) the special case of Ex. 2.5 that I need (the isomorphism case at the generic point), since that exercise is cited in the hint.

Notation. Let R' be a Noetherian domain with fraction field K, $\pi : T \to \operatorname{Spec} R'$ an R'-scheme, $0 \neq h \in R'$. Put $T_h = \pi^{-1}(D(h)) = T \times_{R'} R'_h$ and $T_K = T \times_{R'} K$. By Prop. 1.16 the projection $\iota : T_K \to T$ is a homeomorphism onto $\pi^{-1}(\xi)$, ξ the generic point of $\operatorname{Spec} R'$. All schemes below are of finite type over R' (that is, finite unions of spectra of finitely generated R'-algebras; open subschemes of such schemes are again such, by Ex. 2.2). They are Noetherian (Hilbert's basis theorem, Ex. 2.1), so their topological spaces are Noetherian.

Lemma Z. Let T be of finite type over R' and $F \subseteq T$ closed with $F \cap \pi^{-1}(\xi) = \emptyset$. Then $F \cap T_h = \emptyset$ for some $h \neq 0$.

Proof. Let $F_1, \dots, F_r$ be the irreducible components of F with generic points $z_1, \dots, z_r$. Then $\pi(z_m) \neq \xi$, i.e. $\pi(z_m)$ is a non-zero prime ideal P_m of R'. Choose $0 \neq h_m \in P_m$ and put $h = h_1 \cdots h_r \neq 0$. If $z \in F$, then $z \in F_m$ for some m, so $\pi(z)$ contains $\pi(z_m)$ (since z is a specialization of z_m) and thus contains h_m, hence h. So $\pi(z) \notin D(h)$, i.e. $F \cap T_h = \emptyset$. $\square$

Lemma U (uniqueness). Let T and Y' be of finite type over R' and $f, g : T \to Y'$ morphisms of R'-schemes with $f \circ \iota = g \circ \iota$ (i.e. $f_K = g_K$). Then $f|_{T_h} = g|_{T_h}$ for some $h \neq 0$.

Proof. Let Σ be the set of $x \in T$ having an open neighbourhood N with $f|_N = g|_N$; it is open, and $f|_\Sigma = g|_\Sigma$. We show $\pi^{-1}(\xi) \subseteq \Sigma$. Let $x \in \pi^{-1}(\xi)$. Then $y = f(x) = g(x)$. Choose an affine open $V = \operatorname{Spec} C' \ni y$ of Y' with C' a finitely generated R'-algebra, and an affine open $N = \operatorname{Spec} B' \ni x$ inside $f^{-1}(V) \cap g^{-1}(V)$. The morphisms $f|_N, g|_N : N \to V$ correspond to R'-algebra homomorphisms $u, v : C' \to B'$. The assumption $f_K = g_K$ says that u, v become equal after composition with $B' \to B' \otimes_{R'} K$. Let $c_1, \dots, c_p$ generate C' as R'-algebra. Then $u(c_l) - v(c_l)$ is killed by some $r_l \in R' \setminus 0$. With $r = \prod r_l$, u and v agree after inverting r, i.e. $f = g$ on $N_r = N \cap T_r$, which is an open neighbourhood of x (as $\pi(x) = \xi \in D(r)$). So $x \in \Sigma$. By Lemma Z applied to $F = T \setminus \Sigma$ we get $h \neq 0$ with $T_h \subseteq \Sigma$. $\square$

Lemma E (existence, affine case). Let $T = \operatorname{Spec} B'$ and $V = \operatorname{Spec} C'$ with B', C' finitely generated R'-algebras, and $\psi : T_K \to V_K$ a morphism of K-schemes. Then there are $h \neq 0$ and a morphism of R'_h-schemes $g : T_h \to V_h$ with $g_K = \psi$.

Proof. Write $C' = R'[y_1, \dots, y_p]/(g_1, \dots, g_q)$ (the ideal is finitely generated, $R'[y]$ being Noetherian). The morphism ψ corresponds to a K-algebra homomorphism $\psi^* : C' \otimes K \to B' \otimes K$. Write $\psi^*(y_l) = b_l/r_l$ with $b_l \in B'$, $0 \neq r_l \in R'$, and let $r = \prod r_l$. Define an R'-algebra homomorphism $\Phi : R'[y_1, \dots, y_p] \to B'_r$ by $y_l \mapsto b_l/r_l$. Since ψ^* is well defined, $\Phi(g_m)$ becomes 0 in $B' \otimes K = (R' \setminus 0)^{-1} B'_r$, so $\Phi(g_m)$ is killed by some $t_m \in R' \setminus 0$. Let $h = r \prod_m t_m$. Then Φ induces an R'_h-algebra homomorphism $C'_h \to B'_h$, i.e. a morphism $g : T_h = \operatorname{Spec} B'_h \to V_h = \operatorname{Spec} C'_h$, and $g_K = \psi$ since both correspond to homomorphisms agreeing on the generators y_l. $\square$

Lemma Sp (spreading out an isomorphism). Let X, Y be R'-schemes of finite type and $\varphi : X_K \to Y_K$ an isomorphism of K-schemes. Then there are $h \neq 0$ and an isomorphism of R'_h-schemes $X_h \to Y_h$ whose generic fibre is φ.

Proof. Cover Y by finitely many affine opens $Y_a = \operatorname{Spec} C_a$ and X by finitely many affine opens $X_i = \operatorname{Spec} B_i$ (finitely generated algebras). Fix i. The open sets $X_{iK} \cap \varphi^{-1}(Y_{aK})$ cover $X_{iK} = \operatorname{Spec}(B_i \otimes K)$; each is a union of principal opens $D(\beta)$, $\beta \in B_i \otimes K$, and we may take $\beta = b/1$ with $b \in B_i$ (units of K are invertible). By quasi-compactness, finitely many $D_{X_{iK}}(b_{il})$, $l = 1, \dots, L_i$, cover X_{iK}, with $\varphi(D(b_{il})) \subseteq Y_{a(i,l),K}$. Put $X_{il} = D_{X_i}(b_{il})$, an affine open of X with $X_{il,K} = D_{X_{iK}}(b_{il})$. The open set $X^1 = \bigcup_{i,l} X_{il}$ contains $\pi^{-1}(\xi)$, so by Lemma Z there is $h_1 \neq 0$ with $X_{h_1} \subseteq X^1$. Replacing R' by R'_{h_1} and X by X_{h_1} (and Y by Y_{h_1}), we may assume $X = \bigcup X_{il}$ with $\varphi(X_{il,K}) \subseteq Y_{a(i,l),K}$.

By Lemma E applied to $\varphi|_{X_{il,K}} : X_{il,K} \to Y_{a(i,l),K}$ we get $h_{il} \neq 0$ and $R'_{h_{il}}$-morphisms $g_{il} : (X_{il})_{h_{il}} \to Y_{a(i,l)} \subseteq Y$ with $(g_{il})_K = \varphi|$. For two indices, $T = X_{il} \cap X_{i'l'}$ is of finite type over R' and $g_{il}|_T, g_{i'l'}|_T$ have the same generic fibre $\varphi|_{T_K}$; by Lemma U they agree on T_h for some $h \neq 0$. Let $h' = \prod h_{il}$. Apply Lemma U to the restrictions of g_{il} and $g_{i'l'}$ to $(X_{il} \cap X_{i'l'})_{h'}$, for each of the finitely many pairs, and let h_2 be h' times the product of the elements h so obtained. Then all g_{il} are defined on $(X_{il})_{h_2}$ and agree on the overlaps. They glue to a morphism $g : X_{h_2} \to Y$ of R'-schemes, which lands in Y_{h_2} and satisfies $g_K = \varphi$.

By symmetry applied to φ^{-1}, there are $h_3 \neq 0$ and $g' : Y_{h_3} \to X_{h_3}$ with $g'_K = \varphi^{-1}$. On $X_{h_2h_3}$ the two morphisms $g' \circ g$ and the inclusion $X_{h_2h_3} \to X$ have generic fibre $\varphi^{-1}\varphi = \mathrm{id}$, so by Lemma U they agree on $X_{h_2h_3h_4}$ for some $h_4 \neq 0$; similarly $g \circ g' = \mathrm{id}$ on $Y_{h_2h_3h_5}$. With $h = h_2h_3h_4h_5$, $g : X_h \to Y_h$ is an isomorphism with inverse $g'|_{Y_h}$ and generic fibre φ. □

(This is the case $s = \xi$, S affine, of Ex. 2.5, in the situation we need.)

The finite case. Suppose $X_\xi \to \operatorname{Spec} K$ is finite. Taking $V = \operatorname{Spec} K$ in the definition, $X_\xi = \operatorname{Spec} C$ with C a finite K-algebra; let $e_1, \ldots, e_m$ be a K-basis of C. There is $0 \neq d \in R$ such that 1 and all products e_ie_j lie in $C_0 := \bigoplus_l R_d e_l$ (clear denominators in the finitely many coefficients). Then C_0 is an R_d-subalgebra of C, free of rank m over R_d, and $C_0 \otimes_{R_d} K = C$. Put $Y = \operatorname{Spec} C_0$, a finite scheme over $R' = R_d$ (a Noetherian domain with fraction field K), with $Y_K = \operatorname{Spec} C = X_\xi$. Both X_d and Y are of finite type over R'. By Lemma Sp there are $h \neq 0$ and an R_{dh}-isomorphism $X_{dh} \simeq Y_{dh} = Y \times_{R_d} R_{dh}$, which is finite over R_{dh} by Ex. 3.17(b) (base change). Hence $X \times_S U \to U$ is finite for $U = D(dh)$, which is non-empty since $dh \neq 0$ in the domain R.

The projective case. Suppose $X_\xi \to \operatorname{Spec} K$ is projective, i.e. there is a closed immersion $j : X_\xi \to \mathbb{P}^n_K$ over K.

Lemma M. There is a closed subscheme Y of $\mathbb{P}^n_R$ and a K-isomorphism $Y_K \simeq X_\xi$.

Proof. Let $\varepsilon : \mathbb{P}^n_K \to \mathbb{P}^n_R$ be the projection and $e = \varepsilon \circ j$. Let $Q_i = D_+(T_i) = \operatorname{Spec} A_i$, $A_i = R[T_l/T_i]_l$. Then $\varepsilon^{-1}(Q_i) = \operatorname{Spec}(A_i \otimes K) =: Q_{iK}$, and $j^{-1}(Q_{iK}) \simeq \operatorname{Spec}(A_i \otimes K/J_i)$ for the ideal $J_i = \operatorname{Ker}(A_i \otimes K \to \mathcal{O}(j^{-1}Q_{iK}))$ (Prop. 2.3.20). The scheme X_ξ is Noetherian (closed in $\mathbb{P}^n_K$), so every morphism out of it, in particular e, is quasi-compact. Also $e^{-1}(Q_i) = j^{-1}(Q_{iK})$ is affine. Let $Y = V(\mathcal{J}) \subseteq \mathbb{P}^n_R$ be the scheme-theoretic closure of e (Ex. 2.3.17), a closed subscheme of $\mathbb{P}^n_R$, with $\mathcal{J} = \operatorname{Ker}(\mathcal{O}_{\mathbb{P}^n_R} \to e_*\mathcal{O}_{X_\xi})$. By Prop. 2.3.20, $Y \cap Q_i = \operatorname{Spec} A_i/I_i$ with $I_i = \mathcal{J}(Q_i) = \operatorname{Ker}(A_i \to \mathcal{O}(e^{-1}Q_i)) = A_i \cap J_i$ (the preimage of J_i under $A_i \to A_i \otimes K$). We claim $I_i(A_i \otimes K) = J_i$: the inclusion $\subseteq$ is clear, and for $x \in J_i$ choose $0 \neq r \in R$ with $rx \in A_i$; then $rx \in I_i$ and $x = (rx)/r$. Hence the closed subscheme $(Y \cap Q_i) \times_R K = \operatorname{Spec}((A_i/I_i) \otimes K) = \operatorname{Spec}(A_i \otimes K/J_i)$ of Q_{iK} equals $j(X_\xi) \cap Q_{iK}$. So the closed subschemes Y_K and $j(X_\xi)$ of $\mathbb{P}^n_K$ agree over each Q_{iK}, i.e. they have the same ideal sheaf, and are therefore equal. This gives $Y_K \simeq X_\xi$. □

With Y as in Lemma M, Y is of finite type over R (closed in $\mathbb{P}^n_R$, Cor. 3.10, Prop. 2.4). By Lemma Sp (with $R' = R$) there are $h \neq 0$ and an R_h-isomorphism $X_h \simeq Y_h$. Now Y_h is a closed subscheme of $\mathbb{P}^n_{R_h}$, so $Y_h \to \operatorname{Spec} R_h$ is projective, and therefore so is $X_h \to \operatorname{Spec} R_h$. Thus $X \times_S U \to U$ is projective for $U = D(h) \neq \emptyset$. □

Exercise 3.25

> **Problem.** Let X, Y be schemes of finite type over a locally Noetherian scheme S. Let $f : X \to Y$ be a finite morphism of S-schemes. Let us fix a point $s \in S$ such that $X_s \neq \emptyset$. (a) Let us suppose that $f_s : X_s \to Y_s$ is a closed immersion. Show that there exists an open neighborhood U of s such that $f_U : X_U \to Y_U$ is a closed immersion. (b) Let us suppose that Y is reduced and that f is surjective. Show the analogue of assertion (a) for isomorphisms (compare to Exercise 2.5).

Solution. The statement is false as printed. Let $S = \operatorname{Spec} \mathbb{Z}$, $s = (p)$, $C = \mathbb{Z}[t]$, $Y =$

$\operatorname{Spec} C = \mathbb{A}^1_{\mathbb{Z}}$ and $X = \operatorname{Spec}(C \times C/(1-pt)) = Y \sqcup V(1-pt)$ with f the identity on Y and the closed immersion on $V(1-pt) \simeq \operatorname{Spec} \mathbb{Z}[1/p]$. Then X, Y are of finite type over S, Y is reduced, f is finite and surjective, and $X_s \neq \emptyset$. Since $1 - pt \equiv 1 \pmod p$, the fibre $V(1-pt)_s$ is empty, so f_s is the identity of $\mathbb{A}^1_{\mathbb{F}_p}$: it is an isomorphism. But for every open $U \ni s$ of S there is a prime $q \neq p$ in U; choose $c \in \mathbb{Z}$ with $pc \equiv 1 \pmod q$. The closed point $y = (q, t - c)$ of Y_U lies on $V(1-pt)$ and has two preimages in X_U, one in each component. So f_U is not injective, hence not a closed immersion (nor an isomorphism). The point is that the conclusion concerns a neighbourhood of s in S, whereas the argument only gives a neighbourhood of the fibre Y_s in Y; the passage from one to the other needs that $Y \to S$ is a closed map (e.g. proper). The hypothesis $X_s \neq \emptyset$ and the local Noetherianity of S are not needed below.

Corrected statement. Let $f : X \to Y$ be a finite morphism of S-schemes, $s \in S$. (a′) If f_s is a closed immersion, there is an open subset $W \subseteq Y$ containing the fibre Y_s such that $f^{-1}(W) \to W$ is a closed immersion. If moreover $Y \to S$ is a closed map (for example proper), then there is an open neighbourhood U of s such that $f_U : X_U \to Y_U$ is a closed immersion. (b′) If Y is reduced, f is surjective and f_s is an isomorphism, then in (a′) the morphism $f^{-1}(W) \to W$ is an isomorphism, and when $Y \to S$ is closed, f_U is an isomorphism for a suitable open neighbourhood U of s.

Proof of (a′). Let $S_0 = \operatorname{Spec} R \ni s$ be an affine open, $\mathfrak{p} \subset R$ the prime of s, and let $Y_j = \operatorname{Spec} C_j$ be affine opens covering $\pi_Y^{-1}(S_0) \supseteq Y_s$. Since f is finite, $f^{-1}(Y_j) = \operatorname{Spec} B_j$ with B_j a finite C_j-module. The fibre over s of $f^{-1}(Y_j) \to Y_j$ is $\operatorname{Spec}(B_j \otimes_R k(s)) \to \operatorname{Spec}(C_j \otimes_R k(s))$. It is the restriction of the closed immersion f_s over an affine open set, so (Prop. 2.3.20) $C_j \otimes_R k(s) \to B_j \otimes_R k(s)$ is surjective. Let $Q_j = B_j / \operatorname{Im}(C_j)$, a finite C_j-module. By right exactness of $\otimes$, $Q_j \otimes_R k(s) = 0$.

Let $y \in Y_j$ lie over s, with prime $\mathfrak{q} \subset C_j$. Then $k(y)$ is a $C_j \otimes_R k(s)$-algebra and $Q_j \otimes_{C_j} k(y) = (Q_j \otimes_R k(s)) \otimes_{C_j \otimes k(s)} k(y) = 0$. By Nakayama's lemma for the finite module $(Q_j)_{\mathfrak{q}}$ over the local ring $(C_j)_{\mathfrak{q}}$, $(Q_j)_{\mathfrak{q}} = 0$. Since Q_j is finitely generated, there is $c \in C_j \setminus \mathfrak{q}$ with $cQ_j = 0$. Let W_j be the union of all such open sets $D(c) \subseteq Y_j$ (that is, $W_j = Y_j \setminus \operatorname{Supp} Q_j$); it contains all points of Y_j over s. Over each such $D(c)$ the map $(C_j)_c \to (B_j)_c$ is surjective ($(Q_j)_c = 0$), so $f^{-1}(D(c)) = \operatorname{Spec}(B_j)_c \to D(c)$ is a closed immersion (Lemma 2.3.17). Being a closed immersion is local on the target, so $f^{-1}(W) \to W$ is a closed immersion for $W = \bigcup_j W_j$, which contains Y_s.

If $\pi_Y : Y \to S$ is closed, then $F = \pi_Y(Y \setminus W)$ is closed and does not contain s. Let $U = S \setminus F$. Then $Y_U = \pi_Y^{-1}(U) \subseteq W$, and f_U is the restriction of the closed immersion $f^{-1}(W) \to W$ over Y_U, hence a closed immersion.

Proof of (b′). If f_s is an isomorphism it is a closed immersion, so (a′) applies. The closed immersion $f^{-1}(W) \to W$ is surjective because f is. Over an affine open $\operatorname{Spec} C \subseteq W$ it is $\operatorname{Spec} C/J \to \operatorname{Spec} C$ with $V(J) = \operatorname{Spec} C$, so J is contained in the nilradical, which is 0 as $W \subseteq Y$ is reduced. Thus $J = 0$ and the closed immersion is an isomorphism. The second assertion follows as in (a′). □

Exercise 3.26

Problem. We will study the example (kindly communicated by Florian Pop) of a valuation ring having 'very many' prime ideals. Let k be a field. Let I be a totally ordered set. Let

$K = k(T_i)_{i\in I}$ be the field of rational fractions whose variables are indexed by I. (a) Let $\Gamma = \mathbb{Z}^{(I)}$ be the direct sum of copies of $\mathbb{Z}$ indexed by I. This is a group which is totally ordered by the lexicographical order. Let us denote the canonical basis of $\mathbb{Z}^{(I)}$ by $\{e_i\}_i$. Show that there exists a unique valuation (up to isomorphism) $\nu : K^* \to \Gamma$ such that $\nu(T_i) = e_i$ and $\nu(k^*) = 0$. (b) Let $i \in I$. Show that the radical of the set $\mathfrak{p}(i) := \{a \in K \mid \nu(a) \geq e_i\}$ is a prime ideal of $\mathcal{O}_\nu$. Show that the map $I \to \operatorname{Spec}\mathcal{O}_\nu$ defined in this way is injective and increasing, if we endow $\operatorname{Spec}\mathcal{O}_\nu$ with the order relation determined by the inclusion. (c) Let $\theta : \mathbb{Z}^{(I)} \setminus \{0\} \to I$ be the map defined by $\theta((n_i)_i) =$ the smallest i such that $n_i \neq 0$. Let $\mathfrak{p} \in \operatorname{Spec}\mathcal{O}_\nu$. Show that if $\theta(\nu(\mathfrak{p}))$ is bounded from above by an index i, then for every $j > i$, we have $\mathfrak{p} \subseteq \mathfrak{p}(j)$. Show that if $\theta(\nu(\mathfrak{p}))$ is not bounded, then $\mathfrak{p} = \mathfrak{m}_\nu$.

Solution. The order on Γ. For $\gamma = (n_i) \neq 0$ in Γ (finite support), $\theta(\gamma)$ is the smallest index in the support. The lexicographic order is: $\gamma > 0$ iff $n_{\theta(\gamma)} > 0$; and $\gamma > \delta$ iff $\gamma - \delta > 0$. Let $\Gamma_+ = \{\gamma \neq 0 : n_{\theta(\gamma)} > 0\}$. Then $\Gamma = \Gamma_+ \sqcup \{0\} \sqcup (-\Gamma_+)$, and Γ_+ is closed under addition: for $\gamma, \gamma' \in \Gamma_+$ with $\theta(\gamma) < \theta(\gamma')$ the sum has smallest index $\theta(\gamma)$ with positive coefficient; if $\theta(\gamma) = \theta(\gamma')$ the coefficient there is the sum of two positive integers. So $\leq$ is a total order compatible with addition. We use three comparison facts, with $i \in I$: (F1) if $\gamma > 0$ and $\theta(\gamma) < i$, then $\gamma > e_i$ (the difference $\gamma - e_i$ has smallest index $\theta(\gamma)$ with positive coefficient); (F2) if $\gamma > 0$ and $\theta(\gamma) = i$, then $2\gamma > e_i$ (the coefficient of $2\gamma - e_i$ at i is $2n_i - 1 \geq 1$ and lower indices vanish); (F3) if $\gamma > 0$ and $\theta(\gamma) > i$, then $m\gamma < e_i$ for all $m \geq 1$ (the coefficient of $e_i - m\gamma$ at i is 1 and lower indices vanish).

(a) Existence. Every non-zero polynomial $P = \sum_\alpha c_\alpha T^\alpha \in k[T_i]_{i\in I}$ is a finite sum over exponents $\alpha \in \mathbb{N}^{(I)} \subseteq \Gamma$. Set $\nu(P) = \min\{\alpha : c_\alpha \neq 0\} \in \Gamma$ (a minimum in the totally ordered group Γ, over a finite set). If $P, Q \neq 0$, let α_0, β_0 be the minimal exponents of P, Q. In PQ the monomial $T^{\alpha_0+\beta_0}$ has coefficient $c_{\alpha_0} d_{\beta_0} \neq 0$ (all other products of monomials $T^\alpha T^\beta$ with $\alpha \geq \alpha_0$, $\beta \geq \beta_0$, $(\alpha, \beta) \neq (\alpha_0, \beta_0)$ have exponent $\alpha + \beta > \alpha_0 + \beta_0$, so they do not contribute to it), and every monomial of PQ has exponent $\geq \alpha_0 + \beta_0$. Hence $\nu(PQ) = \nu(P) + \nu(Q)$. Also all exponents occurring in $P + Q$ occur in P or in Q, so $\nu(P + Q) \geq \min(\nu(P), \nu(Q))$. Now define $\nu(P/Q) = \nu(P) - \nu(Q)$ on K^*; this does not depend on the representation ($P/Q = P'/Q'$ gives $\nu(P) + \nu(Q') = \nu(P') + \nu(Q)$), ν is a homomorphism, and $\nu(a + b) \geq \min(\nu(a), \nu(b))$ follows from the case of polynomials by writing $a + b = (PQ' + P'Q)/(QQ')$. We have $\nu(T_i) = e_i$ and $\nu(c) = 0$ for $c \in k^*$ (exponent 0). So ν is a valuation as in Def. 3.22.

Uniqueness. Let ν' be a valuation with $\nu'(T_i) = e_i$ and $\nu'(k^*) = 0$. For a monomial, $\nu'(cT^\alpha) = \alpha$. The monomials of a polynomial P have pairwise distinct exponents, hence pairwise distinct values of ν'. For a sum of elements with pairwise distinct valuations, ν' equals the minimum (if $\nu'(x) < \nu'(y)$ then $\nu'(x + y) \geq \nu'(x)$ and $\nu'(x) = \nu'(x + y - y) \geq \min(\nu'(x + y), \nu'(y))$ forces $\nu'(x + y) = \nu'(x)$). So $\nu'(P) = \nu(P)$ for polynomials, hence $\nu' = \nu$ on K^*.

(b) Let $\mathcal{O}_\nu = \{\nu \geq 0\}$ and $\mathfrak{m}_\nu = \{\nu > 0\}$. Since $e_i > 0$, the set $\mathfrak{p}(i)$ is contained in $\mathfrak{m}_\nu$, and it is an ideal of $\mathcal{O}_\nu$ (closed under sums since $\nu(a+b) \geq \min$, and under multiplication by elements of $\mathcal{O}_\nu$). Let $\mathfrak{q}(i) = \sqrt{\mathfrak{p}(i)} = \{a \in \mathcal{O}_\nu : a^m \in \mathfrak{p}(i) \text{ for some } m \geq 1\} = \{a : m\,\nu(a) \geq e_i \text{ for some } m \geq 1\}$. By (F1)–(F3), for $a \neq 0$ with $\nu(a) = \gamma > 0$:

$$a \in \mathfrak{q}(i) \iff \theta(\gamma) \leq i.$$

Indeed if $\theta(\gamma) < i$ then $\gamma > e_i$ by (F1); if $\theta(\gamma) = i$, then $2\gamma > e_i$ by (F2); if $\theta(\gamma) > i$ then $m\gamma < e_i$ for all m by (F3). (If $\nu(a) = 0$ then $m\nu(a) = 0 < e_i$, so $a \notin \mathfrak{q}(i)$.) Thus $\mathfrak{q}(i) = \{0\} \cup \{a : \nu(a) > 0,\ \theta(\nu(a)) \leq i\}$. It is a proper ideal (the radical of an ideal), and it is

prime: let $a, b \in \mathcal{O}_\nu \setminus \mathfrak{q}(i)$. Then $\nu(a), \nu(b)$ are each 0 or have $\theta > i$ (all indices in the support $> i$). So $\nu(ab) = \nu(a) + \nu(b)$ is 0 or has its support within indices $> i$, i.e. $\theta(\nu(ab)) > i$. So $ab \notin \mathfrak{q}(i)$.

Injective and increasing. If $i < j$, then $\{\theta \le i\} \subseteq \{\theta \le j\}$ gives $\mathfrak{q}(i) \subseteq \mathfrak{q}(j)$. Moreover $T_j \in \mathfrak{q}(j) \setminus \mathfrak{q}(i)$, since $\nu(T_j) = e_j$ has $\theta = j > i$. So $i < j \Rightarrow \mathfrak{q}(i) \subsetneq \mathfrak{q}(j)$, and the map $i \mapsto \mathfrak{q}(i)$ is injective and strictly increasing.

(c) Let $\mathfrak{p} \in \operatorname{Spec} \mathcal{O}_\nu$. As $\mathcal{O}_\nu$ is local with maximal ideal $\mathfrak{m}_\nu$, $\mathfrak{p} \subseteq \mathfrak{m}_\nu$, so $\nu(a) > 0$ for $0 \neq a \in \mathfrak{p}$ and $\theta(\nu(a))$ is defined. (In (c) the set $\mathfrak{p}(j)$ may be taken to be the set itself or its radical $\mathfrak{q}(j) \supseteq \mathfrak{p}(j)$; we prove the stronger inclusion in the set $\mathfrak{p}(j)$.)

Bounded case. Suppose $\theta(\nu(a)) \le i$ for all $0 \neq a \in \mathfrak{p}$. Let $j > i$ and $0 \neq a \in \mathfrak{p}$. Then $\theta(\nu(a)) \le i < j$, so $\nu(a) > e_j$ by (F1), i.e. $a \in \mathfrak{p}(j)$. Hence $\mathfrak{p} \subseteq \mathfrak{p}(j) \subseteq \mathfrak{q}(j)$.

Unbounded case. Suppose $\theta(\nu(\mathfrak{p} \setminus 0))$ has no upper bound in I. Let $0 \neq b \in \mathfrak{m}_\nu$, $\beta = \nu(b) > 0$, $i_0 = \theta(\beta)$. As $\theta(\nu(\mathfrak{p} \setminus 0))$ is not bounded above by i_0 and I is totally ordered, there is $a \in \mathfrak{p}$, $a \neq 0$, with $l := \theta(\nu(a)) > i_0$. Then $\beta - \nu(a)$ has smallest index i_0 (here $\nu(a)$ vanishes at indices $\le i_0$) with coefficient $n_{i_0}(\beta) > 0$, so $\nu(b) > \nu(a)$, i.e. $b/a \in \mathcal{O}_\nu$. Then $b = a \cdot (b/a) \in \mathfrak{p}$. Hence $\mathfrak{m}_\nu \subseteq \mathfrak{p}$, and $\mathfrak{p} = \mathfrak{m}_\nu$. □

Exercise 3.27

> **Problem.** (A scheme without closed point) Let I be a totally ordered set without maximal element. Let $\mathcal{O}_\nu$ be the valuation ring defined in the preceding exercise, with maximal ideal $\mathfrak{m}_\nu$. Show that the scheme $\operatorname{Spec} \mathcal{O}_\nu \setminus \{\mathfrak{m}_\nu\}$ has no closed point.

Solution. Let $U = \operatorname{Spec} \mathcal{O}_\nu \setminus \{\mathfrak{m}_\nu\}$; it is an open subscheme since $\{\mathfrak{m}_\nu\}$ is closed. If $I = \emptyset$ then $\mathcal{O}_\nu = k$ and $U = \emptyset$, and there is nothing to prove; so let $I \neq \emptyset$. Let $\mathfrak{p} \in U$, so $\mathfrak{p} \neq \mathfrak{m}_\nu$. We show that $\{\mathfrak{p}\}$ is not closed in U, by exhibiting a prime $\mathfrak{q}$ with $\mathfrak{p} \subsetneq \mathfrak{q} \subsetneq \mathfrak{m}_\nu$. Then $\mathfrak{q} \in U$ and $\mathfrak{q} \in V(\mathfrak{p}) \cap U$, which is the closure of $\{\mathfrak{p}\}$ in U, with $\mathfrak{q} \neq \mathfrak{p}$.

By Ex. 3.26(c), since $\mathfrak{p} \neq \mathfrak{m}_\nu$, the set $\theta(\nu(\mathfrak{p} \setminus 0))$ is bounded above by some $i \in I$ (if $\mathfrak{p} = 0$ this holds trivially). As I has no maximal element, there are $j, j' \in I$ with $i < j < j'$. By Ex. 3.26(c), $\mathfrak{p} \subseteq \mathfrak{p}(j) \subseteq \mathfrak{q}(j)$, where $\mathfrak{q}(j) = \sqrt{\mathfrak{p}(j)}$ is the prime of Ex. 3.26(b). By Ex. 3.26(b), $\mathfrak{q}(j) \subsetneq \mathfrak{q}(j')$. Finally, $\mathfrak{q}(j') \neq \mathfrak{m}_\nu$: pick $l > j'$ in I; then $T_l \in \mathfrak{m}_\nu$ ($\nu(T_l) = e_l > 0$) and $T_l \notin \mathfrak{q}(j')$ because $\theta(e_l) = l > j'$. So

$$\mathfrak{p} \subseteq \mathfrak{q}(j) \subsetneq \mathfrak{q}(j') \subsetneq \mathfrak{m}_\nu,$$

and $\mathfrak{q} = \mathfrak{q}(j')$ satisfies $\mathfrak{p} \subsetneq \mathfrak{q} \subsetneq \mathfrak{m}_\nu$. Hence no point of U is closed in U. □